VOLUME 1 OF TWO · SESSION 1

Clinical Reasoning Manual

NBEO Part II PAM / TMOD. An emergency gate, condition cards, and a full 175-item simulation with a generated, auditable key.

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175	35	3.5	105
ITEMS IN THIS VOLUME	CASES, OF 3 TO 6 ITEMS	HOURS, ONE SITTING	PAGES, NO FILLER

WHAT CHANGED IN THIS EDITION

The previous key was 82% option A, and 95% in Session 2: guessing A scored 272 of 333 items with no stem read. Letters are now generated to a distribution and audited on page 4, options are length-matched, and every rationale explains the tempting wrong answer as well as the right one.

HOW THE TWO VOLUMES DIVIDE

Each volume is complete: the same reference apparatus, then one session. Cases come first and the teaching keys follow in Part 9, so a session can be sat cold. Volume Two carries the other 175 items, on the same letter distribution, so the two scores are comparable.

Six passes

Reading this manual is not studying it. Retrieval under time pressure is what transfers to the exam room.

PASS	WORK	PROOF IT WORKED
1 · Systems	Learn the algorithm, the emergency gate, the condition cards, the drug anchors.	You can state recognition, danger, next test, treatment, contraindication, follow-up, and failure trigger without looking.
2 · Retrieval	Closed-book recall at roughly 1, 3, 7, 14, and 30 days, always from a changed prompt.	Successful recalls are logged. Hours are not.
3 · Session 1	175 items, 210 minutes, uninterrupted, confidence marked on every item.	Scored by item type as well as total, with pacing and error categories written down.
4 · Repair	High-confidence misses first. Each becomes one if-then rule.	Every false rule has a replacement you can recite.
5 · Session 2	Second 175 items under identical conditions.	Weak item types improve; late-session accuracy holds.
6 · Taper	Red flags, drugs, formulas, image algorithms, logistics.	No new content in the final 48 hours. Sleep protected.

Error log: seven fields, one line each

Concept	Short disease or decision label.
Answer contrast	What you chose against what was correct.
Error type	Knowledge, discrimination, image reading, management sequence, calculation, or misreading the stem.
False rule	The exact belief that produced the miss, written as a sentence.
New rule	One executable if-then replacement.
Next recall	Tomorrow for a high-confidence miss, three days for low-confidence, seven days after a success.
Transfer test	A changed patient or the competing diagnosis.

WHY SCORE BY ITEM TYPE
The blueprint weights treatment and management far above basic science, and TMOD is decided separately. A 74% total can hide a failing treatment score. Every item here is tagged, so scoring by tag takes two minutes and tells you what to repair.

INTERLEAVE, DO NOT BLOCK
Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital. The exam tests the discriminator, never the textbook description.

What is checked, and what is not

A guide that labels its own uncertainty is more useful than one that asserts it was verified. Every factual claim carries one of three states.

Verification table

CLAIM	STATE	BASIS
350 items, presented as cases of 3-6 items	VERIFIED	NBEO exam page, fetched 15 Aug 2026
Two sessions of 175 items, 3.5 hours each	VERIFIED	NBEO exam page
Approximately 120 TMOD items, separate breakout decision	VERIFIED	NBEO exam page
Delivered at Pearson centres; tutorial being updated	VERIFIED	NBEO exam page
Per-domain item ranges, 17 domains	UNCONFIRMED	Transcribed from prior edition. Content Matrix PDF resolves but was not machine-read
Item-type percentage bands	UNCONFIRMED	As above
Deep link to the Pearson tutorial	REMOVED	URL did not resolve; reach it from the exam page

The defect this edition exists to fix

ANSWER KEY	SESSION 1	SESSION 2	BOTH
Single-answer items	161	172	333
Correct answer was A	67.1%	95.3%	81.7%
Longest run of one letter	30	140	-
Letters never correct	E	D, E	E

WHAT THAT MEANT IN PRACTICE

Answering A to every single-answer item scored 272/333, about 82%, without reading one stem. Session 2 alone scored 95%. The two sessions were also biased by different amounts, so a Session 1 score could not be compared with a Session 2 score, which broke the guide's own repair-and-confirm workflow.

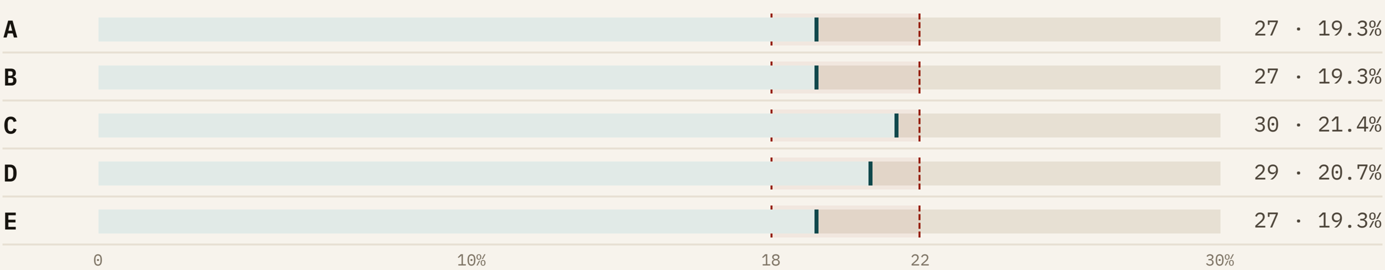
REBUILD RULE, AUDITABLE

Each letter A to E is correct on 18-22% of single-answer items in each session, no letter repeats more than three times consecutively, all five letters appear in both sessions, and option length is matched within each item. The build fails if any of that breaks. Re-run `tools/build.py --audit` after any edit to the bank.

The key, audited

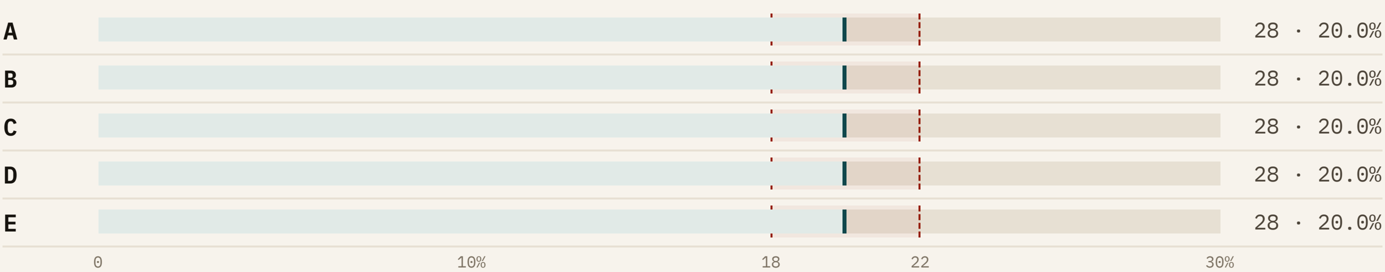
A biased key trains the wrong skill and then flatters you for it. This page is the evidence that the bank you are about to sit does not.

Session 1, 140 single-answer items



Share of single-answer items on which each letter is correct. Dashed band is the 18-22% target. Longest run of one letter: 3. The best single-letter guess scores 30 of 140, 21%.

Session 2, 140 single-answer items



Share of single-answer items on which each letter is correct. Dashed band is the 18-22% target. Longest run of one letter: 3. The best single-letter guess scores 28 of 140, 20%.

How the letters were chosen

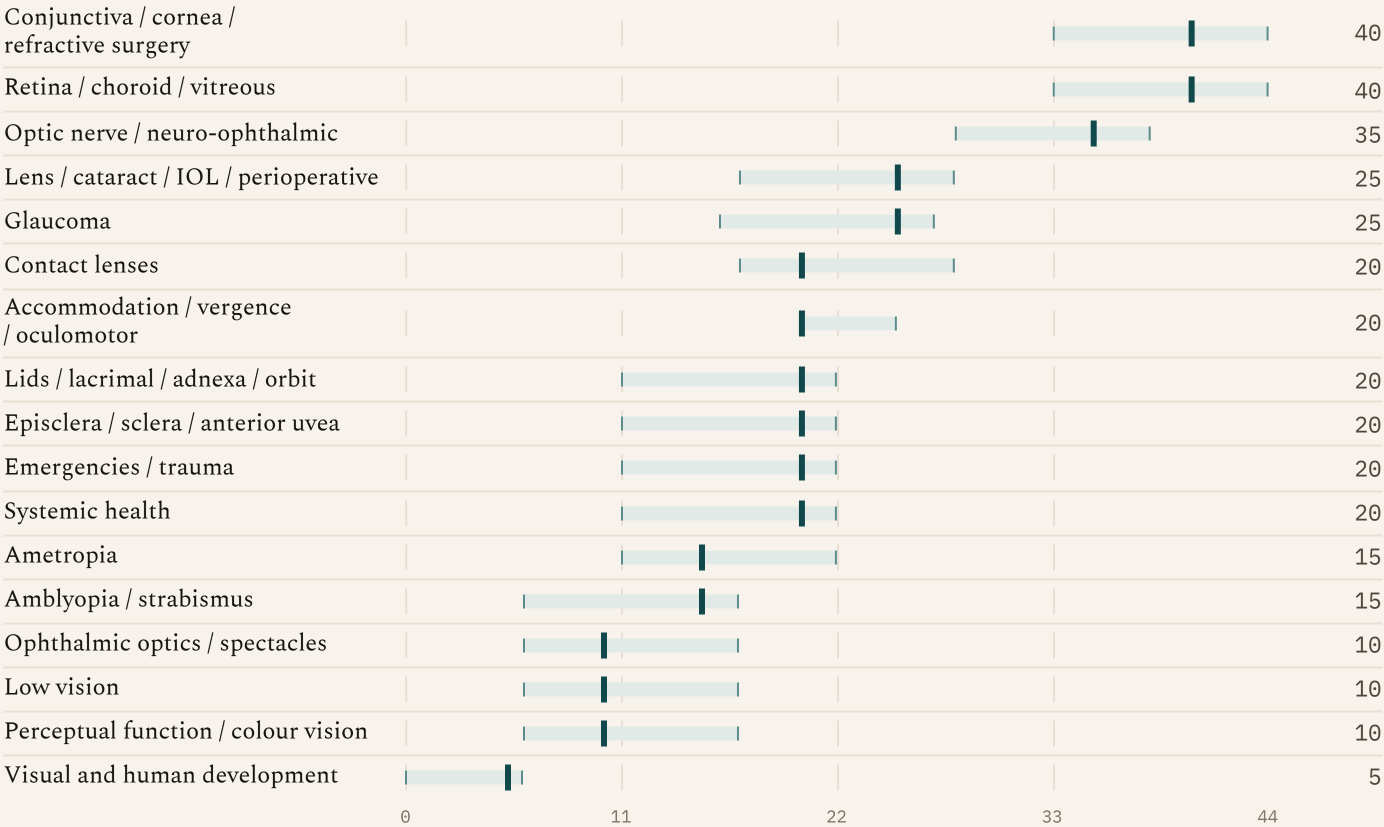
Not by the author	Item writers mark the correct option in place and never see a letter. Letters are assigned afterwards by a generator, so a habit of putting the right answer first cannot survive into print.
Even by construction	Each session's single-answer items are dealt an even multiset of A to E, so every letter lands on 19 to 21% before any adjustment.
No streaks	The sequence is rejected and redealt until no letter is correct more than three times in a row.
Order preserved	Options are rotated, not shuffled, so graded or paired options stay adjacent and a numeric ladder stays in order.
Length controlled	Within each item, the longest option is at most about twice the shortest, and across a session the correct option is the longest no more often than a third of the time.
Checked on build	The manual will not build if any of those conditions fails. The numbers above are printed from the same run that produced these pages.

TEST IT YOURSELF

Mark A for every single-answer item in either session and score it against the answer key in Part 10. You should land near a fifth. The previous edition scored 82% for exactly that, and 95% in Session 2, which is set out on page 3.

Know the target

Band shows the published range. Bar shows this manual's 350-item allocation. Scale runs 0 to 44 items.



READ IT THIS WAY

Cornea and retina alone can carry 88 items. Revision split evenly across 17 domains is misallocated by a factor of eight at the extremes. Development items usually arrive inside paediatric cases.

Case conventions worth memorising

BVA	Best achievable or best-corrected acuity. If it is reduced, pinhole improvement has already been accounted for, so do not spend an item hunting refractive causes.
Near acuity	Assume 16 inches unless the case says otherwise.
Fields	Side by side: OD right, OS left. Stacked: OD above OS.
Images	Fundus, OCT, fields, angiography, ultrasound, radiology, anterior segment. Describe before diagnosing.

245 ITEMS · NORMAL HEALTH, DISEASE, TRAUMA	105 ITEMS · REFRACTIVE, SENSORY, OCULOMOTOR

Where the marks are

Four item types, weighted very unevenly. Every item here is tagged, and the score sheets total by tag.

Item-type weighting and this manual's allocation

ITEM TYPE	BAND	HERE	WHAT IT ACTUALLY ASKS
Treatment and management	40-57%	155 · 44%	Choice, mechanism, dose, schedule, duration, follow-up, education, escalation.
Diagnosis	26-34%	119 · 34%	Most likely diagnosis, the supporting or missing datum, the next test.
Basic science, applied	11-17%	58 · 17%	Anatomy, physiology, microbiology, optics, pharmacology, tied to this patient.
Law, ethics, public health	4-10%	18 · 5%	Duty, consent, refusal, records, delegation, telehealth, screening arithmetic.

Bands transcribed, marked unconfirmed on page 3. The right-hand column is the real allocation across 350 items and 17 domains, counted at build time. 70 items are all-or-none, a format rather than a type.

Two item formats, two disciplines

Single answer

ONE OF FIVE

READ	The question before the options. Decide what you think, then look.
COMPARE	Options in pairs. The pair differing on one variable shows what is tested.
COMMIT	Once. Changed management answers usually move to the more cautious wrong option.
TRAP	Length is not a signal in this manual. Correct options are the longest in roughly a fifth of items, which is chance.

Select all that apply

ALL OR NONE

COUNT	Underline the requested number before reading the options.
JUDGE	Each option alone, true or false for this patient.
PREFER	Patient-specific over merely true.
SCORE	Every correct choice and no incorrect choice. A partial set earns zero.
TRAP	Findings that prove the diagnosis are not the same as findings that measure organ function. Read which the item wants.

What a complete management answer contains

Immediate action	What happens in the next minutes, before any prescription.
Agent and route	Drug or procedure, form, concentration, site of action.
Dose and duration	Loading and maintenance, frequency, taper, stop point.
Contraindication	The host factor in the stem that changes the choice.
Follow-up interval	A number, not as needed. Often the only difference between two options.
Measurable response	What you will remeasure to know it worked.
Escalation trigger	What failure looks like, and what you do then.

HOW THIS IS EXAMINED

Wrong options are right actions in the wrong order, right drugs at the wrong interval, or plans missing the contraindication the stem planted. An option with no interval and no failure trigger is incomplete.

Pacing, and the seven-step algorithm

175 items in 210 minutes is 72 seconds per item. The cost of running late is not the last case, it is the rereading you do in the middle.

Budget by case size

CASE	BUDGET	MOVE
3 items	4 min	Read once, answer the direct items, flag one uncertainty.
4 items	5 min	Commit before option-hunting for a rescue.
5 items	6 min	The typical unit. Do not reread the stem for each item.
6 items	7 min	Diagnosis, threat, test, treatment, follow-up, then the science.

Checkpoints: item 50 by 60 minutes, item 100 by 120, item 150 by 180. That leaves 30 minutes for the last 25 items and the flags. Behind schedule means cutting rereading, not standards.

SITUATION	DECISION
Two options survive and you have spent 90 seconds	Choose the one that acts sooner on the more dangerous possibility, flag, move.
The stem contains a number you have not used	Reread. Numbers are planted to be used: pulse, weight, dose, pressure, age.
An image is described but not shown	Answer from the described findings. Do not invent what is not stated.
You do not know the disease	Answer the management item anyway. Threat level is often decidable without the name.
15 minutes left, six flags	Spend it on flagged management items, not on flagged basic science.

Seven-step case algorithm

1 Frame	Age, laterality, tempo, pain, vision loss, systemic context, medications, risk behaviour.
2 Threat gate	Could delay cost life, the eye, or the fellow eye? Run the eighteen patterns in Part 3 before anything else.
3 Localise	Media, cornea and anterior segment, retina, optic nerve, post-chiasmal pathway, neuromuscular junction, orbit, binocular motor system.
4 Build three	Most likely, most dangerous plausible, best mimic. The mimic is usually one of the five options.
5 Discriminate	The one finding or test that most changes probability or management. Not a panel.
6 Treat in order	Stabilise, protect tissue and vision, treat the cause, prevent recurrence and fellow-eye injury.
7 Close the loop	Interval, measurable response, adverse effects, red flags, referral urgency, and what failure triggers.

INTERLEAVE, DO NOT BLOCK

Study competing pairs together: episcleritis against scleritis, optic neuritis against ischaemic neuropathy, AMD against central serous, HSV against zoster, preseptal against orbital, pupillary block against ciliochoroidal effusion. Part 7 sets them out side by side, because the exam tests the discriminator and never the textbook description.

Six patterns that cannot wait

Recognise, act, and know the specific thing that must not be done. The prohibition is examined as often as the treatment.

Chemical injury

SURFACE

RECOGNISE
Exposure history. Do not wait for acuity or pH before acting.

DO FIRST
Irrigate immediately and copiously. Evert lids, sweep fornices for retained particles, continue until pH is neutral and stays neutral after a pause. Then grade limbal ischaemia and epithelial loss.

NEVER
Neutralise alkali with acid. Delay irrigation to find equipment.

Open globe

TRAUMA

RECOGNISE
Full-thickness wound, Seidel, peaked pupil, uveal prolapse, low IOP with a mechanism.

DO FIRST
Rigid shield, nil by mouth, antiemetic and analgesia, tetanus and systemic antibiotics per protocol, CT orbits when appropriate, immediate surgical referral.

NEVER
Tonometry, gonioscopy, ultrasound, or any pressure on the globe.

Orbital cellulitis

ORBIT

RECOGNISE
Lid swelling plus any of: painful or restricted motility, proptosis, reduced acuity or colour, RAPD, raised IOP, fever, severe chemosis.

DO FIRST
Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ophthalmology and ENT. Drain a subperiosteal abscess when vision or the apex is threatened.

NEVER
Call it preseptal until orbital signs have been actively excluded.

Microbial keratitis

CORNEA

RECOGNISE
Contact lens wear, epithelial defect over a stromal infiltrate, chamber reaction, pain, central location.

DO FIRST
Measure and photograph. Culture central, large, deep, atypical, post-surgical, or unresponsive ulcers. Intensive topical antimicrobial, review inside 24 hours.

NEVER
Send a central ulcer home on conjunctivitis dosing, or patch it.

Acute angle closure

GLAUCOMA

RECOGNISE
Pain, halos, nausea, corneal oedema, shallow chamber, mid-dilated fixed pupil, very high IOP.

DO FIRST
Aqueous suppressants the patient can safely take, systemic acetazolamide if appropriate, control inflammation and nausea, then definitive iridotomy. Assess the fellow eye.

NEVER
Lead with pilocarpine at extreme pressure: an ischaemic sphincter will not respond.

Giant cell arteritis

SYSTEMIC

RECOGNISE
Over 50 with new headache, jaw claudication, scalp tenderness, polymyalgia or constitutional symptoms, chalky pallid disc oedema.

DO FIRST
ESR, CRP, and platelets, and start high-dose systemic corticosteroid immediately. Temporal artery imaging or biopsy follows treatment.

NEVER
Let a normal ESR reassure you, or wait for biopsy. The fellow eye is what is at stake.

CONTINUED ON PAGE 9
Retinal artery occlusion, tear and detachment, endophthalmitis, papilloedema, acute third-nerve palsy and painful Horner syndrome, then six more on page 10 that arrive looking mild.

And six more

Retinal artery occlusion
VASCULAR

RECOGNISE
Sudden painless profound monocular loss, dense RAPD, retinal whitening or cherry-red spot.

DO FIRST
Treat as acute stroke: activate the emergency stroke pathway now, with carotid and cardiac source evaluation and a GCA screen by age.

NEVER
Spend the treatment window on ocular massage or other office manoeuvres.

Tear and detachment
RETINA

RECOGNISE
New flashes, shower of floaters, pigment in the vitreous, vitreous haemorrhage, curtain or field loss.

DO FIRST
Dilated peripheral examination with indentation, B-scan if the view is blocked, urgent retina referral. Macula-on detachment is the most time-critical.

NEVER
Accept a normal central OCT as evidence against a peripheral break.

Endophthalmitis
POSTOPERATIVE

RECOGNISE
Pain and severe acuity loss after surgery or injection, hypopyon, fibrin, vitritis, lost red reflex.

DO FIRST
Same-hour retina-level care: intraocular sampling then intravitreal antibiotics, with vitrectomy by severity.

NEVER
Treat it as routine postoperative inflammation, or wait for culture before treating.

Papilloedema
NEURO

RECOGNISE
Bilateral disc oedema with headache, transient obscurations, pulsatile tinnitus, diplopia, possible sixth-nerve palsy.

DO FIRST
Urgent brain imaging including venous evaluation, then lumbar puncture with opening pressure when imaging makes it safe.

NEVER
Puncture before imaging, or dismiss true oedema as crowded discs without evidence.

Third-nerve palsy
NEURO-VASCULAR

RECOGNISE
Ptosis with the eye down and out; pupil involvement, pain, or an incomplete palsy raises compression.

DO FIRST
Emergency aneurysm-focused CTA or MRA and neurovascular referral. Modern practice images acute isolated palsies.

NEVER
Treat pupil sparing as proof against aneurysm in an acute palsy.

Painful Horner syndrome
NEURO-VASCULAR

RECOGNISE
Acute ptosis and miosis, anisocoria greater in darkness, dilation lag, with head or neck pain.

DO FIRST
Emergency head and neck vascular imaging for carotid dissection.

NEVER
Delay imaging to complete pharmacologic localisation.

TREATMENT GRAMMAR, EVERY TIME

Immediate action, drug or procedure, dose and form and schedule and duration, contraindication, patient education, follow-up interval, the measurable response, and the escalation trigger. An answer that names a drug but no interval or failure trigger is incomplete, and incomplete options are how these items are written.

Six that are missed because they look mild

These are the ones that arrive looking like something routine. The recognition line is the whole point.

Acute retinal necrosis
VIRAL RETINA

RECOGNISE
Peripheral confluent whitening, occlusive arteritis, dense vitritis, rapid circumferential spread, often with pain and a red eye.

DO FIRST
Same-day retina referral. Systemic antiviral therapy starts on clinical suspicion, with intravitreal antiviral cover for posterior pole threat.

NEVER
Treat as idiopathic intermediate uveitis with steroid alone. Steroid without antiviral cover accelerates it.

Sympathetic ophthalmia
UVEA

RECOGNISE
Bilateral granulomatous panuveitis weeks to years after penetrating injury or intraocular surgery to the other eye. The fellow eye is the eye at risk.

DO FIRST
Urgent high-dose systemic corticosteroid with immunomodulatory planning, and specialist co-management.

NEVER
Dismiss mild fellow-eye inflammation after trauma as sympathetic irritation.

Ciliochoroidal effusion angle closure
GLAUCOMA

RECOGNISE
Bilateral acute myopic shift with uniformly shallow chambers, days after starting a sulfa-derived drug such as topiramate.

DO FIRST
Stop the drug with the prescriber, suppress aqueous, cycloplege to deepen the chamber, treat pressure medically.

NEVER
Treat it as pupillary block. Iridotomy does not help and miotics make the anterior rotation worse.

Non-accidental injury
PAEDIATRIC

RECOGNISE
Multi-layered retinal haemorrhages extending to the periphery, retinoschisis, in an infant with a history that does not fit the findings.

DO FIRST
Admit through the paediatric safeguarding pathway the same day, document findings and images verbatim, notify per jurisdiction.

NEVER
Arrange outpatient follow-up and let the child leave, or record an opinion about who caused it.

Retinoblastoma
PAEDIATRIC TUMOUR

RECOGNISE
Leukocoria, new strabismus, or unexplained poor fixation under three years. Any white reflex is abnormal until proven otherwise.

DO FIRST
Same-week referral to ocular oncology for examination under anaesthesia, with B-scan and MRI. Examine the parents and siblings.

NEVER
Biopsy an intraocular mass in a child, and never book routine review for leukocoria.

Pituitary apoplexy
CHIASM

RECOGNISE
Sudden severe headache with bitemporal field loss, ophthalmoplegia, and reduced consciousness or hypotension.

DO FIRST
Emergency department now: urgent MRI, endocrine support for adrenal insufficiency, neurosurgical assessment.

NEVER
Book an outpatient field to confirm a chiasmal defect in an acutely unwell patient.

THE GATE IN ONE SENTENCE
Before you name the disease, decide whether delay costs life, the eye, or the fellow eye, because that decision is answerable from the stem even when the diagnosis is not.

Ocular surface and adnexa

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Preseptal cellulitis
LIDS / ORBIT

PATTERN Lid erythema and oedema with normal acuity, pupils, colour, motility, and globe position.

CONFIRM Clinical. Image when the examination is limited, severe, atypical, or orbital spread is possible.

ACT Oral antibiotic covering skin and sinus organisms, review inside 24-48 hours, written orbital red flags.

TRAP In a febrile child with sinusitis this is a diagnosis of exclusion. Motility and colour are the gate.

Dacryocystitis
LACRIMAL

PATTERN Tender erythematous swelling below the medial canthal tendon, with reflux or tearing.

CONFIRM Clinical, culture the drainage, assess for orbital extension.

ACT Systemic antibiotics, warm compresses, drainage of an abscess, definitive lacrimal surgery once settled.

TRAP Do not probe or irrigate through acutely infected tissue.

Recurrent chalazion
LIDS

PATTERN Firm lesion returning to the same site, madarosis, yellow tarsal thickening, unilateral chronic blepharoconjunctivitis.

CONFIRM Full-thickness biopsy with conjunctival mapping, not a third curettage.

ACT Urgent oculoplastic referral. Evert both lids, examine conjunctiva and regional nodes.

TRAP Sebaceous carcinoma masquerades as chalazion and as blepharitis, and spreads pagetoid.

Thyroid eye disease
ORBIT

PATTERN Lid retraction, proptosis, exposure, restrictive diplopia, apical crowding.

CONFIRM Thyroid function, colour and pupil testing, fields, exophthalmometry; imaging when atypical or neuropathy is suspected.

ACT Surface protection, smoking cessation, urgent specialist care for optic neuropathy or corneal breakdown.

TRAP Activity and severity are separate axes. Muscle belly enlargement with tendon sparing is the signature.

Adenoviral keratoconjunctivitis
CONJUNCTIVA

PATTERN Follicles, watery discharge, preauricular node, high contagion; subepithelial infiltrates may follow.

CONFIRM Clinical; rapid antigen testing helps in selected settings.

ACT Supportive care and strict hygiene. Corticosteroid only for visually significant infiltrates, monitored.

TRAP Contagious period and instrument disinfection matter as much as the drop.

Episcleritis vs scleritis
EPISCLERA / SCLERA

PATTERN Sectoral mobile vessels that blanch with phenylephrine, against deep violaceous non-blanching injection with boring pain and globe tenderness.

CONFIRM Phenylephrine, palpation, B-scan for posterior disease, phenotype-directed systemic workup.

ACT Episcleritis: lubricant, cool compress, NSAID. Scleritis: systemic anti-inflammatory or immunomodulatory therapy, urgent specialist.

TRAP Topical steroid alone is inadequate in scleritis and dangerous if the cause is infectious.

Cornea and refractive surgery

Recall each card in one order: decisive pattern, confirming test, first action, trap. Then name the most dangerous mimic.

Bacterial keratitis

CORNEA

PATTERN Epithelial defect over a dense stromal infiltrate, pain, chamber reaction, lens wear or trauma.

CONFIRM Measure and photograph. Scrape for culture when central, large, deep, atypical, post-surgical or unresponsive.

ACT Intensive topical antimicrobial with a loading phase, no lens wear, review inside 24 hours.

TRAP Never patch, and never treat a central ulcer on conjunctivitis dosing.

HSV keratitis

CORNEA

PATTERN Dendrite with terminal bulbs and reduced corneal sensation for epithelial disease; stromal haze or endothelial oedema with keratic precipitates for immune disease.

CONFIRM Clinical, with sensation testing. PCR when atypical.

ACT Oral or topical antiviral for epithelial disease. Add steroid only for immune stromal or endothelial disease, always under antiviral cover.

TRAP Steroid over an active dendrite converts a treatable ulcer into a geographic one.

Acanthamoeba keratitis

CORNEA

PATTERN Pain far out of proportion to signs, water exposure or poor lens hygiene, radial keratoneuritis, poor response to antibacterials and antivirals.

CONFIRM Culture on non-nutrient agar with bacterial overlay, PCR, or confocal. Repeat if suspicion persists.

ACT Prolonged intensive biguanide therapy, often with a diamidine, cornea specialist involved from the start.

TRAP A ring infiltrate is late and not required for the diagnosis. Steroid worsens it.

Keratoconus

ECTASIA

PATTERN Rising irregular astigmatism, scissoring reflex, Vogt striae, Fleischer ring, inferior steepening and thinning on tomography.

CONFIRM Serial tomography for progression: thinnest point, maximum keratometry, posterior elevation.

ACT Cross-linking for documented progression, specialty lenses for vision. Treat the eye rubbing and the atopy.

TRAP Cross-linking stabilises, it does not correct. Progression is a tomographic judgement, not a refraction change.

Post-LASIK DLK

REFRACTIVE SURGERY

PATTERN Diffuse sterile interface haze in the first days, no epithelial defect, no chamber reaction, minimal pain.

CONFIRM Clinical, staged by extent and central involvement.

ACT Intensive topical steroid, close review. Flap lift and irrigation for stage three or four.

TRAP Focal infiltrate, pain, discharge or chamber reaction is infectious keratitis, where steroid is the wrong answer.

Neurotrophic keratopathy

SURFACE

PATTERN Persistent epithelial defect with rolled edges, minimal pain, reduced sensation, after zoster, HSV, surgery or a fifth-nerve lesion.

CONFIRM Corneal sensation testing before anaesthetic, staged by epithelial and stromal loss.

ACT Preservative-free lubrication, punctal occlusion, autologous serum or nerve growth factor therapy, tarsorrhaphy or bandage lens for progression.

TRAP The absence of pain is why it presents late and perforates. Sensation is the test that names it.

Retina, choroid and vitreous

Neovascular AMD
MACULA

PATTERN New distortion or a central scotoma over days to weeks, subretinal or intraretinal fluid, haemorrhage, pigment epithelial elevation.

CONFIRM OCT for fluid and its compartment, angiography or OCT angiography when the lesion type is unclear.

ACT Urgent retina referral for anti-VEGF therapy, with a loading course and interval-based review. Amsler self-monitoring and fellow-eye counselling.

TRAP Good acuity is a reason to treat quickly, not a reason to wait. Drusen do not explain new fluid.

Central serous chorioretinopathy
MACULA

PATTERN Younger patient, hyperopic shift, mild distortion, a shallow serous detachment with a thick choroid, steroid exposure or high stress.

CONFIRM OCT for subretinal fluid and choroidal thickness. Angiography for chronic or atypical disease.

ACT Remove exogenous corticosteroid where possible, observe acute episodes, treat persistent disease with photodynamic therapy or a mineralocorticoid antagonist per specialist.

TRAP Anti-VEGF is not first-line here, and the corticosteroid in the stem is the answer to the management item.

Proliferative diabetic retinopathy
RETINA

PATTERN New vessels at the disc or elsewhere, vitreous or preretinal haemorrhage, with or without centre-involving oedema.

CONFIRM Dilated examination, OCT for the macula, wide-field imaging or angiography for the periphery.

ACT Panretinal photocoagulation or anti-VEGF per specialist, treat centre-involving oedema, and drive systemic control: glycaemia, blood pressure, lipids, renal review.

TRAP Anti-VEGF without a reliable follow-up plan is unsafe in proliferative disease. Ask who is coming back.

Retinal vein occlusion
VASCULAR

PATTERN Sector or four-quadrant haemorrhage with dilated tortuous veins, cotton wool spots, macular oedema.

CONFIRM OCT for oedema, RAPD and field for ischaemia, blood pressure, glucose, lipids, and a clotting or inflammatory screen in the young.

ACT Anti-VEGF for centre-involving oedema, treat neovascular complications, and manage the vascular risk factors that caused it.

TRAP An ischaemic central occlusion threatens the angle. Undilated review at 4 to 6 weeks is not enough.

Rhegmatogenous detachment
VITREORETINAL

PATTERN Flashes, a shower of floaters, a curtain, relative field loss, pigment or haemorrhage in the vitreous.

CONFIRM Dilated peripheral examination with indentation. B-scan when the view is obscured.

ACT Urgent retina referral. Macula-on disease is the most time-critical thing on this page after an artery occlusion.

TRAP A normal central OCT says nothing about the periphery. Positioning advice is not treatment.

Hydroxychloroquine retinopathy
TOXIC

PATTERN Often asymptomatic early. Parafoveal outer retinal thinning, later a bull's-eye. Asian eyes may show a pericentral pattern.

CONFIRM Baseline then annual screening after five years, earlier with risk: dose above roughly 5 mg/kg, renal impairment, tamoxifen. Fields plus OCT, with mfERG or FAF when equivocal.

ACT Discuss stopping with the prescriber when toxicity is confirmed. Damage does not reverse.

TRAP Fundus appearance is a late sign. Calculate the dose per real body weight and test the pericentral field in Asian patients.

Glaucoma, uvea and the optic nerve

Primary open angle glaucoma
GLAUCOMA

PATTERN Open angle, structural rim and RNFL loss matching a repeatable field defect, often asymmetric, usually asymptomatic.

CONFIRM Gonioscopy, corrected pressure with central thickness in mind, disc photography, OCT, and at least two reliable fields before committing.

ACT Set a target pressure from the damage, the baseline and life expectancy. First-line selective laser trabeculoplasty or a prostaglandin analogue, then reassess against the target.

TRAP Pressure is a risk factor, not the diagnosis. Treating a number without structure and function is how normal-tension disease is missed.

Angle closure mechanisms
GLAUCOMA

PATTERN Pupillary block gives a convex iris that flattens after iridotomy. Plateau iris persists after iridotomy. Effusion gives bilateral myopic shift and uniform shallowing.

CONFIRM Gonioscopy with indentation, anterior segment OCT or ultrasound, and reassessment after any iridotomy.

ACT Iridotomy for block, iridoplasty or long-term miotic for plateau, drug withdrawal with cycloplegia for effusion.

TRAP One treatment does not fit the three mechanisms. Naming the mechanism is the item.

Anterior uveitis
UVEA

PATTERN Pain, photophobia, circumlimbal injection, cells and flare, keratic precipitates, posterior synechiae, pressure high or low.

CONFIRM Grade cells and flare, dilate to check the posterior segment, and phenotype before ordering: granulomatous, bilateral, recurrent, or systemic features.

ACT Topical steroid at a frequency matched to the grade with a real taper, cycloplegia for pain and synechiae, monitor pressure, treat the cause.

TRAP Stopping steroid when the eye looks comfortable causes the rebound. The taper is the treatment plan.

Optic neuritis
OPTIC NERVE

PATTERN Younger patient, subacute monocular loss over days, pain on eye movement, RAPD, dyschromatopsia, disc normal or swollen.

CONFIRM MRI of brain and orbits with contrast, and antibody testing for MOG and aquaporin-4 when the picture is atypical or severe.

ACT Neurology referral. Intravenous corticosteroid speeds recovery without changing final acuity, and is used for severe or bilateral loss and for MRI-defined risk.

TRAP Oral prednisone at standard dose alone is the wrong answer. Painless loss over 50 with a pallid swollen disc is arteritic, not this.

Ischaemic optic neuropathy
OPTIC NERVE

PATTERN Sudden painless loss with an altitudinal defect and a swollen disc. Chalky pallid oedema, jaw claudication and constitutional symptoms mean arteritic.

CONFIRM ESR, CRP and platelets immediately for any suspicion over 50; temporal artery imaging or biopsy after treatment starts.

ACT High-dose systemic corticosteroid at once for arteritic disease. Non-arteritic disease gets vascular risk management and sleep apnoea screening.

TRAP A normal ESR does not exclude arteritis. The fellow eye is what the treatment protects.

Compressive and chiasmal disease
PATHWAY

PATTERN Slowly progressive loss, disc pallor without prior swelling, bitemporal defects respecting the vertical midline, or a junctional defect.

CONFIRM Formal fields both eyes, OCT for a band or bow-tie pattern, and MRI with attention to the sella and chiasm.

ACT Neurosurgery and endocrinology referral by urgency, with acute presentations treated as apoplexy.

TRAP A field defect that respects the vertical midline is never retinal. Pallor with no history of swelling means chronic compression.

Anchor, caveat, monitor

A memorised dose used in the wrong tissue or the wrong host is a wrong answer.

Antimicrobial anchors

INDICATION	ANCHOR	CAVEAT THAT GETS EXAMINED
Bacterial keratitis	Fluoroquinolone for small peripheral ulcers; fortified therapy for central, deep or resistant disease. Load every 5-15 minutes, then hourly.	Culture first when central, large, deep or unresponsive. Do not wait for results to treat.
HSV epithelial keratitis	Aciclovir 400 mg five times daily, or valaciclovir 500 mg three times daily, 7-10 days. Topical ganciclovir 0.15% is an alternative.	No steroid over an active epithelial ulcer. Steroid belongs to immune stromal or endothelial disease, with antiviral cover.
Herpes zoster ophthalmicus	Valaciclovir 1 g three times daily, or aciclovir 800 mg five times daily, 7-10 days, renally adjusted.	Ocular disease lags the rash. A quiet first examination does not end surveillance.
Fungal keratitis	Natamycin 5% hourly initially for filamentous organisms, prolonged course, cornea specialist involved.	Corticosteroid accelerates progression. Vegetative trauma and feathery margins are the clues.
Acanthamoeba keratitis	Intensive biguanide therapy, PHMB or chlorhexidine 0.02%, often with a diamidine, for months.	Pain out of proportion, water exposure. Early disease is mislabelled HSV.
Hyperacute conjunctivitis	Systemic ceftriaxone-based treatment per current CDC guidance, saline lavage, STI testing, partner care.	Copious purulence with corneal risk is an emergency, not a drop problem.

Contraindication anchors

Topical beta-blocker
CAUTION

AVOID Bronchospastic asthma or COPD, bradycardia, heart block, decompensated heart failure.

REDUCE Punctal occlusion and lid closure lower systemic absorption.

TRAP A resting pulse in the low 50s is in the stem for a reason.

Alpha-2 agonist
CAUTION

AVOID Brimonidine in infants and very young children: CNS depression and apnoea.

EXPECT Fatigue, dry mouth, late follicular allergy, hypotension.

TRAP Check interacting systemic drugs before prescribing.

Miotic
CAUTION

AVOID Susceptible peripheral retina, active inflammation, ciliochoroidal effusion mechanisms.

EXPECT Brow ache, induced myopia, poor night vision.

TRAP Useless with an ischaemic sphincter, harmful in topiramate-type effusion.

Topical corticosteroid
CAUTION

AVOID Active HSV epithelial disease, fungal and Acanthamoeba keratitis, uncontrolled bacterial infection.

MONITOR IOP, epithelial healing, cataract, and the taper.

TRAP Name tissue, cause, dose, and monitoring, or the answer is incomplete.

Lowering pressure

Pressure-lowering classes

CLASS	MECHANISM	ANCHOR, AND THE CAVEAT THAT GETS EXAMINED
Prostaglandin analogue	Increases uveoscleral outflow	Once nightly, the largest single-agent reduction. Hyperaemia, lash growth, iris and periocular pigmentation, orbital fat atrophy. Caution with active uveitis or a history of herpetic keratitis.
Beta-blocker	Reduces aqueous production	Twice daily, or once for gel-forming. Avoid in bronchospastic disease, bradycardia, heart block and decompensated failure. A resting pulse in the low 50s is in the stem for a reason.
Alpha-2 agonist	Reduces production, some outflow	Two or three times daily. Late follicular allergy is common. Contraindicated in infants and young children: CNS depression and apnoea.
Topical carbonic anhydrase inhibitor	Reduces production	Two or three times daily. Stinging, metallic taste. Caution with sulfonamide reactions and low endothelial reserve.
Systemic acetazolamide	Reduces production strongly	For acute pressure crises. Paraesthesia, malaise, metabolic acidosis, renal stones, hypokalaemia. Avoid in sickle disease and significant renal impairment.
Rho kinase inhibitor	Increases trabecular outflow	Once nightly, additive to other classes. Hyperaemia is very common, with subconjunctival haemorrhage and corneal verticillata.
Cholinergic agonist	Increases trabecular outflow	Useful in plateau iris and pseudophakic eyes. Brow ache, induced myopia, poor night vision. Useless against an ischaemic sphincter and harmful in ciliochoroidal effusion.

> **CHOOSING THE SECOND AGENT**
> Add a class with a different mechanism, not a second drug of the same one. Check adherence and instillation technique before escalating, and count the preservative load on a compromised surface. Every added bottle costs adherence, which is the commonest reason a target pressure is missed.

Controlling inflammation

Four steps, and the specific thing each one fails at. Naming the step is usually easier than naming where it stops working, which is what the item asks.

Anti-inflammatory ladder, and where each step fails

Topical NSAID
SURFACE AND MACULA

USE	Postoperative pain and macular oedema prophylaxis, episcleritis, some allergic disease.
FAILS	Not adequate for significant intraocular inflammation.
TRAP	Prolonged use on a compromised surface, particularly a neurotrophic or post-refractive cornea, causes melt.

Topical steroid
ANTERIOR SEGMENT

USE	Anterior uveitis, immune stromal keratitis with antiviral cover, significant postoperative inflammation.
FAILS	Does not reach the posterior segment reliably, and does not control scleritis.
TRAP	Contraindicated over active HSV epithelial disease, fungal and Acanthamoeba keratitis, and uncontrolled bacterial infection. Monitor pressure and the taper.

Periocular and intravitreal steroid
POSTERIOR

USE	Intermediate and posterior uveitis, refractory macular oedema.
FAILS	Not for undiagnosed infectious disease, and not a substitute for systemic control in systemic vasculitis.
TRAP	Steroid response and cataract are expected, not surprises. Exclude infection first, every time.

Systemic steroid and steroid sparing
HOST

USE	Scleritis, arteritic ischaemic neuropathy, sympathetic ophthalmia, sight-threatening systemic disease.
FAILS	Long-term control. Immunomodulatory therapy is planned from the start, co-managed.
TRAP	Glucose, bone, blood pressure, mood, infection risk and vaccination status are all part of the correct answer.

HOST FACTORS THAT CHANGE THE DRUG

Pregnancy and breastfeeding, age at either extreme, renal and hepatic function, asthma and cardiac conduction, sulfonamide reactions, sickle disease, immunosuppression, and the systemic drug list. When a stem gives you a weight, a pulse, a creatinine or a pregnancy, it is the discriminator, not decoration.

Arithmetic, fast

The fastest marks on the exam, and the easiest to lose to a sign error.

Formulas

NAME	FORM	USE AND SIGN DISCIPLINE
Spherical equivalent	`sph + cyl/2`	Halve the cylinder with its own sign. -6.00 -2.00 x 180 gives -7.00 D.
Transposition	`sph + cyl, flip, 90°`	New sphere is old sphere plus old cylinder, reverse the cylinder sign, rotate the axis by 90 degrees.
Prentice rule	`Δ = cm × D`	Decentration in centimetres. Base follows the thick edge for plus lenses, the thin edge for minus.
Vertex conversion	`F2 = F1/(1 - dF1)`	Distance in metres, signs preserved. Matters beyond about ±4.00 D.
Kestenbaum estimate	`1 / acuity fraction`	Starting magnification only. Refine by task, working distance, contrast, field, endurance.
Equivalent viewing power	`1 / metres`	10 cm working distance is 10 D.
Hofstetter	`18.5 - 0.30 age`	Average. Minimum is 15 - 0.25 age, maximum 25 - 0.40 age. A screening comparison, not a diagnosis.
Gradient AC/A	`Δphoria / ΔD`	State the sign convention before you calculate.
Screening arithmetic	`TP/(TP+FN)`	Sensitivity. Specificity is TN/(TN+FP). PPV rises with prevalence; NPV falls.

Drills: cover the right column

PROMPT	WORKING AND ANSWER
Transpose +2.00 -3.00 x 180	`-1.00 +3.00 x 090`
Spherical equivalent of -4.00 -2.00 x 090	`-4.00 + (-1.00) = -5.00 D`
4 mm decentration through +5.00 D	`0.4 × 5 = 2Δ`
-10.00 D spectacle at 12 mm, to the corneal plane	`-10/(1 - (0.012 × -10)) = -8.93 D`
Starting magnification for 20/160	`160/20 = 8×`
Hofstetter average amplitude at age 20	`18.5 - 6.0 = 12.5 D`
Hydroxychloroquine 400 mg daily at 60 kg	`6.67 mg/kg, above the 5 mg/kg threshold`
Soft toric rotated 10° left of a 180 axis	`LARS: order 010`
GP steepened 0.50 D, over-refraction -0.50 D, from -3.00 D	`SAM: -3.00 -0.50 -0.50 = -4.00 D`
Anisometropia -2.00 OD, -5.00 OS, 10 mm below centre	`2Δ and 5Δ base down: 3Δ imbalance`

Marks are lost to sign errors in transposition and vertex, to forgetting that Prentice needs centimetres, and to quoting a magnification estimate as a prescription. Write one line of working every time.

Lenses, and the arithmetic that follows

Fitting decisions and screening arithmetic. Both are fast marks, and both are lost to sign errors rather than to ignorance.

Contact lens and refractive decisions that get examined

SITUATION	THE RULE, AND THE ARITHMETIC IF THERE IS ANY
Soft toric rotation	Left add, right subtract, from the lens marking as observed. Rotated 10 degrees left of a 180 axis means order axis 010.
Rigid lens fit change	Steeper add minus, flatter add plus. Steepening 0.50 D from -3.00 D with a -0.50 D over-refraction gives -4.00 D.
Lens-related red eye	Stop wear, exclude infiltrative keratitis, and decide sterile against infectious on pain, epithelial defect, chamber reaction and location.
Vertex distance	Convert beyond about four dioptres. F2 = F1 divided by (1 minus dF1), distance in metres, signs preserved.
Induced prism	Prentice: prism dioptres equal decentration in centimetres times power. Anisometropia through a reading position is the classic item.
Myopia control	Discuss the options with their evidence, expected effect size and rebound, and document the discussion, including the option of doing nothing.

Screening arithmetic, worked once

QUANTITY	FORMULA	WORKED EXAMPLE: 100 WITH DISEASE, 900 WITHOUT, 80 DETECTED, 90 FALSE POSITIVES
Sensitivity	TP/(TP+FN)	80/100 = 80%
Specificity	TN/(TN+FP)	810/900 = 90%
Positive predictive value	TP/(TP+FP)	80/170 = 47%
Negative predictive value	TN/(TN+FN)	810/830 = 98%

THE POINT OF THAT TABLE

Sensitivity and specificity are properties of the test. Predictive values are properties of the population. Halve the prevalence and the positive predictive value falls while the test does not change, which is exactly what the item is testing.

The cheapest marks on the paper

Fifteen or so items turn on duty, consent and documentation. They are the least prepared part of the exam, and the answers are almost always the same shape: do the clinically safe thing, say so plainly, write it down, and keep the patient in the system.

Law, ethics and public health

ITEM TYPE	WHAT THE CORRECT ANSWER LOOKS LIKE
Informed refusal	Explain the risk in plain terms, offer the alternative, document the refusal and the specific risks discussed, arrange the shortest safe review, and keep the door open. Do not discharge the patient for refusing.
Capacity and consent	Capacity is decision-specific. Assess understanding, retention, weighing and communication. Involve a surrogate only when capacity is genuinely absent.
Minors	Consent from a parent or guardian, assent from the child, and jurisdiction-specific exceptions. Safeguarding concerns override confidentiality.
Records	Contemporaneous, factual, timed and attributed. Amend by addendum, never overwrite. Record what you were told, what you found, what you advised and what was declined.
Delegation	Delegate tasks, never clinical judgement. The delegating clinician remains responsible for the decision.
Telehealth	Establish that the standard of care can be met remotely. If the decision needs a finding you cannot obtain, arrange in-person examination.
Driving and work	Advise against driving where the standard is not met, document the advice, and follow the local reporting duty exactly.
Notifiable and public health	Report as jurisdiction requires, treat the patient, and address contacts and partners where the disease demands it.

> **THE SHAPE OF A CORRECT ANSWER**
> Act on the clinical risk first. Explain in plain language. Document what you found, advised and were refused. Set the shortest safe review interval. Follow the local duty exactly, and never make the patient leave the system as a punishment for disagreeing with you.

Competing pairs

Each pair is two conditions that trade places under time pressure. The decider row is the single finding that separates them, which is the finding the item will turn on.

Orbital cellulitis	vs	Cavernous sinus thrombosis
LATERALITY	Unilateral, stays unilateral	Crosses to the other eye within hours to days
CRANIAL NERVES	Motility limited by swelling and pain	III, IV, VI and V1-V2 out of proportion to swelling
PUPIL	RAPD if the apex is crowded	RAPD plus a poorly reactive mid-dilated pupil
SYSTEMIC	Fever with sinus disease	Fever, rigors, obtunded, meningism
DECIDER	Bilateral signs, or multiple cranial neuropathies out of proportion to orbital swelling, mean cavernous sinus and neurosurgical admission rather than orbital drainage.	

Acanthamoeba keratitis	vs	Herpes simplex stromal keratitis
PAIN	Severe, out of proportion to the signs	Proportionate, often modest
INFILTRATE	Radial perineural, then an incomplete ring	Diffuse or disciform central haze
ON ANTIVIRALS	Unchanged or worse over weeks	Settles, then recurs on withdrawal
RISK FACTOR	Tap water, swimming, poor lens hygiene	Previous cold sore or previous dendrite
DECIDER	Radial perineural infiltrates or a ring infiltrate in a lens wearer exposed to water outweigh any pseudodendrite. Scrape and culture for amoeba before more steroid goes in.	

Scleritis	vs	Episcleritis
PAIN	Deep and boring, wakes the patient	Gritty or uncomfortable, sleeps through it
PHENYLEPHRINE	Deep injection does not blanch	Redness blanches within minutes
DAYLIGHT	Blue-violet hue with an oedematous sclera	Bright red vessels over white sclera
SYSTEMIC LINK	Rheumatoid disease, vasculitis, infection	Usually none, most often idiopathic
DECIDER	Redness that will not blanch with phenylephrine, in an eye tender to palpation, means scleritis, systemic work-up and systemic treatment rather than a topical prescription.	

Neovascular AMD	vs	Polypoidal choroidal vasculopathy
PATIENT	Older, soft drusen in both eyes	Often Asian ethnicity, fewer soft drusen
LESION	Grey-green subretinal membrane	Orange-red nodules at the posterior pole
HAEMORRHAGE	Small bleed beside the lesion	Large and recurrent, sometimes massive
IMAGING	Fluid on OCT, leakage on angiography	Polyps on indocyanine green angiography
DECIDER	Recurrent large haemorrhage or orange nodules justify indocyanine green angiography, because polypoidal disease may need photodynamic therapy as well as anti-VEGF.	

Competing pairs, continued

Central retinal vein occlusion	vs	Ocular ischaemic syndrome
ONSET	Abrupt, often noticed on waking	Gradual blur over weeks, with ache in some eyes
HAEMORRHAGES	Dense, four quadrants, up to the disc	Sparse, mid-peripheral, few at the posterior pole
ARTERIES	Normal calibre or mildly narrowed	Markedly narrowed, spontaneous pulsation on gentle pressure
PRESSURE	Normal or raised early	Often low, because ciliary perfusion has fallen
DECIDER	Sparse mid-peripheral haemorrhage with narrowed arteries and a soft eye points to carotid disease, which sends the patient for urgent vascular imaging rather than for injections alone.	

Central serous chorioretinopathy	vs	Neovascular age-related maculopathy
AGE AND MACULA	Thirties to fifties, no drusen	Usually past sixty, drusen in both maculae
OCT FLUID	Clear subretinal dome, thick choroid	Fluid with hyper-reflective tissue, thin choroid
BLOOD	Absent	Present in most eyes at some stage
REFRACTION	Hyperopic shift, micropsia	No consistent shift, metamorphopsia
DECIDER	Hyper-reflective material above the pigment epithelium, or any haemorrhage, means a membrane and same-week anti-VEGF. Its absence with a thick choroid buys three months of observation.	

Neovascular glaucoma	vs	Acute primary angle closure
ONSET	Days to weeks of increasing ache	Hours, severe, with nausea and vomiting
CENTRAL CHAMBER	Deep, until the membrane contracts	Very shallow in both eyes
IRIS AND ANGLE	Vessels at the pupil margin, membrane in the angle	No vessels, angle closed by apposition
FELLOW EYE	Normal depth, open angle	Narrow, at risk, needs iridotomy
DECIDER	A deep central chamber with vessels at the pupil margin means the closure is neovascular, so the treatment is anti-VEGF and retinal laser rather than iridotomy to either eye.	

Optic neuritis	vs	Non-arteritic ischaemic optic neuropathy
AGE AND ONSET	Twenties to forties, worse over days	Past fifty, noticed on waking, then static
PAIN	Ache on eye movement in most cases	Painless
DISC	Normal in retrobulbar cases	Swollen, often segmental, with splinter haemorrhage
FIELD	Central depression	Altitudinal, respecting the horizontal
DECIDER	A swollen disc with an altitudinal field in an older patient moves the work to vascular risk and the fellow eye. A flat disc with pain on movement sends the patient for MRI and neurology.	

Competing pairs, continued

Arteritic AION	vs	Non-arteritic AION
DISC	Chalky pallid swelling, cup often spared	Hyperaemic segmental swelling in a small crowded disc
ACUITY	Counting fingers or worse in most eyes	Often 20/60 or better at onset
SYSTEMIC	Jaw ache, scalp tenderness, weight loss	Nocturnal hypotension, sleep apnoea, diabetes
MARKERS	ESR and CRP raised, platelets high	ESR and CRP normal for age
DECIDER	Raised ESR and CRP with jaw claudication move the case to same-day steroid and biopsy, whatever the disc looks like. Normal markers with a crowded fellow disc do not.	

Horner syndrome	vs	Physiological anisocoria
ANISOCORIA	Larger in dim light, with dilation lag	Equal in light and dark, no lag
LIDS	Mild ptosis with the lower lid raised	Lid position normal or symmetrically lax
APRACLONIDINE	Reverses the anisocoria	No change in either pupil
COURSE	New, with neck, chest or vascular symptoms	Long-standing, visible in old photographs
DECIDER	Anisocoria that grows in the dark with dilation lag is denervation and needs imaging of the whole sympathetic chain. Anisocoria equal in light and dark needs nothing.	

Acquired fourth nerve palsy	vs	Skew deviation
TORSION	Higher eye extorted, often symptomatic	Higher eye intorted, part of a tilt reaction
HEAD TILT	Away from the palsy, relieves the torsion	Towards the lower eye, with a body lean
THREE STEP	Fits one muscle throughout	Often fits no single muscle
COMPANY	Isolated, follows the trauma	Brainstem signs, nystagmus, gait ataxia
DECIDER	The direction of torsion separates them: extorsion of the higher eye is a superior oblique palsy, intorsion is a skew and needs posterior fossa imaging.	

Sterile contact lens infiltrate	vs	Microbial keratitis
PAIN	Gritty, eases within an hour of lens removal	Severe, out of proportion, persists after removal
SITE	Peripheral, with a clear zone to the limbus	Central or paracentral, no clear zone
EPITHELIUM	Intact, or a stain smaller than the infiltrate	Defect wider than the infiltrate beneath it
CHAMBER	Quiet, no cells	Cells, flare, sometimes a hypopyon
DECIDER	An epithelial defect larger than the infiltrate under it, or any anterior chamber reaction, moves the eye to culture and hourly therapy whatever the size of the lesion.	

Competing pairs, continued

	Ethambutol optic neuropathy vs	Nutritional optic neuropathy
ONSET	Weeks to months into a dose above 15 mg/kg	Months to years of poor diet or alcohol
FIELDS	Central depression, sometimes bitemporal	Central or centrocaecal scotomata
DECIDING TEST	Dose per kg, with renal function	B12, folate, red cell thiamine
COURSE	Improves over months once the drug goes	Improves with repletion, not withdrawal
DECIDER	The weight-based dose read against renal function separates them, and it is the one thing that can be acted on the same day. Both need the blood tests, only one needs a drug stopped.	

PART 8

Session 1

Thirty-five cases, 175 items, 210 minutes. Sit it once, cold, in one block, with confidence marked on every item.

BEFORE YOU START

Phone away, timer visible, no notes. Answer on the capture line under each item, and mark L, M or H for confidence. The confidence mark is what makes the review efficient.

PACING

Item 50 by 60 minutes, item 100 by 120, item 150 by 180. If you are behind, stop rereading stems, not standards.

WHEN YOU FINISH

Score on the sheet overleaf, by item type as well as total. Then read the teaching keys in Part 9, in the order given there: wrong answers first, then correct answers you were unsure of, then high-confidence misses last.

WHAT NOT TO DO

Do not turn to Part 9 before the whole session is answered. Do not sit Volume Two on the same day. Do not score only the total.

Session 1 · case index

Sorted as sat. Dx diagnosis, Tx treatment, Sci basic science, Law law and ethics, Sel select all that apply. Use the item-type column to work a single weakness across the whole session.

CASE	PRESENTATION	DOMAIN	ITEMS	CASE	KEY
S1-01	The swollen eyelid that hurts to move	Lids, lacrimal, adnexa, orbit	Dx Dx* Tx Sci Tx	28	64
S1-02	The pain that woke a lens wearer at four in the morning	Contact lenses	Dx Tx* Tx Tx Sci	29	65
S1-03	The eye that has been red twice before	Cornea and refractive surgery	Dx Tx Dx Tx Sci*	30	66
S1-04	Pain out of proportion to the slit lamp findings	Cornea and refractive surgery	Dx Dx* Tx Tx Sci	31	67
S1-05	The teenager whose glasses change at every visit	Cornea and refractive surgery	Dx Tx Tx* Dx Sci	32	68
S1-06	The rash that stopped at the middle of the forehead	Systemic health	Dx Tx Tx Tx* Law	33	69
S1-07	Vision that faded again two years after surgery	Lens, cataract, IOL, perioperative	Dx Tx Dx Tx Law*	34	70
S1-08	A red aching eye that cannot bear the sunlight	Episclera, sclera, anterior uvea	Dx Tx Dx* Tx Sci	35	71
S1-09	A boring pain that wakes her at three in the morning	Episclera, sclera, anterior uvea	Dx Dx* Tx Tx Sci	36	72
S1-10	Door frames that started to bend six weeks ago	Retina, choroid, vitreous	Dx Tx Tx* Tx Sci	37	73
S1-11	Reading vision that faded over two months	Retina, choroid, vitreous	Dx Tx Tx* Sci Tx	38	74
S1-12	Vision clouded over in one eye on waking	Retina, choroid, vitreous	Dx Dx* Tx Tx Sci	39	75
S1-13	Flashes on Friday, and spots ever since	Retina, choroid, vitreous	Dx Tx Tx* Sci Law	40	76
S1-14	Everything went black in one eye at breakfast	Emergencies and trauma	Dx Tx Tx* Dx Sci	41	77
S1-15	A grey smudge in the middle of the page	Retina, choroid, vitreous	Dx Dx* Tx Tx Sci	42	78
S1-16	A drug the rheumatologist started five years ago	Systemic health	Tx Sci* Dx Tx Law	43	79
S1-17	Pressures that crept up between two visits	Glaucoma	Dx Tx Tx* Tx Dx	44	80
S1-18	A red eye, a headache, and vomiting since evening	Glaucoma	Dx Tx Tx* Tx Dx	45	81
S1-19	Ache and redness in the eye that failed in spring	Glaucoma	Dx Dx* Tx Tx Sci	46	82
S1-20	Ache behind the eye, then the colours went flat	Optic nerve and neuro-ophthalmic	Dx Dx Tx Tx* Sci	47	83
S1-21	Vision gone on waking, and the scalp hurts to comb	Optic nerve and neuro-ophthalmic	Dx Dx* Tx Sci Tx	48	84
S1-22	Headaches worst on waking, with a whoosh in both ears	Optic nerve and neuro-ophthalmic	Dx Tx* Sci Tx Dx	49	85
S1-23	The lid that dropped with pain behind the eye	Optic nerve and neuro-ophthalmic	Dx Tx Dx* Sci Tx	50	86
S1-24	The smaller pupil that hides in the dark	Optic nerve and neuro-ophthalmic	Dx Sci* Tx Tx Dx	51	87
S1-25	The lid that sits lower by the evening	Systemic health	Dx Tx* Tx Sci Dx	52	88
S1-26	The print that slides together after twenty minutes	Accommodation, vergence, oculomotor	Dx Tx* Tx Tx Law	53	89
S1-27	The eye that turns in over a picture book	Amblyopia and strabismus	Dx Tx Dx* Sci Tx	54	90
S1-28	The head tilt that started after the bicycle fall	Accommodation, vergence, oculomotor	Dx Tx* Tx Sci Dx	55	91
S1-29	The prescription that has moved twice this year	Ametropia	Dx Tx Tx Tx Law*	56	92
S1-30	The new spectacles that double the print	Ophthalmic optics and spectacles	Sci Dx Tx* Tx Tx	57	93
S1-31	Clearer when the frame slides down her nose	Ophthalmic optics and spectacles	Dx Sci Dx* Tx Tx	58	94
S1-32	Gritty red eye on waking in a lens wearer	Contact lenses	Dx Dx* Tx Tx Law	59	95
S1-33	The post she can no longer read	Low vision	Sci Dx Tx* Tx Law	60	96
S1-34	Both eyes, dimmer colours and harder print	Perceptual function and colour vision	Dx Sci Tx Tx* Dx	61	97
S1-35	The after-hours call about flashes and floaters	Emergencies and trauma	Tx Sci Law* Law Tx	62	98

Session 1 · score by item type

Each cell carries its item type. Mark it right or wrong, then total the five columns on the right.

CASE	1	2	3	4	5	6	DX	TX	SCI	LAW
S1-01	Dx	Dx*	Tx	Sci	Tx					
S1-02	Dx	Tx*	Tx	Tx	Sci					
S1-03	Dx	Tx	Dx	Tx	Sci*					
S1-04	Dx	Dx*	Tx	Tx	Sci					
S1-05	Dx	Tx	Tx*	Dx	Sci					
S1-06	Dx	Tx	Tx	Tx*	Law					
S1-07	Dx	Tx	Dx	Tx	Law*					
S1-08	Dx	Tx	Dx*	Tx	Sci					
S1-09	Dx	Dx*	Tx	Tx	Sci					
S1-10	Dx	Tx	Tx*	Tx	Sci					
S1-11	Dx	Tx	Tx*	Sci	Tx					
S1-12	Dx	Dx*	Tx	Tx	Sci					
S1-13	Dx	Tx	Tx*	Sci	Law					
S1-14	Dx	Tx	Tx*	Dx	Sci					
S1-15	Dx	Dx*	Tx	Tx	Sci					
S1-16	Tx	Sci*	Dx	Tx	Law					
S1-17	Dx	Tx	Tx*	Tx	Dx					
S1-18	Dx	Tx	Tx*	Tx	Dx					
S1-19	Dx	Dx*	Tx	Tx	Sci					
S1-20	Dx	Dx	Tx	Tx*	Sci					
S1-21	Dx	Dx*	Tx	Sci	Tx					
S1-22	Dx	Tx*	Sci	Tx	Dx					
S1-23	Dx	Tx	Dx*	Sci	Tx					
S1-24	Dx	Sci*	Tx	Tx	Dx					
S1-25	Dx	Tx*	Tx	Sci	Dx					
S1-26	Dx	Tx*	Tx	Tx	Law					
S1-27	Dx	Tx	Dx*	Sci	Tx					
S1-28	Dx	Tx*	Tx	Sci	Dx					
S1-29	Dx	Tx	Tx	Tx	Law*					
S1-30	Sci	Dx	Tx*	Tx	Tx					
S1-31	Dx	Sci	Dx*	Tx	Tx					
S1-32	Dx	Dx*	Tx	Tx	Law					
S1-33	Sci	Dx	Tx*	Tx	Law					
S1-34	Dx	Sci	Tx	Tx*	Dx					
S1-35	Tx	Sci	Law*	Law	Tx					
Totals	Mark each item R or W. A star means all or none						Then total the right-hand columns by item type			

READING YOUR OWN TOTALS

Under 60% in any single column is a study plan, not a bad day. Diagnosis misses send you to the condition cards. Treatment misses send you to pharmacology and the emergency gate. Basic science misses are the cheapest to fix and the easiest to postpone.

CASE S1-01 · LIDS, LACRIMAL, ADNEXA, ORBIT · 5 ITEMS

The swollen eyelid that hurts to move

A 13-year-old boy has three days of worsening left eyelid swelling and frontal headache, after a week of purulent nasal discharge. Temperature 38.4 °C. BVA 20/20 OD, 20/40 OS. Mild left RAPD, red saturation 60% OS, 3 mm proptosis, tense chemosis, painful restriction of elevation and abduction. IOP 17 OD, 27 OS. Cornea clear, chamber quiet, fundus normal.

1 What is the most likely diagnosis? DIAGNOSIS

A Cavernous sinus thrombosis complicating sinusitis
B Orbital cellulitis of ethmoid sinus origin
C Orbital rhabdomyosarcoma with secondary inflammation
D Preseptal cellulitis with reactive chemosis
E Idiopathic orbital inflammatory syndrome

ANSWER ______ CONFIDENCE L · M · H

2 Which three findings measure optic nerve function, rather than supporting the infective diagnosis? DIAGNOSIS · SELECT 3

A Red saturation reduced to 60% OS
B Purulent nasal discharge for one week
C Intraocular pressure 27 mmHg OS
D Best corrected acuity 20/40 OS
E Temperature 38.4 °C
F Left relative afferent pupillary defect

ANSWER ______ CONFIDENCE L · M · H

3 What is the most appropriate next step? TREATMENT

A Outpatient MRI within 48 hours, then decide about admission
B Intravenous antibiotics booked for the morning, oral cover overnight
C Oral corticosteroid with a topical antibiotic to settle the swelling
D Oral amoxicillin-clavulanate, nasal decongestant, review in 24 hours
E Same-day referral for contrast CT, intravenous antibiotics, ophthalmology and ENT

ANSWER ______ CONFIDENCE L · M · H

4 Which anatomic route most commonly explains this presentation? BASIC SCIENCE

A Direct spread across the lamina papyracea from the ethmoid sinus
B Retrograde thrombophlebitis through valveless facial and orbital veins
C Extension through the optic canal from the sphenoid sinus
D Ascending infection along the nasolacrimal duct from the inferior meatus
E Haematogenous seeding of the orbit from an occult bacteraemia

ANSWER ______ CONFIDENCE L · M · H

5 He is admitted on intravenous antibiotics. Which change over 18 hours most supports drainage? TREATMENT

A Acuity falls to 20/80 with worse colour and an abscess crowding the apex
B Blood cultures grow an organism fully sensitive to current therapy
C Chemosis increases while pupil, colour and acuity stay unchanged
D Temperature settles after two doses while the lid oedema persists
E CT shows a 4 mm medial subperiosteal collection, acuity improving

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-02 · CONTACT LENSES · 5 ITEMS

The pain that woke a lens wearer at four in the morning

A 24-year-old woman wears monthly silicone hydrogel lenses, sleeps in them twice a week, and tops up the solution in her case rather than discarding it. Pain, photophobia and green discharge for 36 hours OS. BVA 20/25 OD, 20/200 OS. A 2.2 mm paracentral epithelial defect over a dense stromal infiltrate, 1.2 mm hypopyon, circumlimbal injection, stromal oedema. IOP 14 OD, 11 OS.

1 **What is the most likely diagnosis?** DIAGNOSIS

A Marginal staphylococcal hypersensitivity keratitis
B Contact lens induced sterile peripheral ulcer
C Pseudomonas keratitis related to overnight wear
D Adenoviral keratitis with subepithelial infiltrates
E Acanthamoeba keratitis in a topping-up lens user

ANSWER ______ CONFIDENCE L · M · H

2 **Which three steps belong in the first hour of management?** TREATMENT · SELECT 3

A Topical dexamethasone four times daily to limit scarring
B Topical antifungal cover pending microscopy results
C Pressure patch after cycloplegia for pain relief
D Corneal scrape for Gram stain and culture before drops start
E Hourly topical fluoroquinolone monotherapy, day and night
F Send the lens, case and solution for culture as well

ANSWER ______ CONFIDENCE L · M · H

3 **She is reviewed at 48 hours. Which finding most justifies changing therapy?** TREATMENT

A The infiltrate is denser with 0.4 mm more stromal thinning
B Culture grows Pseudomonas aeruginosa fully sensitive to ciprofloxacin
C Pain is worse although the epithelial defect has shrunk by half
D The hypopyon has settled and the anterior chamber is quieter
E Injection persists and she complains of stinging on the drops

ANSWER ______ CONFIDENCE L · M · H

4 **Which addition best manages her pain and anterior chamber reaction?** TREATMENT

A Oral prednisolone 40 mg daily for three days
B Topical cyclopentolate 1% twice daily with oral analgesia
C Topical ketorolac four times daily with oral analgesia
D Subconjunctival gentamicin with topical anaesthetic
E Topical atropine 1% twice daily and a bandage lens

ANSWER ______ CONFIDENCE L · M · H

5 **Which organism factor best explains the speed of stromal destruction?** BASIC SCIENCE

A Endotoxin release driving neutrophil chemotaxis into the stroma
B Adhesion to compromised epithelium through pili and flagella
C Cyst formation that resists disinfection by the solution
D Secreted proteases and elastase that digest stromal collagen
E Biofilm formation on the lens surface and in the storage case

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-03 · CORNEA AND REFRACTIVE SURGERY · 5 ITEMS

The eye that has been red twice before

A 41-year-old man has four days of watering, mild photophobia and blur OD, and two similar episodes in the past two years. BVA 20/40 OD. A branching epithelial lesion with terminal bulbs stains with fluorescein, its bed stains faintly with rose bengal, corneal sensation is reduced OD, and two small anterior stromal opacities remain from an earlier episode. Chamber quiet, IOP 16 both eyes.

1 What is the most likely diagnosis? DIAGNOSIS

- A Thygeson superficial punctate keratitis
- B Herpes zoster pseudodendritic keratopathy
- C Herpes simplex epithelial keratitis, recurrent
- D Healing abrasion with a dendritic epithelial ridge
- E Acanthamoeba radial keratoneuritis, early

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate treatment now? TREATMENT

- A Topical chloramphenicol with a bandage contact lens
- B Topical ganciclovir 0.15% gel five times daily
- C Topical ganciclovir gel with dexamethasone twice daily
- D Oral aciclovir 400 mg twice daily as sole therapy
- E Preservative-free lubricants and review in one week

ANSWER ______ CONFIDENCE L · M · H

3 At two weeks the epithelium is intact but acuity is 20/60 with central stromal haze and cells, and no keratic precipitates. What has happened? DIAGNOSIS

- A Herpetic stromal keratitis without ulceration
- B Neurotrophic keratopathy from lost sensation
- C Herpetic endotheliitis with secondary oedema
- D Bacterial superinfection of a healed dendrite
- E Antiviral epithelial toxicity with pseudodendrites

ANSWER ______ CONFIDENCE L · M · H

4 Which regimen is most appropriate for this stage? TREATMENT

- A Topical ciclosporin 0.05% twice daily as sole therapy
- B Topical prednisolone with antiviral cover, slow taper
- C Topical prednisolone hourly without antiviral cover
- D Oral aciclovir 400 mg five times daily as sole therapy
- E Topical ganciclovir gel five times daily as sole therapy

ANSWER ______ CONFIDENCE L · M · H

5 Which two statements about latency and recurrence are correct? BASIC SCIENCE · SELECT 2

- A Recurrence reflects fresh exogenous reinfection
- B Corneal sensation recovers fully between episodes
- C Latency is maintained in corneal stromal keratocytes
- D Latent virus resides in the trigeminal ganglion
- E Recurrence follows reactivation and axonal transport

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-04 · CORNEA AND REFRACTIVE SURGERY · 5 ITEMS

Pain out of proportion to the slit lamp findings

A 27-year-old man rinses his monthly lenses under the tap and swims in them. Three weeks of worsening OS pain, now severe and disturbing sleep, with photophobia and blur. Two courses of topical antiviral and one of a fluoroquinolone have not helped. BVA 20/80 OS. Patchy anterior stromal infiltrates in an incomplete ring, radial perineural infiltrates, epithelial pseudodendrites, limbitis. IOP 18 both eyes.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Acanthamoeba keratitis from tap water exposure
- B Fungal keratitis with feathery stromal extension
- C Atypical mycobacterial keratitis after lens trauma
- D Pseudomonas keratitis, partially treated
- E Herpes simplex stromal keratitis, unresponsive

ANSWER ______ CONFIDENCE L · M · H

2 **Which three findings most strongly separate this from herpes simplex?** DIAGNOSIS · SELECT 3

- A A unilateral red eye with photophobia
- B Radial perineural stromal infiltrates
- C Pain far in excess of the slit lamp signs
- D An incomplete ring infiltrate in the stroma
- E Reduced corneal sensation in the affected eye
- F Epithelial irregularity with pseudodendrites

ANSWER ______ CONFIDENCE L · M · H

3 **What is the most appropriate initial treatment?** TREATMENT

- A Therapeutic penetrating keratoplasty within the fortnight
- B Oral itraconazole with topical natamycin drops
- C Topical polyhexanide and propamidine, hourly at first
- D Hourly fortified vancomycin and amikacin drops
- E Topical prednisolone with the antiviral continued

ANSWER ______ CONFIDENCE L · M · H

4 **At six weeks the infiltrate has cleared but a sterile ring inflammation persists with acuity 20/100. What now?** TREATMENT

- A Add topical steroid, continuing the biguanide
- B Stop the biguanide and start topical steroid
- C Add oral voriconazole to the biguanide
- D Restart hourly propamidine for two weeks
- E Proceed to therapeutic keratoplasty this month

ANSWER ______ CONFIDENCE L · M · H

5 **Which property of the organism explains why treatment runs for months?** BASIC SCIENCE

- A It forms biofilm on the corneal surface
- B It survives inside keratocytes as a latent form
- C It seeds the trigeminal nerve and reactivates later
- D The cyst wall resists drugs and disinfectants
- E Trophozoites divide faster than epithelium heals

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-05 · CORNEA AND REFRACTIVE SURGERY · 5 ITEMS

The teenager whose glasses change at every visit

A 17-year-old boy with atopic eczema rubs his eyes daily. Over 14 months OD refraction moved from -2.75 -1.50 x 175 to -4.00 -3.25 x 160, BVA 20/25. Tomography OD: maximum keratometry 51.8 D, up 1.9 D over those 14 months, thinnest point 442 µm, inferior steepening with a displaced thin point. OS Kmax 46.2 D, unchanged. No scarring, no oedema, endothelium normal.

1 **Which finding establishes progression?** DIAGNOSIS

- A Maximum keratometry has risen 1.9 D in 14 months
- B The thinnest point measures 442 µm
- C Inferior steepening with a displaced thin point
- D Best corrected acuity is still 20/25 in that eye
- E Cylinder has increased by 1.75 D with axis rotation

ANSWER ______ CONFIDENCE L · M · H

2 **What is the most appropriate management?** TREATMENT

- A Cross-linking of both eyes at the same session
- B Repeat tomography in 12 months, then decide
- C Intrastromal ring segments OD with spectacles
- D Epithelium-off cross-linking OD, monitor OS
- E Rigid gas permeable fitting OD, no cross-linking

ANSWER ______ CONFIDENCE L · M · H

3 **Which two factors would defer standard epithelium-off cross-linking?** TREATMENT · SELECT 2

- A Active atopic keratoconjunctivitis with lid disease
- B Maximum keratometry above 50 D
- C Age 17 rather than over 18
- D Apical scarring visible on the slit lamp
- E Corrected acuity better than 20/30
- F Stromal thickness under 400 µm after debridement

ANSWER ______ CONFIDENCE L · M · H

4 **Three months after treatment acuity is 20/60 with central stromal haze and no epithelial defect. What is the most likely cause?** DIAGNOSIS

- A Endothelial decompensation from irradiation
- B Sterile infiltrate from the bandage lens
- C Microbial keratitis after epithelial removal
- D Expected post-cross-linking stromal haze
- E Continued progression despite treatment

ANSWER ______ CONFIDENCE L · M · H

5 **How does the treatment stiffen the cornea?** BASIC SCIENCE

- A Riboflavin absorbs UVA and new collagen bonds form
- B UVA denatures collagen and shrinks the stroma thermally
- C Riboflavin scavenges radicals and prevents enzymatic loss
- D UVA kills keratocytes so the stroma stops remodelling
- E Riboflavin blocks UVA from reaching the endothelium

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-06 · SYSTEMIC HEALTH · 5 ITEMS

The rash that stopped at the middle of the forehead

A 68-year-old woman taking methotrexate has four days of right brow pain, then a vesicular rash over the forehead and upper lid that stops at the midline, with lesions on the tip of the nose. BVA 20/30 OD. Pseudodendritic epithelial plaques, reduced corneal sensation, 1+ cells, a few pigmented keratic precipitates, sectoral iris atrophy. IOP 15 OS, 26 OD. Discs normal.

1 What is the most likely diagnosis? DIAGNOSIS

A Primary herpes simplex with a periocular rash
B Herpes zoster ophthalmicus with anterior uveitis
C Impetiginised contact dermatitis of the lids
D Bacterial preseptal cellulitis with lid vesicles
E Eczema herpeticum with secondary keratitis

ANSWER ______ CONFIDENCE L · M · H

2 Which antiviral treatment is most appropriate? TREATMENT

A Intravenous aciclovir for seven days as an inpatient
B Oral valaciclovir 500 mg twice daily for five days
C Oral aciclovir 800 mg five times daily, seven days
D Oral aciclovir 400 mg five times daily for five days
E Topical ganciclovir 0.15% gel five times daily

ANSWER ______ CONFIDENCE L · M · H

3 How should the pressure of 26 mmHg in that eye be managed? TREATMENT

A Oral acetazolamide 250 mg four times daily
B Topical pilocarpine 2% four times daily
C Topical steroid for the uveitis plus a beta-blocker
D Topical prostaglandin analogue as first-line control
E Observe the pressure until the rash has resolved

ANSWER ______ CONFIDENCE L · M · H

4 Which three actions belong in her care over the next month? TREATMENT · SELECT 3

A Stop the methotrexate permanently to prevent recurrence
B Prescribe a topical antibiotic for the crusting lesions
C Arrange varicella serology to confirm the diagnosis
D Discuss zoster vaccination once this episode settles
E Review at intervals for a year for late stromal disease
F Tell her rheumatology team about the acute infection

ANSWER ______ CONFIDENCE L · M · H

5 She asks whether she can work in a nursery while the rash is active. What is the correct advice? LAW AND ETHICS

A Return when the rash is covered by a dressing
B No restriction, since zoster is not transmissible
C Stay away for 21 days from the rash appearing
D Stay away until every lesion has crusted over
E Return once oral antiviral therapy has started

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-07 · LENS, CATARACT, IOL, PERIOPERATIVE · 5 ITEMS

Vision that faded again two years after surgery

A 71-year-old man had uncomplicated phacoemulsification OS 26 months ago and reached 20/20. He now reports 20/50 OS, glare in sunlight and difficulty with print. Refraction is unchanged from his one-month postoperative result. The implant is centred in the bag, with a wrinkled pearly membrane behind it and pearls at its edge. Cornea clear, IOP 16, no cells, macula normal on OCT.

1 **What is the most likely cause of the drop in vision?** DIAGNOSIS

A Refractive change from posterior corneal astigmatism
B Pseudophakic cystoid macular oedema
C Posterior capsule opacification behind the implant
D Late in-the-bag intraocular lens dislocation
E Fuchs endothelial dystrophy, decompensating

ANSWER ______ CONFIDENCE L · M · H

2 **What is the most appropriate treatment?** TREATMENT

A Intravitreal triamcinolone for the membrane
B New spectacles with an anti-glare coating
C Lens exchange with polishing of the capsule
D Nd:YAG laser posterior capsulotomy in clinic
E Surgical membrane peel by a pars plana approach

ANSWER ______ CONFIDENCE L · M · H

3 **Two days after the laser he reports new floaters and a shadow in the upper field. What is the priority?** DIAGNOSIS

A Repeat the laser to enlarge the opening
B Dilated peripheral retinal examination today
C Reassurance, since floaters are expected after laser
D Measure the pressure and review in one month
E Macular OCT to look for postoperative oedema

ANSWER ______ CONFIDENCE L · M · H

4 **Which pressure finding one hour after the laser needs treatment before he leaves?** TREATMENT

A Pressure risen from 16 to 21 mmHg
B Pressure unchanged with pigment in the chamber
C Pressure fallen to 12 with a shallow chamber
D Pressure 18 with a few red cells behind the iris
E Pressure risen from 16 to 31 mmHg

ANSWER ______ CONFIDENCE L · M · H

5 **Which two duties apply when consenting him for the laser?** LAW AND ETHICS · SELECT 2

A Record that alternatives, including no treatment, were discussed
B Obtain consent from a relative as well as from him
C Warn him that the implant will be replaced
D Guarantee the acuity he had at one month
E Disclose the raised risk of retinal detachment

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-08 · EPISCLERA, SCLERA, ANTERIOR UVEA · 5 ITEMS

A red aching eye that cannot bear the sunlight

A 29-year-old man has three days of aching OD with photophobia and blur, and six months of early morning low back stiffness that eases as he moves. BVA 20/40 OD. Circumlimbal injection, 3+ cells with 1+ flare, fine keratic precipitates inferiorly, a strand of fibrin, posterior synechiae at 7 o'clock. IOP 11 OD, 15 OS. No vitreous cells, fundus normal. A similar episode affected OS last year.

1 What is the most likely diagnosis? DIAGNOSIS

- A Intermediate uveitis with anterior spillover
- B Toxoplasma retinochoroiditis with iritis
- C Herpetic anterior uveitis with trabeculitis
- D Acute anterior uveitis, HLA-B27 associated
- E Fuchs uveitis syndrome, first presentation

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate initial treatment? TREATMENT

- A Oral prednisolone 40 mg daily with a topical antibiotic
- B Topical ketorolac four times daily with cyclopentolate
- C Sub-Tenon triamcinolone injection at this visit
- D Hourly topical prednisolone 1% with cyclopentolate
- E Topical prednisolone four times daily with lubricants

ANSWER ______ CONFIDENCE L · M · H

3 Which two features point to a systemic association worth investigating? DIAGNOSIS · SELECT 2

- A Inflammatory back pain for six months
- B Recurrence alternating between the eyes
- C Fine keratic precipitates inferiorly
- D Intraocular pressure of 11 mmHg
- E Photophobia with circumlimbal injection

ANSWER ______ CONFIDENCE L · M · H

4 Which referral and test plan fits him best? TREATMENT

- A Rheumatology referral, HLA-B27 and sacroiliac imaging
- B Rheumatology referral with a full autoimmune screen
- C Chest imaging with serum ACE and calcium
- D Syphilis and tuberculosis testing before any steroid
- E No investigation, since first-line therapy is unchanged

ANSWER ______ CONFIDENCE L · M · H

5 Which mechanism best explains the low pressure in the inflamed eye? BASIC SCIENCE

- A Reduced episcleral venous pressure from injection
- B Reduced aqueous production by an inflamed ciliary body
- C Increased outflow through an inflamed trabecular meshwork
- D Fibrin plugging the trabecular meshwork
- E Posterior synechiae obstructing the pupil

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-09 · EPISCLERA, SCLERA, ANTERIOR UVEA · 5 ITEMS

A boring pain that wakes her at three in the morning

A 54-year-old woman with nine years of seropositive rheumatoid arthritis has 10 days of deep boring OS pain that wakes her, with tenderness on palpation of the globe. BVA 20/30 OS. A violaceous patch of deep injection nasally that does not blanch with topical phenylephrine, an oedematous sclera in daylight, and a 1 mm peripheral corneal gutter with intact epithelium. IOP 14 both eyes.

1 **What is the most likely diagnosis?** DIAGNOSIS

A Nodular episcleritis with adjacent corneal drying
B Diffuse anterior scleritis with a corneal gutter
C Peripheral ulcerative keratitis without scleritis
D Necrotising scleritis with established thinning
E Sclerosing keratitis from long-standing dry eye

ANSWER ______ CONFIDENCE L · M · H

2 **Which three findings separate this from episcleritis?** DIAGNOSIS · SELECT 3

A Watering with mild photophobia
B Symptoms present for 10 days
C Injection that phenylephrine does not blanch
D Pain that wakes her from sleep
E Tenderness of the globe on palpation
F Sectoral rather than diffuse redness

ANSWER ______ CONFIDENCE L · M · H

3 **What is the most appropriate initial treatment?** TREATMENT

A Oral NSAID with gastric cover, rheumatology review
B Topical prednisolone four times daily with lubricants
C Oral prednisolone 1 mg per kg starting today
D Topical NSAID with a bandage contact lens
E Subconjunctival triamcinolone to the affected quadrant

ANSWER ______ CONFIDENCE L · M · H

4 **Three weeks later the pain is worse, the gutter is deeper, and an avascular scleral patch has appeared. What now?** TREATMENT

A Increase the oral NSAID to the maximum dose
B Topical ciclosporin with punctal occlusion
C Scleral patch graft before any medical change
D Oral prednisolone alone, tapered over four weeks
E Urgent systemic immunosuppression, shared care

ANSWER ______ CONFIDENCE L · M · H

5 **Why does the cornea thin at the periphery in this disease?** BASIC SCIENCE

A Peripheral stroma holds fewer keratocytes for repair
B Limbal vessels deliver immune complexes and proteases
C Tear film breaks up faster over the peripheral cornea
D Peripheral cornea is thinner than the centre to begin with
E Limbal stem cells fail and the epithelium breaks down

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-10 · RETINA, CHOROID, VITREOUS · 5 ITEMS

Door frames that started to bend six weeks ago

A 78-year-old woman reports six weeks of distortion OD, with door frames bowed and print she managed in June now hard to read. She smokes 15 cigarettes a day. BVA was 20/25 OD in June and is 20/60 today. Amsler shows central metamorphopsia OD. Fundus OD: soft confluent drusen, a grey-green subretinal lesion and a small subretinal haemorrhage temporal to fixation. OS drusen only, 20/25.

1 What is the most likely diagnosis? DIAGNOSIS

- A Chronic central serous chorioretinopathy
- B Retinal macroaneurysm with macular exudation
- C Vitreomacular traction with a lamellar defect
- D Geographic atrophy involving the foveal centre
- E Neovascular age-related macular degeneration

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate next step? TREATMENT

- A Refer for focal laser photocoagulation of the lesion
- B Refer for macular OCT and anti-VEGF within one week
- C Refer for macular OCT within one week, then observe
- D Routine referral for assessment in six to eight weeks
- E Start high-dose antioxidant supplements and review

ANSWER ______ CONFIDENCE L · M · H

3 Which three belong in her care alongside the injections? TREATMENT · SELECT 3

- A Smoking cessation support offered at this visit
- B Amsler self-monitoring of the fellow eye
- C Low vision assessment and reading aids
- D A supplement containing beta-carotene
- E Surrender of her driving licence this week
- F Laser prophylaxis to drusen in the fellow eye

ANSWER ______ CONFIDENCE L · M · H

4 After three monthly injections OD is 20/40 with a little persistent subretinal fluid. What now? TREATMENT

- A Stop injections and monitor with monthly OCT
- B Switch to a different anti-VEGF agent today
- C Add photodynamic therapy to the next injection
- D Extend to four-monthly injections immediately
- E Continue injections and set the treatment interval

ANSWER ______ CONFIDENCE L · M · H

5 Which process most directly produces the leakage in this eye? BASIC SCIENCE

- A VEGF driven vessel growth through Bruch membrane
- B Lipofuscin accumulation in the pigment epithelium
- C Choriocapillaris dropout with photoreceptor loss
- D Vitreous traction on the internal limiting membrane
- E Complement activation forming drusen deposits

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-11 · RETINA, CHOROID, VITREOUS · 5 ITEMS

Reading vision that faded over two months

A 52-year-old man with type 2 diabetes of 18 years, HbA1c 9.8%, reports two months of blur in the right eye. BVA 20/60 OD, 20/25 OS. OD: flat new vessel fronds at the disc and on the superior arcade, faint vitreous haze, dot-blot haemorrhages in four quadrants, venous beading. OCT OD shows cystoid spaces through the fovea, central subfield 445 µm. IOP 16 both eyes, angles open, iris without vessels.

1 Which classification best fits the right eye? DIAGNOSIS

A Moderate non-proliferative retinopathy, oedema off the centre
B Proliferative retinopathy with vitreous haemorrhage, no oedema
C Ocular ischaemic syndrome on a background of retinopathy
D Severe non-proliferative retinopathy with centre-involving oedema
E Proliferative retinopathy with centre-involving macular oedema

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate initial treatment of the right eye? TREATMENT

A Vitrectomy with endolaser and anti-VEGF given at the end of surgery
B Panretinal photocoagulation at this visit, macular review in three months
C Intravitreal anti-VEGF now, panretinal laser once the centre is drier
D Focal macular laser to the leaking points, then panretinal laser
E Dexamethasone implant now, with panretinal laser deferred six months

ANSWER ______ CONFIDENCE L · M · H

3 Which three steps belong in the plan alongside intravitreal treatment? TREATMENT · SELECT 3

A OCT of both maculae at each treatment visit to guide retreatment
B Dilated assessment of the left eye for proliferation and its own need
C Stopping aspirin because the vitreous is already showing some blood
D Deferring the next dilated examination for six months of injections
E Advising that all reading be done with a stronger near addition
F Same-week diabetes team review of glycaemic control and blood pressure

ANSWER ______ CONFIDENCE L · M · H

4 Which mechanism most directly explains the thickening at the fovea? BASIC SCIENCE

A Breakdown of the inner blood-retinal barrier at capillary endothelium
B Failure of the pigment epithelial pump at the outer blood-retinal barrier
C Traction on the fovea by a taut thickened posterior hyaloid membrane
D Loss of choriocapillaris perfusion beneath the central macula
E Osmotic swelling of Muller cells driven by hyperglycaemia alone

ANSWER ______ CONFIDENCE L · M · H

5 After three monthly injections the central subfield is 402 µm and acuity 20/60. What next? TREATMENT

A Keep monthly injections and add a dexamethasone implant now
B Stop injections and complete panretinal photocoagulation instead
C Refer for vitrectomy with peeling of the internal limiting membrane
D Continue the same agent monthly for three further doses
E Switch to a different anti-VEGF agent and reassess after three doses

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-12 · RETINA, CHOROID, VITREOUS · 5 ITEMS

Vision clouded over in one eye on waking

A 68-year-old man noticed painless blur in the left eye on waking three days ago. Treated hypertension, home readings 158/96. No diabetes. BVA 20/25 OD, 20/200 OS. Trace RAPD OS. IOP 15 both eyes, angles open, iris without vessels. OS: veins dilated and tortuous in four quadrants, flame and dot haemorrhages to the periphery, several cotton wool spots, blurred hyperaemic disc, central subfield 620 µm.

1 What is the most likely diagnosis? DIAGNOSIS

- A Undiagnosed severe non-proliferative diabetic retinopathy
- B Central retinal vein occlusion with macular oedema
- C Hemiretinal vein occlusion with macular oedema
- D Ocular ischaemic syndrome from carotid disease
- E Papilloedema with bilateral venous stasis

ANSWER ______ CONFIDENCE L · M · H

2 Which three findings raise the concern that this occlusion is ischaemic? DIAGNOSIS · SELECT 3

- A Best corrected acuity of 20/200 in the affected eye
- B Several cotton wool spots at the posterior pole
- C Veins dilated and tortuous in all four quadrants
- D Central subfield thickness measured at 620 µm
- E Blurred disc margin with a hyperaemic disc
- F Relative afferent pupillary defect in the affected eye

ANSWER ______ CONFIDENCE L · M · H

3 What is the most appropriate management of the macular oedema? TREATMENT

- A Observation for three months, since many occlusions settle alone
- B Intravitreal dexamethasone implant as the first-line agent
- C Oral acetazolamide with aspirin and review at six weeks
- D Macular grid laser to the thickened area, review at three months
- E Intravitreal anti-VEGF now, with monthly acuity and OCT review

ANSWER ______ CONFIDENCE L · M · H

4 Which systemic step matters most at this visit? TREATMENT

- A Erythrocyte sedimentation rate and C-reactive protein today
- B Same-week review of blood pressure and glucose in general practice
- C Carotid duplex within the week to exclude a stenosis on the affected side
- D Thrombophilia screen, given the inherited risk of thrombosis
- E Immediate stroke unit referral for antiplatelet loading today

ANSWER ______ CONFIDENCE L · M · H

5 Where does the obstruction sit anatomically? BASIC SCIENCE

- A In the superior ophthalmic vein behind the apex, draining intracranially
- B At the choriocapillaris, where venules collect into the vortex veins
- C In the short posterior ciliary vessels around the optic nerve head
- D At an arteriovenous crossing under a stiffened arteriole, superotemporally
- E In the vein at the lamina cribrosa, sharing a sheath with the artery

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-13 · RETINA, CHOROID, VITREOUS · 5 ITEMS

Flashes on Friday, and spots ever since

A 58-year-old woman, -5.50 D in both eyes, has four days of new floaters and arcs of light in the right eye, worse in dim rooms. No shadow in the field. BVA 20/20 both eyes, IOP 14. OD: pigment granules in the anterior vitreous, a Weiss ring, no cells, settled blood inferiorly. With indentation, a horseshoe tear at 11 o'clock with a vitreous strand on the flap, retina flat, no subretinal fluid. OS: inferotemporal lattice, no breaks.

1 Which feature carries the highest risk of progression to detachment? DIAGNOSIS

- A Axial myopia of -5.50 D in both eyes
- B Pigment granules dispersed in the anterior vitreous
- C A horseshoe tear with traction on its flap
- D Settled vitreous blood inferior to the break
- E Lattice degeneration in the inferotemporal fellow eye

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate management of the right eye? TREATMENT

- A Vitrectomy with endolaser and a gas tamponade
- B Laser retinopexy surrounding the tear within 24 hours
- C Cryotherapy to the tear and laser to the fellow lattice
- D Review in two weeks with a drawing of the periphery
- E Scleral buckle to relieve the traction on the flap

ANSWER ______ CONFIDENCE L · M · H

3 Which two symptoms after retinopexy require same-day return? TREATMENT · SELECT 2

- A A shadow spreading from the periphery towards the centre
- B A sudden fresh increase in floaters with new flashes
- C Mild ache around the brow for two days after laser
- D Faint glare in bright sunlight lasting about a week
- E Slight morning blur that clears after some blinking

ANSWER ______ CONFIDENCE L · M · H

4 Why do tears of this kind form where they do? BASIC SCIENCE

- A Vitreous collagen inserts firmly at the posterior vitreous base, and separation tears there
- B The internal limiting membrane is thinnest at the equator, so it splits first under traction
- C Pigment epithelial adhesion is weakest superiorly, so the retina lifts there first
- D Choroidal vessels tether the retina at the equator and resist vitreous movement
- E The subretinal space is widest at the equator, so fluid enters most easily there

ANSWER ______ CONFIDENCE L · M · H

5 What does the standard of care require of the practitioner at this visit? LAW AND ETHICS

- A A dilated record made now, with the patient asked to telephone in a week
- B Referral held until the vitreous blood clears enough to grade the break
- C A referral letter posted today, with review arranged in one month
- D Same-day documented referral, the break located, warning symptoms in writing
- E Advice to attend an emergency department if the vision changes

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-14 · EMERGENCIES AND TRAUMA · 5 ITEMS

Everything went black in one eye at breakfast

A 74-year-old man lost the vision of his right eye painlessly 90 minutes ago. No headache, no scalp tenderness, no pain on chewing. Atrial fibrillation, on no anticoagulant. Blood pressure 176/92, pulse irregular at 96. BVA hand movements OD, 20/25 OS. Dense RAPD OD. OD: pale oedematous posterior retina, a cherry red spot, segmented columns in the arterioles, no plaque at the disc. IOP 16 both eyes.

1 What is the most likely diagnosis? DIAGNOSIS

- A Non-arteritic anterior ischaemic optic neuropathy
- B Non-arteritic central retinal artery occlusion
- C Ophthalmic artery occlusion with choroidal ischaemia
- D Arteritic retinal occlusion in giant cell arteritis
- E Ischaemic central retinal vein occlusion

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate immediate action? TREATMENT

- A Carotid duplex in clinic today, cardiology review next week
- B High dose oral prednisolone now, pending inflammatory markers
- C Intravenous acetazolamide with rebreathing to raise carbon dioxide
- D Ocular massage and paracentesis, with stroke referral tomorrow
- E Immediate transfer to a stroke unit for assessment and imaging

ANSWER ______ CONFIDENCE L · M · H

3 Which three belong in the same-day workup? TREATMENT · SELECT 3

- A Electroretinography to confirm inner retinal ischaemia
- B Thrombophilia screen before any anticoagulation is started
- C Erythrocyte sedimentation rate and C-reactive protein today
- D Blood pressure, rhythm assessment and an electrocardiogram
- E Carotid imaging arranged within the coming week
- F Fluorescein angiography to time the arterial filling delay

ANSWER ______ CONFIDENCE L · M · H

4 Which new finding would move this towards an arteritic cause? DIAGNOSIS

- A Amaurosis fugax in the same eye last month
- B An irregular pulse at 96 with no anticoagulation
- C Scalp tenderness with pain in the jaw on chewing
- D Segmented columns of blood within the arterioles
- E A calcific plaque visible at the disc margin

ANSWER ______ CONFIDENCE L · M · H

5 Which anatomic feature explains the cherry red spot? BASIC SCIENCE

- A The fovea is thin and takes its oxygen from the perfused choroid
- B A cilioretinal artery supplies the fovea in most eyes and spares it
- C Foveal pigment epithelium is denser, so it reads red against pale retina
- D Xanthophyll at the fovea absorbs light scattered by swollen retina
- E Ganglion cells crowd the foveal centre and tolerate ischaemia longer

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-15 · RETINA, CHOROID, VITREOUS · 5 ITEMS

A grey smudge in the middle of the page

A 41-year-old man reports six weeks of a central grey patch in the right eye, with objects on that side looking smaller and further away. He finished a course of oral corticosteroid for back pain three weeks ago and works night shifts. BVA 20/40 OD, improving to 20/30 with +1.00, 20/20 OS. OD: shallow serous elevation at the macula, no blood, no drusen, fine pigment flecks. OCT: subfoveal fluid, choroid 470 µm.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Pachychoroid neovasculopathy with a type 1 membrane
- B Acute central serous chorioretinopathy
- C Optic disc pit maculopathy with subretinal fluid
- D Inflammatory choroidal neovascular membrane
- E Cystoid macular oedema of uncertain cause

ANSWER ______ CONFIDENCE L · M · H

2 **Which two features argue against a neovascular membrane?** DIAGNOSIS · SELECT 2

- A No drusen in either macula at the age of 41
- B Micropsia with distortion on the Amsler chart
- C A hyperopic shift of +1.00 in the affected eye
- D Choroidal thickness measured at 470 µm
- E Subretinal fluid with no hyper-reflective material

ANSWER ______ CONFIDENCE L · M · H

3 **What is the most appropriate management now?** TREATMENT

- A Focal laser to the leakage point identified on angiography
- B Observation to three months, with steroid avoidance and sleep advice
- C Half-dose photodynamic therapy to the leaking area within two weeks
- D Intravitreal anti-VEGF monthly for three doses, then reassessment
- E Oral eplerenone for three months with electrolyte monitoring

ANSWER ______ CONFIDENCE L · M · H

4 **Four months on, fluid persists and acuity has fallen to 20/50. What next?** TREATMENT

- A Monthly intravitreal anti-VEGF for three doses
- B Oral eplerenone with two-weekly electrolytes
- C Full-fluence photodynamic therapy to the leak
- D Half-dose photodynamic therapy guided by angiography
- E Continued observation for a further three months

ANSWER ______ CONFIDENCE L · M · H

5 **Which structure is the primary site of the disease?** BASIC SCIENCE

- A Bruch membrane, through defects of which new vessels grow forward
- B Outer plexiform layer, where Muller cells swell and form cysts
- C Vitreoretinal interface, where traction lifts the photoreceptors
- D Choroid, whose hyperpermeable vessels overwhelm the pigment epithelium
- E Retinal capillary endothelium, with inner barrier breakdown at the fovea

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-16 · SYSTEMIC HEALTH · 5 ITEMS

A drug the rheumatologist started five years ago

A 46-year-old woman with systemic lupus erythematosus takes hydroxychloroquine 400 mg daily, started five years ago. Weight 61 kg, eGFR 88, no tamoxifen, no other ocular medication. Reading is unchanged. BVA 20/20 both eyes. Maculae show no pigment disturbance and no ring of depigmentation. Automated 10-2 fields are full, spectral domain OCT shows an intact outer retina, and no baseline record exists.

1 What does the current dose require of the practitioner? TREATMENT

- A Advise stopping the drug today, pending further macular imaging
- B Halve the dose in clinic now and repeat the fields in a year
- C Ask that the dose be recalculated on ideal rather than actual weight
- D Continue as prescribed, since retina, fields and acuity are normal
- E Report to the prescriber that the dose exceeds 5 mg/kg and needs review

ANSWER ______ CONFIDENCE L · M · H

2 Which two tests form the core of annual screening here? BASIC SCIENCE · SELECT 2

- A Fundus autofluorescence of the pole
- B Colour vision with pseudoisochromatic plates
- C Weekly Amsler chart testing at home
- D Spectral domain OCT of the macula
- E Automated 10-2 threshold visual fields

ANSWER ______ CONFIDENCE L · M · H

3 Which pattern would indicate early toxicity? DIAGNOSIS

- A Loss of the ellipsoid zone directly beneath the foveal centre
- B Cystoid spaces in the inner nuclear layer around the fovea
- C Drusenoid elevation of the pigment epithelium temporal to fixation
- D Choroidal thickening beneath the central macula on enhanced imaging
- E Parafoveal outer retinal thinning with the foveal centre intact

ANSWER ______ CONFIDENCE L · M · H

4 Screening at year seven shows parafoveal ellipsoid loss and a 10-2 ring defect. What next? TREATMENT

- A Advise halving the dose and continue annual screening as before
- B Add autofluorescence and electroretinography before any advice
- C Report to the prescriber this week so the drug can be withdrawn
- D Repeat the fields and OCT in six months before advising a change
- E Tell the patient to stop the drug today and inform the prescriber later

ANSWER ______ CONFIDENCE L · M · H

5 What does the duty of care require of the screening record? LAW AND ETHICS

- A The prescriber's dose calculation copied into the record each year
- B A written note that continuing the drug is at the patient's own risk
- C Annual results kept in the practice record and released only on written request
- D Baseline images and fields on file, with every result sent to the prescriber
- E A consent form for screening signed and filed before each annual visit

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-17 · GLAUCOMA · 5 ITEMS

Pressures that crept up between two visits

A 63-year-old woman has no visual symptoms. Her mother went blind from glaucoma. Asthma controlled on an inhaler, no other medication, resting pulse 54. BVA 20/20 both eyes. Goldmann IOP 26 OD, 28 OS. Central corneal thickness 545 and 548 µm. Gonioscopy: open to the ciliary band, no pigment band, no pseudoexfoliation. Cup-disc 0.75 OD and 0.8 OS with inferior rim thinning. Repeatable 24-2: superior nasal step OS, full OD.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Secondary glaucoma from pseudoexfoliation
- B Primary open angle glaucoma, worse in the left eye
- C Ocular hypertension with physiologically large cups
- D Normal tension glaucoma with asymmetric discs
- E Pigmentary glaucoma at an early stage

ANSWER ______ CONFIDENCE L · M · H

2 **Which first-line topical agent fits this patient?** TREATMENT

- A Brinzolamide 1% three times daily in both eyes
- B Brimonidine 0.2% twice daily in both eyes
- C Pilocarpine 2% four times daily in both eyes
- D Timolol 0.5% twice daily in both eyes, with punctal occlusion taught
- E A prostaglandin analogue at night in both eyes

ANSWER ______ CONFIDENCE L · M · H

3 **Which three make up an adequate baseline before treatment starts?** TREATMENT · SELECT 3

- A Gonioscopy of both eyes with the angle graded
- B Central corneal thickness in both eyes
- C Anterior segment imaging used instead of gonioscopy
- D Corneal endothelial cell count in both eyes
- E Fundus autofluorescence of both maculae
- F Two repeatable threshold visual field examinations

ANSWER ______ CONFIDENCE L · M · H

4 **Three months on the prostaglandin, IOP is 21 OD and 23 OS. What next?** TREATMENT

- A Switch to the lowest available concentration of a beta-blocker
- B Refer for trabeculectomy in the left eye on the next available list
- C Repeat the visual fields before changing any treatment
- D Continue the same drop and review again in six months
- E Add a topical carbonic anhydrase inhibitor and review in six weeks

ANSWER ______ CONFIDENCE L · M · H

5 **Which finding decides that this is glaucoma rather than ocular hypertension?** DIAGNOSIS

- A An intraocular pressure of 28 in the left eye by Goldmann
- B A cup-disc ratio of 0.8 in the left eye
- C A mother blinded by glaucoma in later life
- D Central corneal thickness of 548 µm
- E Inferior rim thinning matched by a nasal field defect

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-18 · GLAUCOMA · 5 ITEMS

A red eye, a headache, and vomiting since evening

A 66-year-old woman, +3.25 D in both eyes, has six hours of severe right brow ache with blurred vision and haloes around lights. She has vomited twice and took a cold remedy yesterday. BVA counting fingers OD, 20/25 OS. OD: injected conjunctiva, hazy oedematous cornea, mid-dilated oval unreactive pupil, very shallow chamber, IOP 54. No cells, no keratic precipitates, no lens opacity flecks. OS: IOP 18, narrow angle, no synechiae.

1 What is the most likely diagnosis? DIAGNOSIS

- A Phacomorphic closure from an intumescent lens
- B Plateau iris syndrome after pharmacological dilation
- C Acute primary angle closure with pupil block
- D Neovascular glaucoma with an angle membrane
- E Acute anterior uveitis with a raised pressure

ANSWER ______ CONFIDENCE L · M · H

2 What is the most appropriate immediate treatment? TREATMENT

- A Laser peripheral iridotomy at once, through the oedematous cornea, with no drops
- B Pilocarpine 2% now, repeated every five minutes for three doses
- C Intravenous mannitol as the first step, before any topical agent
- D Urgent lens extraction on the emergency list this evening
- E Topical pressure lowering with oral acetazolamide, supine, review in an hour

ANSWER ______ CONFIDENCE L · M · H

3 Which two steps belong in the definitive plan once the pressure falls? TREATMENT · SELECT 2

- A Laser peripheral iridotomy to the affected eye
- B Prophylactic iridotomy to the fellow eye
- C Long-term prostaglandin drops in both eyes
- D Selective laser trabeculoplasty to the angle
- E Iridoplasty repeated at monthly intervals

ANSWER ______ CONFIDENCE L · M · H

4 One week after iridotomy, IOP is 24 with synechiae over 20% of the angle. What next? TREATMENT

- A Observe without treatment and review in three months
- B Start a topical pressure lowering agent and monitor the angle
- C Repeat the iridotomy at a second site in the same eye
- D Add pilocarpine long term to hold the angle open
- E Refer for trabeculectomy with an antimetabolite

ANSWER ______ CONFIDENCE L · M · H

5 Which finding shows that pupil block, not a plateau configuration, drives the closure? DIAGNOSIS

- A A mid-dilated oval pupil that does not react to light
- B Corneal oedema at a pressure of 54 in that eye
- C A narrow angle in the fellow eye with no synechiae and normal pressure
- D Onset in the evening after a dose of cold remedy
- E A convex iris with a very shallow central chamber in a hyperopic eye

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-19 · GLAUCOMA · 5 ITEMS

Ache and redness in the eye that failed in spring

A 71-year-old man had an ischaemic central retinal vein occlusion in the right eye four months ago and now has three weeks of ache, redness and haloes. BVA counting fingers OD, 20/30 OS. OD: oedematous cornea, IOP 44, deep central chamber, fine vessels crossing the pupil margin onto the iris, gonioscopy shows fibrovascular tissue bridging the angle with synechiae over 180°. Disc pale, old haemorrhages. OS: IOP 15, angle open.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Steroid response glaucoma after treatment of the occlusion
- B Ghost cell glaucoma from degenerate red blood cells
- C Uveitic glaucoma with trabeculitis and iris vessels
- D Neovascular glaucoma with a fibrovascular angle membrane
- E Acute angle closure in an eye with an old occlusion

ANSWER ______ CONFIDENCE L · M · H

2 **Which two findings establish that the angle is closed by membrane, not apposition?** DIAGNOSIS · SELECT 2

- A Fibrovascular tissue bridging the angle on gonioscopy
- B Peripheral anterior synechiae over 180° of the angle
- C Fine vessels crossing the pupil margin onto the iris
- D Corneal oedema at a pressure of 44 in that eye
- E Three weeks of increasing ache, redness and haloes

ANSWER ______ CONFIDENCE L · M · H

3 **What is the most appropriate management now?** TREATMENT

- A Trabeculectomy with mitomycin C on the next available list
- B Cyclodiode laser now, with no retinal treatment planned
- C Retrobulbar alcohol for pain, since the eye sees very little
- D Intravitreal anti-VEGF with topical pressure lowering, panretinal laser to follow
- E Panretinal photocoagulation alone now, with pressure drops and review in four weeks

ANSWER ______ CONFIDENCE L · M · H

4 **Pressure stays 38 on maximal drops, the eye is painful and sees hand movements. What next?** TREATMENT

- A Glaucoma drainage device with a tube to the chamber
- B Enucleation, since the eye has no useful vision left
- C Retrobulbar chlorpromazine with the drops continued
- D Cyclodiode laser to lower the pressure and settle the pain
- E Trabeculectomy with mitomycin C and a releasable suture

ANSWER ______ CONFIDENCE L · M · H

5 **Which mediator most directly drives the angle change?** BASIC SCIENCE

- A Prostaglandin released from a chronically inflamed uvea
- B Transforming growth factor beta from lens epithelium
- C Complement activated on the trabecular endothelium
- D Ferritin deposited from degraded intraocular blood
- E Vascular endothelial growth factor from ischaemic retina

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-20 · OPTIC NERVE AND NEURO-OPHTHALMIC · 5 ITEMS

Ache behind the eye, then the colours went flat

A 29-year-old woman has four days of worsening blur in the left eye with ache on eye movement, and says colours look washed out on that side. No headache. Two years ago she had a fortnight of numbness in the right leg that resolved without investigation. BVA 20/20 OD, 20/80 OS. RAPD OS. Ishihara 14 of 14 OD, 4 of 14 OS. Central depression on 24-2 OS. Both discs flat and normal in colour. IOP 14 both eyes.

1 What is the most likely diagnosis? DIAGNOSIS

A Leber hereditary optic neuropathy, first eye
B Posterior ischaemic optic neuropathy after hypotension
C Demyelinating retrobulbar optic neuritis
D Non-arteritic anterior ischaemic optic neuropathy
E Compressive optic neuropathy at the orbital apex

ANSWER ______ CONFIDENCE L · M · H

2 Which feature best separates this from an ischaemic optic neuropathy? DIAGNOSIS

A Pain on eye movement with a normal disc at 29
B A relative afferent pupillary defect on that side
C Acuity of 20/80 with a central field depression
D Four of fourteen Ishihara plates read correctly
E An intraocular pressure of 14 in both eyes

ANSWER ______ CONFIDENCE L · M · H

3 What is the most appropriate management? TREATMENT

A Oral prednisolone 1 mg/kg for eleven days with a taper
B Observation alone, since most eyes recover their acuity
C Same-day inflammatory markers and a temporal artery biopsy
D Contrast MRI of brain and orbits, with neurology referral
E Intravenous methylprednisolone started today, imaging later

ANSWER ______ CONFIDENCE L · M · H

4 Which two statements belong in the counselling? TREATMENT · SELECT 2

A Colour vision returns fully in every case by six weeks
B Recovery is unlikely once acuity has fallen to 20/80
C Most eyes recover useful acuity over two to three months
D Steroids speed recovery but do not change the final acuity
E Low dose oral steroid prevents a second attack later

ANSWER ______ CONFIDENCE L · M · H

5 Which mechanism explains the ache on eye movement? BASIC SCIENCE

A Ischaemia of the short posterior ciliary arteries at the disc
B Stretch of the ciliary nerves crossing the posterior sclera
C Raised orbital pressure from oedema within the muscle cone
D Meningeal irritation transmitted along the optic canal
E Traction on the inflamed nerve sheath by the muscle origins

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-21 · OPTIC NERVE AND NEURO-OPHTHALMIC · 5 ITEMS

Vision gone on waking, and the scalp hurts to comb

A 74-year-old woman woke with painless loss of vision in the right eye. Six weeks of scalp tenderness on combing, jaw ache after chewing, weight loss and night sweats. BVA counting fingers OD, 20/25 OS. Right RAPD. Chalky pallid disc oedema OD with two flame haemorrhages, cup 0.2 OS. Macula and retinal vessels normal OD. ESR 78 mm/h, CRP 42 mg/L, platelets 486. Right temporal artery firm and non-pulsatile.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Non-arteritic anterior ischaemic optic neuropathy of the right optic nerve
- B Arteritic anterior ischaemic optic neuropathy, giant cell arteritis
- C Central retinal artery occlusion with secondary disc swelling
- D Demyelinating optic neuritis of the right optic nerve
- E Diabetic papillopathy with segmental oedema of the disc

ANSWER ______ CONFIDENCE L · M · H

2 **Which three findings separate an arteritic cause from a non-arteritic one?** DIAGNOSIS · SELECT 3

- A Right relative afferent pupillary defect
- B Painless onset noticed on waking
- C Jaw ache after chewing for six weeks
- D Chalky pallid oedema of the swollen disc
- E ESR 78 mm/h with CRP 42 mg/L
- F Cup-to-disc ratio 0.2 in the fellow eye

ANSWER ______ CONFIDENCE L · M · H

3 **What is the most appropriate immediate management?** TREATMENT

- A Repeat ESR and CRP in 48 hours, then decide about steroid
- B High-dose systemic corticosteroid today, biopsy within two weeks
- C Withhold systemic steroid until the temporal artery biopsy is reported
- D Oral prednisolone 20 mg daily with rheumatology review next week
- E Aspirin 300 mg daily and rheumatology review within 48 hours

ANSWER ______ CONFIDENCE L · M · H

4 **Which vascular event best explains the appearance of this disc?** BASIC SCIENCE

- A Stenosis of the internal carotid artery reducing perfusion
- B Occlusion of the short posterior ciliary arteries
- C Thrombosis of the central retinal vein at the lamina
- D Embolic occlusion of a cilioretinal artery at the disc
- E Vasculitis of the pial vessels of the canalicular nerve

ANSWER ______ CONFIDENCE L · M · H

5 **Ten days into high-dose steroid, which change most needs escalation?** TREATMENT

- A Insomnia, tremor and low mood since the steroid started
- B ESR fallen to 40 mm/h with the jaw ache still present
- C Right acuity unchanged at counting fingers since day one
- D Transient greying of the left eye with CRP risen to 38
- E Fasting glucose 9.2 mmol/L with nocturia since starting steroid

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-22 · OPTIC NERVE AND NEURO-OPHTHALMIC · 5 ITEMS

Headaches worst on waking, with a whoosh in both ears

A 28-year-old woman has three months of daily headache, worst on waking and on coughing, with brief greyouts of vision on standing and a whooshing noise in both ears. Horizontal diplopia at distance. BMI 38. Doxycycline for four months. BVA 20/20 each, no RAPD. Bilateral disc oedema with obscured crossing vessels, no spontaneous venous pulsation. Abduction limited OS. Perimetry: enlarged blind spots, nasal step OS. BP 128/78.

1 **What is the most likely diagnosis?** DIAGNOSIS

A Malignant hypertension with bilateral disc swelling and exudates
B Bilateral non-arteritic ischaemic optic neuropathy
C Papilloedema from raised intracranial pressure
D Bilateral optic neuritis of demyelinating origin
E Buried optic disc drusen with pseudo-oedema

ANSWER ______ CONFIDENCE L · M · H

2 **Which three steps belong in the initial workup?** TREATMENT · SELECT 3

A Dispense base-out prism for the distance diplopia
B MRI of the brain with venography to exclude venous thrombosis
C Lumbar puncture with an opening pressure, after imaging
D Stop the doxycycline and review the medication list
E Start an oral corticosteroid taper for the disc oedema
F Order carotid duplex ultrasound of both sides

ANSWER ______ CONFIDENCE L · M · H

3 **Which mechanism explains the abduction deficit?** BASIC SCIENCE

A Stretch of the sixth nerve over the petrous ridge
B Infarction of the sixth nerve nucleus in the dorsal pons
C Restriction of the left medial rectus by an infiltrative myopathy
D Compression of the nerve at the orbital apex by swollen muscle
E Demyelinating plaque in the paramedian pontine reticular formation

ANSWER ______ CONFIDENCE L · M · H

4 **Which change would move care to urgent surgical decompression?** TREATMENT

A Weight unchanged after three months of dietary advice
B Acuity falling to 20/60 with the field closing in ten days
C Headache still present after two weeks of acetazolamide
D Discs unchanged at six weeks with stable fields and acuity
E Pulsatile tinnitus louder while acuity stays at 20/20

ANSWER ______ CONFIDENCE L · M · H

5 **Which finding would indicate buried disc drusen rather than true oedema?** DIAGNOSIS

A Peripapillary haemorrhage at the inferior disc margin
B Transient greyouts of vision lasting a few seconds
C Lumpy margin with autofluorescence and clear crossing vessels
D Absent spontaneous venous pulsation on careful observation
E Small crowded disc with no physiological cup and blurred margins

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-23 · OPTIC NERVE AND NEURO-OPHTHALMIC · 5 ITEMS

The lid that dropped with pain behind the eye

A 58-year-old man has six hours of a drooping right lid, vertical and horizontal double vision, and severe pain behind the right eye. Smoker, hypertensive. Right ptosis 5 mm. The right eye rests down and out, with limited elevation, depression and adduction, and full abduction. Right pupil 6 mm, sluggish to light; left 3.5 mm, brisk. No RAPD. Deficit unchanged after 60 seconds of sustained upgaze. BP 168/96, glucose 6.1.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Pupil-involving third nerve palsy from an aneurysm
- B Microvascular ischaemic third nerve palsy sparing the pupil
- C Ocular myasthenia gravis with variable ptosis and diplopia
- D Cavernous sinus lesion involving the third and fourth nerves
- E Giant cell arteritis presenting as an ocular motor palsy

ANSWER ______ CONFIDENCE L · M · H

2 **What is the most appropriate next step?** TREATMENT

- A ESR, CRP and high-dose steroid started today
- B Fresnel prism now with review in two weeks
- C Same-day CT angiography with neurosurgical referral
- D MRI of the brain booked within the next week
- E Observation for four weeks for spontaneous recovery

ANSWER ______ CONFIDENCE L · M · H

3 **Which three findings localise this to the nerve rather than muscle or junction?** DIAGNOSIS · SELECT 3

- A Down and out rest position with full abduction
- B Deficit constant through 60 seconds of upgaze
- C Ptosis of 5 mm on the affected side
- D Vertical and horizontal double vision
- E Severe pain behind the affected eye
- F Dilated pupil of 6 mm, sluggish to light

ANSWER ______ CONFIDENCE L · M · H

4 **Why is the pupil involved in a compressive lesion of this nerve?** BASIC SCIENCE

- A Ischaemia of the vasa nervorum strikes the outer fibres first
- B Sympathetic fibres from the carotid plexus join the nerve trunk
- C Pupillomotor fibres run superficially and are compressed first
- D The ciliary ganglion is carried inside the nerve sheath here
- E Parasympathetic axons are the most metabolically active fibres

ANSWER ______ CONFIDENCE L · M · H

5 **Angiography is negative. Which plan is most appropriate now?** TREATMENT

- A Strabismus surgery once the deviation is measurable at six weeks
- B High-dose oral steroid for presumed inflammatory neuropathy
- C Botulinum toxin to the right lateral rectus this week
- D Vascular risk review, occlusion or prism, reassess at six weeks
- E Repeat vascular imaging every week until recovery begins

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-24 · OPTIC NERVE AND NEURO-OPHTHALMIC · 5 ITEMS

The smaller pupil that hides in the dark

A 52-year-old smoker has three weeks of a drooping right upper lid, aching right-sided neck pain and a hoarse voice, with a cough for two months. BVA 20/20 each. Right upper lid sits 2 mm lower, right lower lid 1 mm higher. In dim light pupils are 2.5 mm OD and 4.0 mm OS; in bright light 3.0 mm and 4.2 mm. Right pupil takes over five seconds to dilate. Motility full, no proptosis, no RAPD.

1 What is the most likely diagnosis? DIAGNOSIS

- A Physiological anisocoria with an involutional ptosis
- B Right tonic pupil with segmental iris palsy
- C Partial right third nerve palsy sparing motility
- D Pharmacological miosis from a topical agent
- E Right oculosympathetic paresis, preganglionic

ANSWER ______ CONFIDENCE L · M · H

2 Which three findings place the lesion in the sympathetic pathway? BASIC SCIENCE · SELECT 3

- A Right pupil takes over five seconds to dilate
- B Right lower lid sits 1 mm higher than the left
- C Right upper lid sits 2 mm lower than the left
- D Best corrected acuity of 20/20 in each eye
- E Full motility with no proptosis on either side
- F Anisocoria larger in dim light than in bright light

ANSWER ______ CONFIDENCE L · M · H

3 Which pharmacological test best confirms the diagnosis at this visit? TREATMENT

- A Apraclonidine 0.5% to both eyes, looking for reversal of anisocoria
- B Pilocarpine 0.125% to both eyes, looking for a supersensitive response
- C Phenylephrine 2.5% to both eyes, comparing the dilation achieved
- D Hydroxyamphetamine 1% to both eyes, comparing dilation of the two
- E Tropicamide 1% to both eyes, comparing the speed of dilation

ANSWER ______ CONFIDENCE L · M · H

4 Which imaging plan is most appropriate? TREATMENT

- A Chest radiograph now, with review in three months if normal
- B No imaging now, review in three months to see if the ptosis changes
- C CT of the neck and chest apex with carotid imaging, this week
- D MRI of the brain alone, arranged as an outpatient in one month
- E Carotid duplex ultrasound alone, arranged within one week

ANSWER ______ CONFIDENCE L · M · H

5 Which change would make carotid dissection the leading concern? DIAGNOSIS

- A Dilation lag of eight seconds instead of five in the dark
- B Weight loss of 4 kg with a cough productive of blood
- C Abrupt onset today with severe neck pain and pulsatile tinnitus
- D Gradual onset over three weeks with the voice unchanged throughout
- E Ipsilateral facial anhidrosis with a normal chest radiograph

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-25 · SYSTEMIC HEALTH · 5 ITEMS

The lid that sits lower by the evening

A 34-year-old woman has six weeks of a right lid that sits lower by the evening and intermittent vertical double vision that changes direction from day to day. No pain. BVA 20/20 each. Right ptosis 3 mm at rest, 4.5 mm after 45 seconds of sustained upgaze, with an upshoot of the lid on refixation. Ptosis 1 mm after a two-minute ice pack. Pupils equal and brisk. Orthophoric at one measurement, 6 Δ left hypertropia at the next.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Aponeurotic ptosis with a decompensating phoria
- B Chronic progressive external ophthalmoplegia
- C Partial right third nerve palsy, pupil sparing
- D Thyroid eye disease with an early restrictive myopathy
- E Ocular myasthenia gravis, no bulbar signs

ANSWER ______ CONFIDENCE L · M · H

2 **Which three investigations are appropriate now?** TREATMENT · SELECT 3

- A Repetitive nerve stimulation or single-fibre EMG
- B MRI of the orbits with fat suppression
- C ESR, CRP and temporal artery ultrasound
- D CT angiography of the circle of Willis
- E Acetylcholine receptor antibody assay
- F CT of the chest to look for a thymoma

ANSWER ______ CONFIDENCE L · M · H

3 **Which action matters most before she leaves today?** TREATMENT

- A Arrange occlusion of the right eye until review in six months
- B Start a trial of pyridostigmine and review in four weeks
- C Safety-net for breathing and swallowing, with urgent routes
- D Dispense press-on prism for the vertical deviation now
- E Order a ptosis crutch to be fitted at the next visit

ANSWER ______ CONFIDENCE L · M · H

4 **Which mechanism explains the fatigue on sustained upgaze?** BASIC SCIENCE

- A Segmental demyelination of the superior division of the third nerve
- B Depletion of acetylcholinesterase in the synaptic cleft
- C Antibodies block postsynaptic acetylcholine receptors
- D Antibodies attack presynaptic voltage-gated calcium channels
- E Deletion of mitochondrial DNA in extraocular muscle fibres

ANSWER ______ CONFIDENCE L · M · H

5 **Which single finding would argue hardest against this diagnosis?** DIAGNOSIS

- A A hypertropia that measures differently on two days
- B Improvement of the ptosis after a two-minute ice pack
- C Normal deep tendon reflexes throughout the four limbs
- D A fixed right pupil of 6 mm with a poor light reaction
- E Weakness of orbicularis oculi on forced lid closure

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-26 · ACCOMMODATION, VERGENCE, OCULOMOTOR · 5 ITEMS

The print that slides together after twenty minutes

A 19-year-old student has four months of print that blurs and slides together after twenty minutes of reading, with frontal headache that eases when one eye is covered. BVA 20/20 each, refraction +0.25 DS each. Cover test orthophoric at distance, 12 Δ exophoria at near. Near point of convergence 14 cm break, 18 cm recovery. Positive fusional vergence at near: blur 8, break 12, recovery 4. Amplitude of accommodation 11 D each.

1 What is the most likely diagnosis? DIAGNOSIS

A Convergence paresis from a dorsal midbrain lesion
B Convergence insufficiency with a remote near point
C Accommodative insufficiency with reduced amplitude
D Basic exophoria equal at distance and at near
E Divergence excess intermittent exotropia

ANSWER ______ CONFIDENCE L · M · H

2 Which three findings support this rather than an accommodative cause? TREATMENT · SELECT 3

A Amplitude of accommodation of 11 D in each eye
B Refraction of +0.25 DS with 20/20 in each eye
C Frontal headache after twenty minutes of reading
D Near point of convergence 14 cm with recovery at 18 cm
E Exophoria of 12 Δ at near with orthophoria at distance
F Positive fusional vergence at near breaking at 12 Δ

ANSWER ______ CONFIDENCE L · M · H

3 Which management has the best evidence as first-line here? TREATMENT

A Office-based vergence therapy with home reinforcement
B Base-in prism at near, prescribed by Sheard's criterion
C Plus addition at near of +1.00 D for reading tasks
D Home pencil push-ups alone, five minutes twice daily
E Referral for bilateral medial rectus resection surgery

ANSWER ______ CONFIDENCE L · M · H

4 Twelve weeks of therapy done well and symptoms persist. What next? TREATMENT

A Prescribe a +1.50 near addition and stop the therapy
B Discharge with reassurance that the condition is benign
C Reassess, then add base-in prism if the near point stays remote
D Repeat the same twelve-week therapy programme without changes
E Refer for surgery on the horizontal muscles at near

ANSWER ______ CONFIDENCE L · M · H

5 Her college asks for a report on her reading difficulty. What is required? LAW AND ETHICS

A Written authorisation from the patient before releasing a report
B Verbal consent recorded in the notes at the time of the request
C Nothing, since the request comes from her educational institution
D Written authorisation from a parent, as she is still a student
E A summary with the identifiers removed, sent without consent

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-27 · AMBLYOPIA AND STRABISMUS · 5 ITEMS

The eye that turns in over a picture book

A 4-year-old boy has six months of an inward turn of the right eye, noticed over picture books and worse when he is tired. Cycloplegic retinoscopy +5.00 DS OD, +4.75 DS OS. Uncorrected cover test 20 Δ esotropia at distance, 30 Δ at near. With the full plus in trial frames, 4 Δ esophoria at distance and 8 Δ esotropia at near. Crowded acuity 20/30 OD, 20/25 OS. Stereopsis 400 seconds of arc. Fundi normal.

1 **What is the most likely diagnosis?** DIAGNOSIS

A Partial right sixth nerve palsy with an incomitant deviation
B Sensory esotropia from untreated unilateral amblyopia
C Refractive accommodative esotropia, convergence excess
D Non-accommodative convergence excess with a high AC/A ratio
E Infantile esotropia that has presented late to care

ANSWER ______ CONFIDENCE L · M · H

2 **What is the most appropriate first management?** TREATMENT

A Full cycloplegic plus worn constantly, review in six weeks
B Half the cycloplegic plus, to preserve some fusional drive
C Full correction with a +2.50 bifocal add from the start
D Part-time occlusion of the left eye for two hours daily
E Referral for bilateral medial rectus recession surgery

ANSWER ______ CONFIDENCE L · M · H

3 **Which three findings show the deviation is accommodative in origin?** DIAGNOSIS · SELECT 3

A Onset noticed six months ago, worse when tired
B Cycloplegic refraction of +5.00 DS in the right eye
C Distance deviation falls to 4 Δ with the full plus
D Esotropia larger at near than at distance untreated
E Crowded acuity of 20/30 in the right eye
F Stereopsis of 400 seconds of arc on testing

ANSWER ______ CONFIDENCE L · M · H

4 **Which mechanism links the refractive error to the deviation?** BASIC SCIENCE

A Bilateral lateral rectus underaction from a mild deficit
B Sensory deprivation from the anisometropia between the two eyes
C Accommodative effort drives convergence by the AC/A link
D Reduced negative fusional vergence range at near fixation
E Immature divergence tone in the developing vergence system

ANSWER ______ CONFIDENCE L · M · H

5 **Six weeks on full plus: aligned at distance, 12 Δ esotropia at near. What next?** TREATMENT

A Refer for surgery on the medial recti for the residual angle
B Add a +2.50 flat-top bifocal with the segment splitting the pupil
C Increase the distance plus by 1.00 D over the cycloplegic finding
D Prescribe 8 Δ base-out prism divided between the two eyes
E Start part-time occlusion of the left eye for the near deviation

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-28 · ACCOMMODATION, VERGENCE, OCULOMOTOR · 5 ITEMS

The head tilt that started after the bicycle fall

A 22-year-old man has three weeks of vertical double vision since a fall from his bicycle with brief loss of consciousness. Worst looking down, and he misjudges stairs. Head tilted to the left. Right hypertropia 8 Δ in primary position, 14 Δ in left gaze, 18 Δ on right head tilt, 4 Δ on left tilt. Depression in adduction limited OD. Excyclotorsion 8 ° on double Maddox rod. Vertical fusion range 4 Δ.

1 What is the most likely diagnosis? DIAGNOSIS

A Decompensated congenital right fourth nerve palsy
B Right skew deviation from a posterior fossa lesion
C Right Brown syndrome limiting elevation in adduction
D Thyroid eye disease restricting the right inferior rectus
E Acquired right fourth nerve palsy after head trauma

ANSWER ______ CONFIDENCE L · M · H

2 Which three findings favour an acquired palsy over a decompensated congenital one? TREATMENT · SELECT 3

A Right hypertropia increasing in left gaze
B Symptomatic excyclotorsion of 8 ° on double Maddox rod
C Vertical fusion range of only 4 Δ
D Onset dated to a head injury three weeks ago
E Right hypertropia increasing on right head tilt
F Habitual head tilt towards the left shoulder

ANSWER ______ CONFIDENCE L · M · H

3 What is the most appropriate management now? TREATMENT

A Surgery on the right inferior oblique within a month
B Botulinum toxin to the right inferior oblique now
C Neurosurgical referral for the head injury this week
D Occlusion or Fresnel prism, reassess at three months
E Ground-in vertical prism in new spectacles today

ANSWER ______ CONFIDENCE L · M · H

4 What does the head tilt test measure in this patient? BASIC SCIENCE

A The vertical fusional range available in each head position
B Vestibulo-ocular gain during a slow head roll manoeuvre
C The comitance of the deviation between the two lateral gazes
D Right intorters compete on right tilt, so the superior rectus elevates
E Right extorters compete on right tilt, so the inferior oblique elevates

ANSWER ______ CONFIDENCE L · M · H

5 Which finding would point to skew deviation instead? DIAGNOSIS

A Hypertropia that increases on head tilt towards the higher eye
B Limited depression of the higher eye in adduction
C Vertical deviation that is larger in contralateral gaze
D Incyclotorsion of the higher eye with a contralateral head tilt
E Excyclotorsion of the higher eye measured on Maddox rod

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-29 · AMETROPIA · 5 ITEMS

The prescription that has moved twice this year

A 9-year-old girl reports distance blur. Her spectacles from a year ago read -1.25 DS each; cycloplegic refraction today is -2.50 DS OD and -2.75 DS OS. Axial length 24.30 mm OD and 24.40 mm OS, up 0.45 mm in the year. Both parents are myopic. She spends under an hour outdoors and about four hours on near work daily. Corrected acuity 20/20 each, orthophoric, accommodative lag +0.75 D. Fundi normal.

1 Which feature best predicts continued progression here? DIAGNOSIS

- A Two myopic parents in the immediate family history
- B Spherical equivalent of -2.50 D at the present visit
- C Four hours of daily near work with little time outdoors
- D Accommodative lag of +0.75 D at the near point
- E Axial elongation of 0.45 mm in the year at age nine

ANSWER ______ CONFIDENCE L · M · H

2 Which single change has been shown to slow axial elongation? TREATMENT

- A Low-dose atropine 0.05% at night to both eyes
- B Undercorrection of the distance prescription by 0.75 D
- C Single vision spectacles worn at the full correction
- D A progressive addition lens for the accommodative lag
- E Vergence therapy to reduce the accommodative demand

ANSWER ______ CONFIDENCE L · M · H

3 What is the most appropriate plan at this visit? TREATMENT

- A Full correction, a control option chosen, axial length at six months
- B Full correction now, with review of the refraction in twelve months
- C Atropine 1% at night with photochromic lenses for the glare
- D Undercorrect by 0.50 D and increase time outdoors to two hours daily
- E Orthokeratology fitted today without discussing the alternatives

ANSWER ______ CONFIDENCE L · M · H

4 At six months on atropine 0.05%, axial length is up 0.25 mm. What next? TREATMENT

- A Switch to single vision spectacles and advise outdoor time
- B Check compliance, then consider adding an optical treatment
- C Stop treatment, since this elongation shows the drop has failed
- D Increase the atropine concentration to 1% each night
- E Continue unchanged and remeasure in a further twelve months

ANSWER ______ CONFIDENCE L · M · H

5 Which three duties apply when starting the atropine off-label? LAW AND ETHICS · SELECT 3

- A Notify the state board of optometry of the prescription
- B Obtain a second practitioner's signature before prescribing
- C Ask the child to countersign the consent entry herself
- D Explain that the use is outside the licensed indication
- E Record the parent's informed consent in the clinical notes
- F Document the alternatives offered and the review interval

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-30 · OPHTHALMIC OPTICS AND SPECTACLES · 5 ITEMS

The new spectacles that double the print

A 58-year-old woman cannot tolerate new single vision spectacles: R +5.00 DS, L +1.00 DS, optical centres set at her distance interpupillary distance. She reads with her eyes 8 mm below the optical centres. Print doubles vertically after a minute of reading. Corrected acuity 20/20 each. Cover test orthophoric at distance, 2 Δ exophoria at near. Vertical fusion range 2 Δ. Frame well adjusted, vertex distance equal.

1 **By Prentice's rule, what vertical imbalance does the reading position induce?** BASIC SCIENCE

A 0.8 Δ, base up before the right eye
B 1.6 Δ, base up before the right eye
C 3.2 Δ, base up before the right eye
D 4.0 Δ, base up before the right eye
E 4.8 Δ, base up before the right eye

ANSWER ______ CONFIDENCE L · M · H

2 **What explains the diplopia at the reading position?** DIAGNOSIS

A Incorrect interpupillary distance creating a horizontal imbalance
B Aniseikonia from the difference in spectacle magnification
C Induced vertical imbalance exceeding her vertical fusion range
D A decompensating vertical phoria unmasked by new spectacles
E A mild right superior oblique palsy revealed by near work

ANSWER ______ CONFIDENCE L · M · H

3 **Which three changes would reduce the imbalance at the reading position?** TREATMENT · SELECT 3

A Contact lens correction of the anisometropia for near work
B A larger frame with a higher fitting cross position
C Reducing the vertex distance by 2 mm at both eyes
D Polycarbonate lenses to reduce the lens thickness
E A separate near pair with centres set at the reading position
F Slab-off on the more plus lens if a bifocal is dispensed

ANSWER ______ CONFIDENCE L · M · H

4 **Which single solution is most appropriate for her reading?** TREATMENT

A Add 3 Δ base in at near, split between the lenses
B A smaller frame with a higher fitting cross for both eyes
C A dedicated near pair centred at the reading position
D Grind 3 Δ base down into the right distance lens
E Reduce the right lens to +4.00 DS to balance the prisms

ANSWER ______ CONFIDENCE L · M · H

5 **Which dispensing check would have caught this before collection?** TREATMENT

A Confirm the interpupillary distance to 1 mm
B Check the base curve matches the previous pair
C Measure the pantoscopic tilt and adjust the pads
D Compute the induced prism at the near gaze position
E Verify the back vertex distance at both eyes

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-31 · OPHTHALMIC OPTICS AND SPECTACLES · 5 ITEMS

Clearer when the frame slides down her nose

A 24-year-old returns three days after collecting new spectacles: distance vision is less crisp than in her old pair and a dull brow ache builds through the afternoon. Subjective refraction two weeks earlier, in a trial frame at a 14 mm vertex, was -12.00 DS each eye giving 20/20. The dispensed frame sits at 9 mm. Acuity is now 20/25 each eye, and clearer when she slides the frame down her nose. Corneae, motility and fundi are normal.

1 What best explains the reduced acuity and the brow ache? DIAGNOSIS

A Accommodative insufficiency unmasked by a full distance correction
B Aniseikonia between the two high-powered lenses
C Over-minus at the shorter vertex distance of the dispensed frame
D Excess pantoscopic tilt introducing unwanted oblique astigmatism
E Base-in prism from optical centres set wider than her pupils

ANSWER ______ CONFIDENCE L · M · H

2 Which power at the 9 mm plane focuses this eye as -12.00 DS did at 14 mm? BASIC SCIENCE

A -13.00 DS
B -12.50 DS
C -12.00 DS
D -11.25 DS
E -10.25 DS

ANSWER ______ CONFIDENCE L · M · H

3 Which two features separate over-minus from an under-corrected refraction here? DIAGNOSIS · SELECT 2

A Best acuity of 20/25 in each eye
B Both eyes carry the same spherical power
C The old spectacles were more comfortable
D Acuity improves as the frame slides away from the eye
E Brow ache that builds with sustained near and distance work
F The refraction was performed at a 14 mm vertex

ANSWER ______ CONFIDENCE L · M · H

4 What is the most appropriate action? TREATMENT

A Order aspheric lenses of the same power to cut edge distortion
B Prescribe a +0.75 D reading addition for the brow ache
C Reassure her that she will adapt, and review in four weeks
D Remake both lenses at -11.25 DS for the 9 mm vertex
E Adjust the pads to hold the frame at the 14 mm refracting vertex

ANSWER ______ CONFIDENCE L · M · H

5 She asks about soft contact lenses. Which trial power fits this refraction? TREATMENT

A -12.00 D
B -11.25 D
C -10.75 D
D -10.25 D
E -9.50 D

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-32 · CONTACT LENSES · 5 ITEMS

Gritty red eye on waking in a lens wearer

A 22-year-old soft lens wearer slept in her monthly lenses for two nights and woke with a gritty red right eye and mild watering. Discomfort eased within an hour of removing the lens. VA 20/20. A single 0.6 mm grey-white infiltrate lies 2 mm inside the temporal limbus with faint punctate stain over it, the cornea between it and the limbus is clear, injection is sectoral, the chamber is quiet and lid margins are clean.

1 **What is the most likely diagnosis?** DIAGNOSIS

- A Staphylococcal marginal keratitis from lid disease
- B Herpes simplex stromal keratitis, first episode
- C Early Acanthamoeba keratitis from tap water rinsing
- D Early bacterial keratitis of the peripheral cornea
- E Contact lens peripheral ulcer, sterile

ANSWER ______ CONFIDENCE L · M · H

2 **Which three findings argue for a sterile infiltrate rather than microbial keratitis?** DIAGNOSIS · SELECT 3

- A Symptoms that began overnight in the lens
- B Sectoral rather than diffuse conjunctival injection
- C Best corrected acuity of 20/20 in that eye
- D Clear cornea between the infiltrate and the limbus
- E An epithelial stain smaller than the infiltrate beneath it
- F An anterior chamber without cells or flare

ANSWER ______ CONFIDENCE L · M · H

3 **What is the most appropriate management today?** TREATMENT

- A Scrape for culture now and start hourly fortified antibiotics through the night
- B Topical dexamethasone four times daily and refitting in a daily disposable
- C Cyclopentolate, a pressure pad overnight, review in 48 hours
- D Stop lens wear, broad-spectrum antibiotic four times daily, review in 24 hours
- E Stop lens wear, unpreserved lubricants only, review in one week

ANSWER ______ CONFIDENCE L · M · H

4 **At review the next day, which change most requires culture and intensive therapy?** TREATMENT

- A Symptoms are worse but the infiltrate is fainter and the stain has gone
- B A second 0.4 mm infiltrate has appeared in the same peripheral quadrant
- C Acuity has dropped to 20/30 from lid oedema and tear film debris
- D Injection has spread to all four quadrants, the infiltrate unchanged at 0.6 mm
- E Infiltrate now 1.8 mm and denser, epithelial defect wider than it, 1+ cells

ANSWER ______ CONFIDENCE L · M · H

5 **She asks for her lens prescription today, to reorder online. What is the right response?** LAW AND ETHICS

- A Record that lens wear is unsafe until the cornea heals, and defer release
- B Release it with a written caution not to wear lenses until the eye settles
- C Decline any further lens supply and move her to spectacles permanently
- D Send her to another practitioner to decide about the prescription
- E Release it, since the fitting was verified and completed last year

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-33 · LOW VISION · 5 ITEMS

The post she can no longer read

A 79-year-old retired teacher with geographic atrophy reads headlines but not her post. Best corrected acuity is 20/125 each eye, stable over a year. She reads 1.6 M print at 25 cm slowly and cannot hold a line of text. Amsler shows a dense scotoma just to the right of fixation in the better eye. Both hands have a mild resting tremor. She still drives locally in daylight. Her kitchen has one ceiling light.

1 **By Kestenbaum's rule, which add starts her on 1 M print, and at what working distance?** BASIC SCIENCE

- A +4.00 D at 25 cm
- B +5.00 D at 20 cm
- C +6.25 D at 16 cm
- D +8.00 D at 12.5 cm
- E +10.00 D at 10 cm

ANSWER ______ CONFIDENCE L · M · H

2 **Which finding best explains why she cannot hold a line of print?** DIAGNOSIS

- A Illumination at the page is far below the level she needs
- B Her acuity reserve for 1 M print is smaller than 2 to 1
- C Uncorrected astigmatism blurs the smaller letters most
- D The scotoma sits to the right of fixation, over the text ahead
- E Contrast sensitivity is reduced across all spatial frequencies

ANSWER ______ CONFIDENCE L · M · H

3 **Which two measures will do most for sustained reading of 1 M print?** TREATMENT · SELECT 2

- A Bring a directable task lamp close to the page
- B Train eccentric viewing to move the scotoma off the text
- C Add a yellow tint to raise apparent contrast
- D Fit a 3x hand magnifier held at arm's length
- E Supply an anti-reflection coating on her spectacles

ANSWER ______ CONFIDENCE L · M · H

4 **Which device suits reading her post at the kitchen table?** TREATMENT

- A A closed-circuit magnifier sited in the sitting room
- B An illuminated stand magnifier at a fixed focal height
- C An illuminated hand magnifier held at its focal distance
- D A +6.25 D half-eye pair used at her habitual 25 cm
- E A 6x monocular telescope braced against the brow

ANSWER ______ CONFIDENCE L · M · H

5 **She mentions that she still drives locally. What is the duty here?** LAW AND ETHICS

- A Tell her she is below the acuity standard, record it, and cite her duty to notify
- B Notify the licensing authority the same day without discussing it with her
- C Advise her to limit driving to daylight and familiar routes, and review in a year
- D Leave the question to her family doctor, who holds the wider record
- E Note the acuity in the record and raise driving at the next visit

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-34 · PERCEPTUAL FUNCTION AND COLOUR VISION · 5 ITEMS

Both eyes, dimmer colours and harder print

A 58-year-old man in month four of treatment for pulmonary tuberculosis reports six weeks of dimmer colours and harder print. He weighs 60 kg and takes ethambutol 1500 mg daily with rifampicin, isoniazid and pyrazinamide; eGFR is 42. Best corrected acuity 20/40 OD and 20/50 OS. Ishihara 2 of 17 plates each eye. No RAPD. Discs and maculae look normal. Fields show symmetric central depression.

1 What is the most likely diagnosis? DIAGNOSIS

A Ethambutol toxic optic neuropathy
B Bilateral simultaneous demyelinating optic neuritis
C Nutritional optic neuropathy of B12 deficiency
D Autosomal dominant optic atrophy presenting late
E Occult macular dystrophy with central loss

ANSWER ______ CONFIDENCE L · M · H

2 By Köllner's rule, which colour loss pairs with disease at this site? BASIC SCIENCE

A Red-green loss, because optic nerve disease affects it first
B Blue-yellow loss, because outer retinal disease affects it first
C Blue-yellow loss, because the S cones are the sparsest type
D Total loss of hue, because all three cone classes fail together
E Red-green loss, because the L and M genes lie on one chromosome

ANSWER ______ CONFIDENCE L · M · H

3 What is the most appropriate next step? TREATMENT

A Reduce ethambutol to 15 mg/kg and reassess in three months
B Start high-dose oral steroid and repeat the fields in a month
C Start hydroxocobalamin and a B-complex, and review in six weeks
D Repeat colour vision in four weeks before contacting anyone
E Speak to the treating physician today to stop ethambutol

ANSWER ______ CONFIDENCE L · M · H

4 Which two actions belong in the plan once the drug is stopped? TREATMENT · SELECT 2

A Genetic testing for OPA1 before accepting the diagnosis
B Monthly full-field electroretinography until recovery
C A course of oral hydroxocobalamin while the nerve recovers
D Written communication to the prescriber with the dose per kg and eGFR
E Recorded acuity, colour and threshold fields now and at six weeks

ANSWER ______ CONFIDENCE L · M · H

5 Which further finding would move the diagnosis away from drug toxicity? DIAGNOSIS

A Temporal disc pallor appearing at three months
B Ishihara errors on every plate in both eyes
C A further fall in eGFR from 42 to 35
D A dense RAPD with pain on movement of one eye
E Bitemporal wedge defects on threshold fields

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

CASE S1-35 · EMERGENCIES AND TRAUMA · 5 ITEMS

The after-hours call about flashes and floaters

An after-hours call. A 61-year-old, -8.00 D myopic and pseudophakic four months, describes six hours of new floaters and arcing flashes in the right eye, with no shadow and no pain. She declines to be seen before a family wedding in four days. A practice audit of 200 such callers found 30 with a tear or detachment; the phone protocol flagged 27 of the 30 and 27 of the other 170.

1 **What should be arranged on this call?** TREATMENT

A Reassurance that this follows surgery, with routine review
B Bed rest with the head still, and a call back in 48 hours
C A dilated fundus examination with indentation today
D A dilated examination at the practice within two weeks
E Emergency department attendance tonight for a B-scan

ANSWER ______ CONFIDENCE L · M · H

2 **From the audit, what is the chance that a flagged caller has a tear or detachment?** BASIC SCIENCE

A 10%
B 15%
C 50%
D 84%
E 90%

ANSWER ______ CONFIDENCE L · M · H

3 **Which three features make her decision to wait an informed refusal?** LAW AND ETHICS · SELECT 3

A The specific risk of a detachment was named, not implied
B She was told what change should make her call back at once
C A signed waiver of liability is held on file
D Her daughter agreed that waiting was reasonable
E The notes were written up the following morning
F She can restate in her own words what delay may cost

ANSWER ______ CONFIDENCE L · M · H

4 **She still declines. What must the record show?** LAW AND ETHICS

A A note to her family doctor recording that the call was made
B The risks named, her refusal, the earliest appointment offered, and return criteria
C That the patient declined advice, with the date and time of the call
D A letter discharging her from the practice for refusing urgent care
E Her signature on a refusal form, to be collected at the next visit

ANSWER ______ CONFIDENCE L · M · H

5 **She calls on day three with a shadow in the upper nasal field. What now?** TREATMENT

A Dilated examination at the practice the next morning
B Reassurance that a shadow follows a large floater load
C Bed rest with the eye positioned, and referral in 48 hours
D Repeat the phone protocol and act on whether it flags
E Same-day vitreoretinal referral, treating it as a detachment

ANSWER ______ CONFIDENCE L · M · H

TARGET 6 MIN Started ______ Finished ______ Flagged ______

PART 9

Teaching keys

Thirty-five keys, one per case, in the order you sat them. They start here and not on the back of each case, so the session can be answered cold.

READ THEM IN THIS ORDER

Wrong answers first. Then correct answers you were unsure of. Then the ones you were sure of and still missed, which are the expensive ones and the last thing anybody wants to look at.

THE DISTRACTOR TABLE IS THE POINT

Knowing why the answer is right is worth one item. Knowing why the trap is tempting is worth every item built the same way, and the exam builds a lot of them the same way.

WHAT EACH KEY CONTAINS

The letter, the answer in a few words, why the findings force it, and a table naming what makes each wrong option tempting and the specific finding that rules it out.

THEN WRITE THE RULE

Each case closes with one or two if-then sentences. Say them aloud. If you cannot, you have read the key rather than learned it.

Case S1-01 · why

1 · B Orbital cellulitis, ethmoid origin

Proptosis, painful restricted motility, RAPD, reduced colour and acuity, and raised IOP all place the disease behind the septum. Fever with a week of purulent rhinorrhoea names the source.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Cavernous sinus thrombosis**	The correct escalation of this infection, and it also gives proptosis and chemosis.	Signs stay strictly unilateral, with no multiple cranial neuropathy and no V1-V2 loss.
D	**Preseptal cellulitis**	The visible finding is a swollen lid, and preseptal disease is commoner at this age.	Preseptal disease spares acuity, pupil, colour, motility and globe position.
C	**Rhabdomyosarcoma**	Rapid proptosis in a child is the classic teaching phrase for it.	Fever and purulent sinus disease over three days fit infection, not tumour.

2 · A, D, F RAPD, red desaturation, reduced acuity

The item asks for afferent function, not for evidence of infection. An RAPD, dyschromatopsia and reduced best acuity each measure optic nerve conduction directly.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**IOP 27 mmHg**	It is genuinely abnormal, and orbital pressure is what threatens the nerve.	It measures orbital pressure, not conduction. Normalising it proves nothing.
B, E	**Fever, purulent discharge**	They confirm infection and drive the diagnosis.	Neither measures the eye. A febrile child can have a normal afferent system.

3 · E Same-day referral, contrast CT, IV antibiotics

Orbital cellulitis with afferent dysfunction threatens the eye and, through intracranial extension, life. Imaging, intravenous therapy and surgical availability are arranged together, today.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**IV antibiotics tomorrow**	It sees that intravenous therapy is needed and feels like a fair compromise.	The overnight delay is where vision goes. The decision point is the RAPD, now.
A	**Outpatient MRI in 48 hours**	MRI is the better soft-tissue study, so it reads as the more thorough answer.	Contrast CT answers today's questions faster: sinus, collection, apex, bone.

4 · A Lamina papyracea, from the ethmoid

The medial orbital wall is paper-thin and perforated by vessels, which makes contiguous ethmoid spread the usual route and explains why abduction and elevation fail first.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Valveless venous spread**	A real mechanism, and how mid-face infection reaches the cavernous sinus.	Not the usual route from sinusitis, and it does not localise medially.
C	**Optic canal, sphenoid**	Sphenoid disease is dangerous and sits against the orbital apex.	Rare in children, and it would show apical signs before medial restriction.

5 · A Falling function with an apical abscess

Drainage is driven by function and location, not by the presence of a collection. Falling afferent function with apical involvement is the indication, as is failure to improve on adequate therapy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Abscess, function improving**	An abscess on imaging feels like a surgical finding by definition.	Small medial collections often resolve on antibiotics. Imaging does not decide.
C	**Chemosis increasing**	It looks like deterioration in front of you.	Surface swelling with unchanged afferent function is no reason to open an orbit.

> **IF-THEN RULES FROM THIS CASE**
>
> If lid swelling comes with any change in acuity, colour, pupil, motility or globe position, treat it as orbital until imaging says otherwise. If function falls, drain.

Case S1-02 · why

1 · C Pseudomonas keratitis, overnight wear

A large epithelial defect over a dense suppurative infiltrate, with hypopyon and mucopurulent discharge inside 36 hours, is bacterial. Sleeping in lenses and topped-up solution point to Pseudomonas.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Sterile CLPU**	Also lens related, also painful, and commoner than infection in this group.	CLPU is small, peripheral, without hypopyon, and the epithelium stays intact.
E	**Acanthamoeba keratitis**	Poor solution hygiene is the classic risk factor quoted for it.	Onset in 36 hours with hypopyon and pus is too fast and too suppurative.

2 · D, E, F Scrape and culture, hourly quinolone, lens culture

Culture is taken before the first drop, because therapy sterilises the plate within hours. Hourly fluoroquinolone covers Pseudomonas, and lens, case and solution often grow the same organism.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Topical steroid now**	Scarring is the long-term risk, and steroid does reduce stromal infiltrate.	Steroid before the organism is known and controlled invites rapid melt.
B, C	**Antifungal, patching**	Both feel cautious: broad cover, and comfort for a very painful eye.	No fungal risk factor here, and patching an infected cornea incubates it.

3 · A Progressive thinning despite hourly cover

Response is judged on infiltrate density, defect size and thinning, not on symptoms or the sensitivity report. Progressive thinning on adequate cover means fortified drops and a perforation plan.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Sensitive on culture**	It confirms the drug choice, which reads like the decision point.	Sensitivity in a dish says nothing about the cornea in front of you.
C	**Worse pain, smaller defect**	Pain is what she reports, so it feels like the outcome measure.	Pain lags healing. A defect halving on hourly drops is progress.

4 · B Cycloplegia plus oral analgesia

Cycloplegia relieves ciliary spasm and limits synechiae without touching epithelial healing or host defence. Analgesia is oral, because topical anaesthetic delays healing and masks progression.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Subconjunctival gentamicin**	It puts potent Gram-negative cover straight at the site.	Hourly drops already give high corneal levels, and this adds needle risk.
C	**Topical NSAID**	It treats pain and inflammation without steroid, which sounds safe.	Topical NSAIDs are linked to melt in a cornea that is already thinning.

5 · D Secreted proteases digest collagen

Pseudomonas secretes proteases and elastase that cleave stromal collagen directly, which is why melting is measured in hours. Biofilm and adhesion explain how it arrived, not the speed.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Biofilm on lens and case**	Biofilm is why topping up solution fails, and it is genuinely present.	It explains inoculation and persistence, not the rate of stromal loss.
C	**Cyst formation**	Resistance to disinfection fits the hygiene history exactly.	Cysts are an amoebic feature, and cysts do not lyse collagen.

IF-THEN RULES FROM THIS CASE

If a lens wearer has a stromal infiltrate under an epithelial defect, scrape and culture before the first drop. If thinning progresses on hourly cover, fortify and plan for perforation.

Case S1-03 · why

1 · C Recurrent HSV epithelial keratitis

A branching lesion with terminal bulbs, an ulcer bed that stains with rose bengal, reduced sensation and old anterior stromal scars from earlier episodes make recurrent herpes simplex the answer.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Zoster pseudodendrite**	Zoster also gives a raised branching lesion with reduced sensation.	Pseudodendrites are heaped mucus with no terminal bulbs and no ulcer bed.
D	**Healing abrasion**	Healing epithelium forms ridges that can look dendritic on staining.	A healing ridge has no terminal bulbs, and it does not recur three times.

2 · B Topical ganciclovir gel, five times daily

Live virus in the epithelium needs an antiviral at treatment dose until the lesion heals. Steroid on a live dendrite drives geographic ulceration, and lubricants do nothing to the virus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Antiviral plus steroid**	Steroid limits the scarring his earlier episodes plainly caused.	Steroid belongs to stromal disease. On a dendrite it enlarges the ulcer.
D	**Oral aciclovir alone**	Oral dosing avoids drop toxicity and reaches the ganglion too.	400 mg twice daily is the prophylactic dose, not a treatment dose.

3 · A Herpetic stromal keratitis, non-necrotising

Stromal haze with chamber cells appearing after the epithelium has healed is an immune stromal reaction, not live epithelial virus. No defect and no precipitates place the disease in the stroma.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Endotheliitis**	Stromal thickening with a chamber reaction is close to its picture.	Endotheliitis needs keratic precipitates under the oedema. There are none.
E	**Antiviral toxicity**	Two weeks of gel five times daily does cause punctate toxicity.	Toxicity is epithelial and stains. This haze sits under an intact surface.

4 · B Steroid under antiviral cover, slow taper

Immune stromal disease needs steroid, and steroid needs antiviral cover to stop epithelial recurrence. Rapid withdrawal produces the rebound that scars the visual axis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Steroid, no cover**	It treats the immune stroma, which is what is costing him vision.	Unopposed steroid reactivates epithelial virus and risks geographic ulcer.
E	**Antiviral alone**	It cleared the dendrite, so continuing it feels consistent.	Antiviral does not touch the immune infiltrate driving the stromal haze.

5 · D, E Ganglionic latency, axonal reactivation

Herpes simplex persists in the trigeminal ganglion and travels down ophthalmic axons when reactivated. Recurrence is endogenous, and sensory loss accumulates with each episode.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Exogenous reinfection**	Repeat episodes suggest repeat exposure, as with conjunctivitis.	Recurrence is reactivation of the patient's own latent virus.
B, C	**Sensation, keratocytes**	Both sound reasonable: nerves recover, keratocytes host stromal disease.	Sensory loss is cumulative, and latency is ganglionic rather than corneal.

IF-THEN RULES FROM THIS CASE

If a branching epithelial lesion has terminal bulbs, treat live virus and hold the steroid. If stromal haze appears after the epithelium heals, add steroid only under antiviral cover.

Case S1-04 · why

1 · A Acanthamoeba keratitis

Pain far beyond the visible signs for three weeks, radial perineural infiltrates, a partial ring infiltrate and failure of antiviral and antibacterial therapy in a tap-rinsing swimmer name the amoeba.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**HSV stromal keratitis**	Pseudodendrites and a chronic course took two clinicians there already.	Simplex gives neither radial perineural infiltrates nor a ring in weeks.
B	**Fungal keratitis**	Also indolent, also unmoved by antibacterials, also lens related.	Water exposure without vegetable trauma, and no feathery satellite edges.

2 · B, C, D Perineural infiltrates, pain, ring infiltrate

Radial keratoneuritis, a ring infiltrate and pain out of proportion are amoebic. Reduced sensation, pseudodendrites and unilateral photophobia occur in both and separate nothing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Reduced sensation**	Sensory loss is the herpetic sign every candidate learns first.	Both diseases blunt sensation, so it cannot tell them apart.
A, F	**Pseudodendrites, redness**	Pseudodendrites are why the antiviral was started twice already.	Amoebic epithelium mimics dendrites, and redness is common to both.

3 · C Hourly biguanide with a diamidine

A biguanide is the agent that kills cysts, and a diamidine added to it covers trophozoites. Intensive hourly dosing in the first days decides whether the ring resolves.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Steroid plus antiviral**	It was the working diagnosis, and steroid does settle the limbitis.	Steroid raises amoebic load and delays cure. Treat the organism first.
A	**Early keratoplasty**	Medical therapy is long and the eye is deteriorating week by week.	Grafting an active amoebic cornea recurs in the graft. Treat first.

4 · A Steroid added, biguanide continued

Inflammation persisting after the organism is controlled is immune, and steroid settles it. The biguanide continues because steroid can wake residual cysts.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Steroid, cover stopped**	The infiltrate has cleared, so the anti-amoebic looks finished.	Cysts survive months. Steroid without cover produces relapse.
E	**Keratoplasty now**	20/100 with a scarred axis looks like a surgical problem.	Graft an inflamed eye and it fails. Wait for quiescence.

5 · D The cyst wall resists drugs

The double-walled cyst is dormant and poorly permeable, so it outlives short courses and multipurpose solutions. Cure means outlasting cysts, not merely clearing trophozoites.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Intracellular latency**	Chronicity and relapse feel like an intracellular hiding place.	The amoeba is extracellular. It hides in a cyst, not in a cell.
C	**Nerve seeding**	Radial perineural infiltrates make the nerves look like a reservoir.	That infiltrate is inflammation around nerves, not neural latency.

IF-THEN RULES FROM THIS CASE

If a lens wearer has pain out of proportion to the signs and no response to antiviral or antibacterial therapy, scrape for amoeba. If steroid is added later, keep the biguanide running.

Case S1-05 · why

1 · A Kmax up 1.9 D over 14 months

Progression is a documented change in corneal shape, most reliably a rise in maximum keratometry with thinning. Refraction varies with fit and effort, and one thin reading is not a change.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Thinnest point 442 µm**	A thin cornea is the hallmark of the disease and looks like severity.	One measurement is a state, not a change. Progression needs two dates.
D	**Acuity still 20/25**	Preserved vision reads as reassurance that nothing has moved.	Acuity holds until the axis distorts. It lags shape by many months.

2 · D Cross-link the progressing eye, watch the other

Cross-linking is indicated by documented progression, and this eye has it at 17 with 442 µm of tissue. The fellow eye is stable, so it is monitored rather than treated.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Rigid lens only**	A rigid lens restores the acuity he actually came about.	A lens corrects the shape, it does not stop the shape changing.
B	**Review in 12 months**	Watchful waiting sounds proportionate in a young seeing eye.	He is already progressing. Another year of steepening costs tissue.

3 · A, F Stroma under 400 µm, active atopic disease

The standard protocol needs 400 µm of stroma after debridement to protect the endothelium, and an inflamed atopic surface heals badly. A high Kmax and young age argue for treating, not waiting.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Age 17**	Consent and cooperation make young age feel like a reason to wait.	Young eyes progress fastest. Age brings the treatment forward.
B, E	**Kmax 50 D, good acuity**	Both read like thresholds: too steep to treat, too good to risk.	Neither defers anything. Documented progression is the indication.

4 · D Expected cross-linking haze

Haze at the demarcation line peaks in the first months and clears over a year as acuity recovers. An intact epithelium with no pain and no cells argues against infection and against decompensation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Microbial keratitis**	Debridement plus a bandage lens is a real route to infection.	Infection brings pain, a defect and cells. None of the three is present.
A	**Endothelial failure**	The stroma was 442 µm thick, close to the safety limit.	Decompensation gives oedema and folds, not isolated stromal haze.

5 · A Riboflavin plus UVA cross-links collagen

UVA excites riboflavin, which releases reactive oxygen species and forms covalent bonds between collagen fibrils and proteoglycans. The stiffened stroma resists further steepening.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Riboflavin as a shield**	It does shield deeper tissue, which is why the 400 µm rule exists.	Shielding is a safety effect. Stiffening comes from bond formation.
B	**Thermal shrinkage**	Older thermokeratoplasty worked that way, so it sounds familiar.	These UVA doses are non-thermal. Collagen is bonded, not shrunk.

IF-THEN RULES FROM THIS CASE

If maximum keratometry rises with thinning across two dates, cross-link rather than refit. If the stroma measures under 400 µm after debridement, change the protocol, not the indication.

Case S1-06 · why

1 · B Herpes zoster ophthalmicus with uveitis

A painful vesicular rash confined to one dermatome, respecting the midline and involving the nose tip, with sectoral iris atrophy, pigmented precipitates and a raised pressure, is zoster.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Primary herpes simplex**	Vesicles and pseudodendrites overlap, and both are herpesviruses.	Simplex respects no dermatome and does not atrophy the iris in sectors.
D	**Preseptal cellulitis**	A hot swollen lid in an immunosuppressed patient reads bacterial.	Dermatomal vesicles with intraocular signs are not preseptal disease.

2 · C Oral aciclovir 800 mg five times daily

Zoster needs the higher oral dose five times daily, started as early as possible, to cut ocular complications and neuralgia. Simplex doses undertreat it and topical gel never reaches the skin.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Aciclovir 400 mg**	It is the familiar dose carried over from simplex keratitis.	Varicella zoster is less sensitive and needs twice that dose.
A	**Intravenous therapy**	She is immunosuppressed, so parenteral cover feels safer.	Reserve it for dissemination or encephalitis. She has neither.

3 · C Treat the trabeculitis, add a beta-blocker

The rise here is trabeculitis, so treating the inflammation lowers it, with aqueous suppression added while it settles. Prostaglandins and miotics aggravate inflammation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Prostaglandin analogue**	It is the strongest single agent and the usual first choice.	Prostaglandins can worsen uveitis and are avoided while it is active.
B	**Pilocarpine**	Miosis and better outflow both sound useful in a pressure spike.	Miotics increase inflammation and pull the iris into synechiae.

4 · D, E, F Vaccine discussion, long review, inform the team

Zoster eye disease relapses for years, so review runs long and vaccination is discussed after recovery. Her immunosuppression is a decision for the team that prescribes it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Stop methotrexate**	Immunosuppression allowed reactivation, so stopping looks protective.	Her joint disease needs it. Any change belongs to the prescriber.
C	**Varicella serology**	Laboratory proof feels rigorous before a year of follow-up.	A dermatomal vesicular rash is the diagnosis. Serology changes nothing.

5 · D Away from the nursery until lesions crust

Vesicle fluid transmits varicella to non-immune contacts, and infectivity ends when the lesions crust. Treatment shortens the illness, and a dressing does not remove the duty to stay away.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Back once treated**	Antivirals reduce viral shedding, which sounds like safety.	Shedding continues until crusting. Treatment does not end exclusion.
B	**No restriction**	Zoster needs contact, unlike chickenpox spread through the air.	Vesicle fluid still gives varicella to non-immune children.

IF-THEN RULES FROM THIS CASE

If a dermatomal rash reaches the tip of the nose, examine the eye and start high-dose oral antiviral that day. If the pressure rises, treat the trabeculitis rather than adding a prostaglandin.

Case S1-07 · why

1 · C Posterior capsule opacification

A wrinkled pearly membrane with Elschnig pearls behind a centred implant accounts for glare and reduced acuity years after surgery. A normal OCT and stable refraction exclude the alternatives.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Pseudophakic CMO**	It is the classic late cause of blur after cataract surgery.	OCT shows a normal macula, and CMO peaks in months, not two years.
A	**Refractive change**	New spectacles are the simplest explanation for gradual blur.	Refraction is unchanged from one month after his surgery.

2 · D Nd:YAG posterior capsulotomy

A YAG capsulotomy clears the axis in minutes without entering the eye, and his symptoms justify it. Surgery is kept for a dense fibrotic plate the laser cannot open.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Lens exchange**	It removes membrane and pearls together in one definitive operation.	It trades a two-minute laser for intraocular surgery and its risks.
B	**Anti-glare spectacles**	Glare is his main complaint, and coatings genuinely reduce it.	The scatter is inside the eye. No coating reaches behind the implant.

3 · B Same-day dilated peripheral examination

A field shadow with new floaters after capsulotomy is a retinal detachment until proven otherwise, and the laser raises that risk. Only a dilated peripheral view answers the question.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Reassure about floaters**	Debris and floaters are genuinely common in the first days.	A field defect is not debris. The periphery has to be seen today.
E	**Macular OCT**	Post-laser oedema is real and OCT is the test that finds it.	OCT misses the peripheral break causing an upper field shadow.

4 · E A spike to 31 mmHg before discharge

Pressure peaks in the first hours after capsulotomy as pigment and debris load the outflow. A rise into the thirties is treated and rechecked before he goes home.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Rise to 21 mmHg**	It is a rise, and any rise after laser feels like a complication.	A few points of transient rise is expected and settles untreated.
B	**Pigment, same pressure**	Visible pigment is exactly the mechanism of the spike.	Pigment without a pressure rise needs recheck, not treatment.

5 · A, E Disclose detachment risk, record alternatives

Consent covers the material risks a patient in his position would want, detachment and pressure rise among them, and the alternatives including doing nothing. Outcomes are explained, not promised.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Relative consents too**	Involving family feels safer for an older patient.	He has capacity. The consent is his alone to give.
D	**Guarantee 20/20**	He reached 20/20 before, so promising it sounds honest.	A guaranteed outcome is not consent. State the expected range.

> **IF-THEN RULES FROM THIS CASE**
>
> If vision falls years after cataract surgery with a clear macula and stable refraction, look behind the implant. If a field shadow follows capsulotomy, dilate and examine the periphery that day.

Case S1-08 · why

1 · D HLA-B27 acute anterior uveitis

Sudden painful unilateral non-granulomatous inflammation with fibrin, a low pressure and recurrence in the other eye, in a young man with inflammatory back pain, is the B27 pattern.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Herpetic uveitis**	It recurs unilaterally too, with fibrin and synechiae of its own.	Pressure is low, not high, with no sectoral iris atrophy or scar.
A	**Intermediate uveitis**	Recurrent uveitis at this age raises it, and spillover is real.	No vitreous cells and no snowbanks. The reaction is purely anterior.

2 · D Hourly topical steroid with cycloplegia

Three plus cells with fibrin needs intensive topical steroid rather than a four times daily dose, and cycloplegia both eases spasm and breaks the synechiae that are forming.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Steroid four times daily**	It is the right drug at a familiar starting frequency.	Underdosing fibrinous uveitis leaves synechiae behind. Start hourly.
C	**Sub-Tenon steroid**	One injection secures delivery and removes any compliance worry.	Kept for resistant or posterior disease. Drops reach the front.

3 · A, B Inflammatory back pain, alternating recurrence

Morning stiffness that eases with movement suggests axial spondyloarthritis, and recurrent alternating acute anterior uveitis is its ocular signature. The rest occurs in any acute iritis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Fine precipitates**	Precipitate character does separate granulomatous disease.	Fine precipitates are non-specific and appear in any acute iritis.
D, E	**Low pressure, glare**	Both are abnormal and both belong to the picture in front of you.	Ciliary hyposecretion and photophobia occur whatever the cause.

4 · A Rheumatology, HLA-B27, sacroiliac imaging

The history points at one disease, so the tests are aimed there rather than scattered. Naming axial disease changes his systemic treatment even though it does not change today's drops.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Full autoimmune screen**	Casting wide feels thorough once a systemic cause is suspected.	Unaimed panels generate false positives and delay the right referral.
E	**No investigation**	Treatment today is identical whichever result comes back.	The result changes his spinal care and his relapse risk, not his drops.

5 · B Ciliary hyposecretion in acute inflammation

Inflamed ciliary epithelium secretes less aqueous, so pressure falls despite debris in the meshwork. When pressure rises instead, suspect trabeculitis or a herpetic cause.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Fibrin in the meshwork**	Fibrin is visible in this chamber and does obstruct outflow.	Obstruction raises pressure. This eye is soft, so inflow has fallen.
E	**Posterior synechiae**	Pupil block is a familiar pressure mechanism in uveitis.	Seclusion raises pressure through iris bombe. It does not lower it.

IF-THEN RULES FROM THIS CASE

If acute anterior uveitis recurs and alternates in a young adult, ask about inflammatory back pain and refer. If fibrin is present, dose the steroid hourly and dilate the pupil.

Case S1-09 · why

1 · B Diffuse anterior scleritis with a gutter

Pain that wakes her, tenderness on palpation, violaceous injection that phenylephrine does not blanch, and an oedematous sclera in daylight define scleritis. The gutter marks corneal involvement.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Nodular episcleritis**	Both are red and sectoral, and episcleritis is far commoner.	Episcleral vessels blanch with phenylephrine and rarely wake anyone.
D	**Necrotising scleritis**	Rheumatoid disease is exactly where necrotising scleritis appears.	No avascular patch and no translucent thinning. The sclera is swollen.

2 · C, D, E No blanching, night pain, globe tenderness

Deep vessels do not blanch, and only scleral inflammation gives boring pain that wakes a patient and tenderness felt through the lid. Sector redness, watering and duration occur in both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	**Sectoral redness**	Nodular episcleritis is sectoral, so a sector feels diagnostic.	Both diseases can be sectoral. The plane of the vessels decides.
A, B	**Watering, 10 days**	Symptoms and duration read like markers of severity.	Episcleritis waters as well, and it can run for several weeks.

3 · A Oral NSAID with gastric cover, refer

Non-necrotising scleritis usually answers a full-dose oral NSAID, and topical steroid does not reach the sclera. The corneal gutter and her rheumatoid disease bring in the physician.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Topical steroid alone**	Steroid is the reflex anti-inflammatory for a painful red eye.	Drops do not penetrate to the sclera. This treatment is systemic.
E	**Subconjunctival steroid**	It places drug exactly where the inflammation can be seen.	Avoided where sclera may thin. The perforation risk is real.

4 · E Urgent systemic immunosuppression

An avascular patch with a deepening gutter is necrotising disease, which threatens the globe and signals active systemic vasculitis. Steroid alone relapses, so a steroid-sparing agent starts now.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**More NSAID**	It is the class that seemed to be controlling her at first.	Necrotising disease has outrun NSAIDs. Change class, not dose.
C	**Patch graft first**	Thinning sclera looks like a structural problem to repair.	Grafts melt in inflamed sclera. Control the disease, then rebuild.

5 · B Limbal vessels bring complexes and proteases

The peripheral cornea abuts limbal vessels, so immune complexes deposit there and recruit neutrophils whose proteases digest stroma. The avascular centre is spared.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Peripheral thinness**	Anatomy makes it sound as though the thinnest tissue goes first.	The peripheral cornea is thicker than the centre, not thinner.
C	**Tear break-up**	Rheumatoid patients are dry, and dryness does harm epithelium.	Dryness gives punctate staining, not a vascularised immune gutter.

IF-THEN RULES FROM THIS CASE

If a red eye is tender to palpation and the redness does not blanch with phenylephrine, treat it systemically and ask why. If a scleral patch turns avascular, escalate immunosuppression.

Case S1-10 · why

1 · E Neovascular AMD, right eye

Six weeks of metamorphopsia with acuity falling from 20/25 to 20/60, a grey-green subretinal lesion, subretinal blood and confluent soft drusen is choroidal neovascularisation on an AMD background.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Geographic atrophy**	Same disease and same drusen, and it also destroys reading vision.	Atrophy is slow, grey-white with visible choroidal vessels, and dry.
A	**Chronic CSC**	Subretinal fluid with distortion belongs to its picture too.	Her age, the drusen and the subretinal blood do not fit CSC.

2 · B Urgent referral for OCT and anti-VEGF

Vision lost to active neovascularisation returns only if injections start within days to weeks. Imaging and treatment are arranged on one pathway rather than one after the other.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Routine referral**	Six weeks feels prompt measured against usual waiting times.	Fibrosis is permanent. Each week of delay costs recoverable acuity.
E	**Antioxidant supplements**	Supplements do have trial evidence in age-related disease.	They lower risk in the fellow eye. They do not treat active disease.

3 · A, B, C Stop smoking, Amsler the fellow eye, low vision aids

Smoking is her one modifiable risk, the fellow eye needs a home test that catches conversion early, and reading aids restore function while treatment runs.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Beta-carotene formula**	It resembles the supplement trial she may have read about.	Beta-carotene raises lung cancer risk in smokers. Use the later formula.
E	**Surrender the licence**	Central loss in one eye reads like the end of driving.	OS is 20/25. She is assessed against the standard, not disqualified.

4 · E Continue injections, then titrate the interval

Acuity improved and fluid is reducing, which is a response. Treatment continues while the interval is titrated to the eye, because stopping after loading invites recurrence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Stop and monitor**	Three injections is the loading course, so it sounds complete.	Loading starts therapy. Fluid remains, so injections continue.
B	**Switch the agent**	Persistent fluid reads as failure of the drug in use.	This eye improved. Switching is for non-response, not partial response.

5 · A VEGF driven growth through Bruch membrane

Hypoxic, inflamed outer retina raises VEGF, which drives choroidal vessels through breaks in Bruch membrane. Those new vessels are permeable, so fluid and blood collect under the retina.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Complement and drusen**	Complement is central to the disease and drusen are visible here.	It explains the background, not the leaking vessel in front of you.
B	**Lipofuscin build-up**	It underlies the pigment epithelial ageing that starts all this.	Lipofuscin drives atrophy over years, not acute subretinal fluid.

IF-THEN RULES FROM THIS CASE

If distortion is new and acuity has fallen with a grey-green lesion or subretinal blood, refer for injection within days. If fluid persists while vision improves, continue and titrate.

Case S1-11 · why

1 · E PDR with centre-involving oedema

Fronds at the disc and on the arcade are new vessels, which places the eye past non-proliferative disease. Cystoid thickening through the fovea at 445 µm makes the oedema centre-involving.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Severe NPDR**	Haemorrhage in four quadrants and venous beading do meet the severe count.	New vessels are present, and their presence alone ends the non-proliferative grade.
B	**PDR without oedema**	The fronds and the vitreous haze dominate, so the macula gets read second.	Cystoid spaces through the fovea at 445 µm are oedema at the centre.

2 · C Anti-VEGF first, panretinal laser after

Anti-VEGF answers both problems at once: new vessels regress and the fovea dries. Panretinal laser applied over a thick centre worsens acuity first, so it follows rather than leads.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Panretinal laser first**	Laser is the definitive treatment for proliferation and this eye needs it.	Scatter laser over centre-involving oedema drives thickness and acuity the wrong way.
A	**Vitrectomy**	Vitreous haze reads as bleeding, and surgery clears the media in one step.	The haze is faint and the view supports laser and injection, so surgery waits.

3 · A, B, F Systemic review, serial OCT, fellow eye

An HbA1c of 9.8% and an untreated fellow eye drive the eye disease as hard as any injection does. Retreatment is decided on measured thickness, so OCT is repeated at each visit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Stopping aspirin**	Antiplatelet therapy with blood in the vitreous feels like an obvious risk.	Aspirin does not worsen diabetic haemorrhage, and stopping it costs cardiac cover.
D	**Six-month dilated review**	Monthly injection visits give the impression the eye is being watched.	Injection visits do not examine the periphery, where proliferation progresses.

4 · A Inner blood-retinal barrier breakdown

Pericyte loss and damaged endothelial tight junctions let plasma into the retina, which collects in the outer plexiform and inner nuclear layers as the cystoid spaces seen here.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Outer barrier pump failure**	The pigment epithelium does clear subretinal fluid, so its failure floods the macula.	That mechanism gives serous elevation, not intraretinal cysts.
C	**Taut posterior hyaloid**	A real diabetic pattern, and traction genuinely thickens the fovea.	No hyaloid deformation of the fovea is described, only cystoid spaces.

5 · E Switch anti-VEGF agent, then reassess

Thickness fell under 10% across a full loading course and acuity has not moved, which defines a poor response rather than a slow one. Changing agent is the next step before changing class.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Same agent, three more**	Some eyes are slow responders, and persistence is often rewarded.	A 43 µm fall over three doses is non-response, not slow response.
C	**Vitrectomy with peel**	Surgery is the recognised answer to oedema that will not dry.	Peeling is for tractional oedema, and no traction is present here.

IF-THEN RULES FROM THIS CASE

If new vessels and centre-involving oedema sit together, dry the centre with anti-VEGF before laying panretinal laser. If thickness barely moves across loading, change the drug.

Case S1-12 · why

1 · B Central retinal vein occlusion

Haemorrhage and venous dilatation occupy all four quadrants of one eye, which places the obstruction at the central trunk. Unilateral disease with a normal fellow eye excludes the systemic patterns.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Hemiretinal occlusion**	It produces the same signs and is easy to accept without mapping the quadrants.	Hemiretinal disease respects the horizontal midline. All four quadrants are involved.
A	**Severe NPDR**	Dot haemorrhages and cotton wool spots are the diabetic vocabulary.	Onset overnight in one eye only, and the fellow eye carries no retinopathy.

2 · A, B, F RAPD, 20/200, cotton wool spots

Perfusion is judged on afferent function and capillary closure, not on how much blood or fluid is visible. An RAPD, acuity at 20/200 and cotton wool spots each point at closure.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Thickness 620 µm**	It is the most abnormal number on the page and it does explain the blur.	Thickness measures leakage. A very thick macula can sit on a perfused bed.
C, E	**Tortuous veins, full disc**	Both look dramatic and both belong to the occlusion.	Both appear in perfused occlusion as well, so neither separates the two.

3 · E Anti-VEGF now, monthly review

Anti-VEGF is first line for oedema after venous occlusion and it also suppresses the drive to anterior segment new vessels. Monthly review catches both non-response and rubeosis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Observation**	Spontaneous improvement is real, and the eye may recover without needles.	Acuity is 20/200 with 620 µm of thickening. Waiting loses recoverable vision.
D	**Macular grid laser**	Grid laser is the older standard and it sounds definitive for leakage.	Grid laser gave no acuity benefit in central occlusion and is not used for it.

4 · B Same-week blood pressure and glucose

Home readings of 158/96 name the modifiable driver, and undetected diabetes is the other. Treating them protects the fellow eye, which is the outcome that matters most here.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Carotid duplex**	It is the right investigation when the retinal problem is arterial.	Carotid disease causes arterial events, not venous thrombosis at the disc.
D	**Thrombophilia screen**	Thrombosis invites a clotting workup and it feels thorough.	At 68 with hypertension the yield is negligible. Reserve it for young patients.

5 · E Vein at the lamina cribrosa

Artery and vein cross the lamina cribrosa inside one adventitial sheath, so arteriolar stiffening compresses the vein there and the resulting turbulence seeds thrombus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Arteriovenous crossing**	It is the correct site for the branch occlusion the signs otherwise resemble.	A crossing lesion drains one sector. Four quadrants means the central trunk.
A	**Superior ophthalmic vein**	It is the next vessel downstream, so obstruction there sounds plausible.	Orbital venous obstruction gives proptosis and dilated lid veins, absent here.

> **IF-THEN RULES FROM THIS CASE**
>
> If haemorrhage and dilated veins cross all four quadrants, the lesion is at the lamina cribrosa. If acuity, pupil or cotton wool spots are bad, treat the eye as ischaemic and watch the angle.

Case S1-13 · why

1 · C Horseshoe tear under traction

A flap tear held open by a vitreous strand keeps pulling with every eye movement, so fluid can enter continuously. That combination carries the highest conversion rate of any break.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Fellow eye lattice**	Lattice is the best known predisposing lesion and it is genuinely present.	Lattice without breaks in a symptom-free eye is low risk and needs no treatment.
B, D	**Pigment and blood**	Both are the classic markers that a break exists somewhere.	They point to the tear. They do not measure the traction that opens it.

2 · B Laser retinopexy within 24 hours

A symptomatic flap tear with the retina still flat is sealed by surrounding retinopexy, which is quick, safe in clinic and highly effective. Delay is what turns the break into a detachment.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Review in two weeks**	Vision is 20/20 with no field loss, so the eye looks stable enough to watch.	Acuity stays normal until the fovea goes. Traction is active now.
A	**Vitrectomy with gas**	Surgery removes the traction itself rather than walling off the break.	The retina is attached. Intraocular surgery adds risk that laser does not carry.

3 · A, B Spreading shadow, or new floaters and flashes

Retinopexy seals one break but does not stop the vitreous separating further. A spreading shadow means fluid under the retina, and a fresh shower of floaters means a new break.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Brow ache**	Pain after any laser feels like the sign that something has gone wrong.	Brief ache follows the contact lens and the burns. It carries no retinal meaning.
D, E	**Glare, morning blur**	Both are new visual symptoms in an eye that has just been treated.	Neither maps to retinal separation. Both settle without any intervention.

4 · A Firm adhesion at the posterior vitreous base

Syneretic vitreous collapses forward and pulls hardest on its firmest attachment, the posterior margin of the vitreous base. Traction concentrated at that line tears full thickness retina.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Thin limiting membrane**	Something must be weakest where the retina breaks, so thickness sounds right.	Break site follows where vitreous is attached, not where retina is thin.
C	**Weak superior adhesion**	Superior tears really are commoner, so a superior weakness sounds plausible.	Superior tears reflect gravity acting on the collapsing gel, not local adhesion.

5 · D Same-day referral, located and documented

A symptomatic flap tear is treated within days, so the referral is made the same day. The record carries the location and the written symptoms that oblige the patient to return at once.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Posted letter, one month**	It refers and it documents, which covers the two obvious duties.	Postal timing does not match a break that can detach within days.
B	**Wait for the blood**	A clearer view would let the whole periphery be graded properly.	The break is already seen and the retina is flat. Waiting only loses the window.

> **IF-THEN RULES FROM THIS CASE**
> If floaters and flashes are new, indent and search the whole periphery. If a flap tear is found with the retina still flat, seal it within a day rather than review it.

Case S1-14 · why

1 · B Non-arteritic central retinal artery occlusion

Sudden painless loss with a pale inner retina, a cherry red spot and segmented arteriolar flow is inner retinal infarction. Untreated fibrillation supplies the embolic source.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Ophthalmic artery occlusion**	Hand movements vision is severe enough to suggest the whole supply has gone.	A cherry red spot needs an intact choroid under the fovea, so the choroid is perfused.
D	**Arteritic occlusion**	At 74 the arteritic form must be excluded, and it changes treatment completely.	No headache, scalp tenderness or jaw pain, and the embolic source is already known.

2 · E Immediate transfer to a stroke unit

Retinal artery occlusion is an acute ischaemic stroke of the eye, and at 90 minutes the patient is inside the window where brain imaging and prevention change outcome.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Massage, then refer**	These manoeuvres are the traditional bedside response and cost nothing to try.	Neither has proven benefit, and the overnight delay wastes the stroke window.
A	**Duplex today, cardiology later**	It investigates the source and looks appropriately urgent.	Investigation without a stroke pathway leaves the brain unprotected today.

3 · C, D, E Inflammatory markers, ECG, carotid imaging

The workup asks where the embolus came from and whether arteritis is hiding. Markers, rhythm and carotid imaging answer that; the ocular tests only restate what the fundus already shows.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	**Fluorescein angiography**	It documents the occlusion objectively and looks like the definitive test.	The diagnosis is already clinical. Angiography delays the stroke referral.
B	**Thrombophilia screen**	It sounds like due diligence before committing to anticoagulation.	At 74 with fibrillation the source is known, and the screen changes nothing.

4 · C Scalp tenderness with jaw claudication

Arteritis is diagnosed on systemic features and markers, not on the fundus, because the retinal picture is identical. Jaw claudication is the most specific symptom in the set.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Calcific plaque**	A visible plaque is dramatic and it proves the artery is diseased.	A plaque names an embolic source, which argues against arteritis.
D	**Segmented columns**	It is the sign of stagnant arterial flow, which fits an inflamed vessel.	Segmentation follows any severe occlusion, whatever closed the artery.

5 · A Thin fovea over a perfused choroid

Infarcted inner retina turns opaque and pale, but the fovea has almost no inner retina and is nourished by the choroid, so the intact red choroidal bed shows through it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Cilioretinal artery**	A cilioretinal artery genuinely spares central vision when one is present.	It exists in a minority of eyes and spares a wedge of retina, not the red spot.
E	**Ganglion cell tolerance**	Survival of the central retina would explain a normal looking centre.	Ganglion cells are absent at the foveal centre and die fastest elsewhere.

IF-THEN RULES FROM THIS CASE

If retinal arterial occlusion is diagnosed inside the stroke window, the destination is the stroke unit, not the eye clinic. If systemic arteritic symptoms are present, treat before biopsy.

Case S1-15 · why

1 · B Acute central serous chorioretinopathy

A shallow serous detachment with a thickened choroid, micropsia and a hyperopic shift in a man of 41 weeks after corticosteroid exposure is the classic presentation. No blood and no drusen.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Pachychoroid neovascular**	It shares the thick choroid and the same age group, and it also leaks fluid.	No subretinal hyper-reflective material and no haemorrhage on OCT or fundus.
E	**Cystoid macular oedema**	Central blur with distortion is the everyday description of macular oedema.	The fluid is under the retina, not in cystoid spaces within it.

2 · A, E No hyper-reflective material, no drusen

A membrane shows as hyper-reflective tissue above the pigment epithelium, usually with blood, and usually on a drusen-bearing macula. Their absence is what argues against it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Thick choroid**	A thick choroid is the signature of serous disease, so it feels exclusive to it.	Pachychoroid neovascular membranes grow on exactly this thick choroid.
B, C	**Micropsia, hyperopic shift**	Both are textbook symptoms of the serous condition being considered.	Both follow from any fluid lifting the fovea, membrane or not.

3 · B Observe to three months, remove steroid

Most acute serous detachments resolve without treatment inside three months, and the recent corticosteroid course is a removable driver. Treatment is held for persistence or recurrence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Photodynamic therapy now**	It is the most effective treatment, so offering it early looks like good care.	At six weeks the natural course is still resolution. Early treatment adds risk.
D	**Anti-VEGF**	Subretinal fluid at the macula is the situation anti-VEGF usually treats.	Without new vessels there is no target, and trials show no benefit.

4 · D Half-dose photodynamic therapy

Fluid past three to four months threatens the photoreceptors, and reduced-dose photodynamic therapy has the best evidence for resolving it while sparing the pigment epithelium.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Observe three months more**	The condition is known for resolving on its own, so patience feels safe.	Past four months the outer retina thins irreversibly. The window is closing.
C	**Full-fluence therapy**	It uses the right modality and full dose sounds more likely to work.	Full fluence carries atrophy and ischaemia risk that the half dose avoids.

5 · D Hyperpermeable choroid, overwhelmed RPE

Choroidal congestion and hyperpermeability drive fluid at the pigment epithelium faster than its pump can clear it, so fluid collects under the retina. The measured choroidal thickness is the clue.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Inner barrier breakdown**	It is the mechanism behind most macular fluid a candidate sees.	Inner barrier failure gives intraretinal cysts, not a smooth serous dome.
A	**Bruch membrane defects**	The pigment epithelium is involved, so its basement complex sounds central.	That is the route for neovascular disease, which this eye does not show.

> **IF-THEN RULES FROM THIS CASE**
>
> If a young adult has serous macular fluid, look for steroid exposure and a thick choroid, and observe to three months. If fluid persists past four months, treat rather than watch.

Case S1-16 · why

1 · E Report the dose above 5 mg/kg

At 61 kg, 400 mg daily is 6.6 mg/kg of actual body weight, above the 5 mg/kg threshold that raises retinopathy risk sharply. Dosing belongs to the prescriber, so the finding is reported with its reason.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Continue as prescribed**	Every test today is normal, so there seems to be nothing to act on.	Risk is set by cumulative dose. Normal screening now does not license the dose.
C	**Use ideal body weight**	Ideal weight dosing was the old rule and it is still widely quoted.	Actual body weight replaced it, because it predicts toxicity better.

2 · D, E SD-OCT with 10-2 automated fields

Screening pairs one structural and one functional test, because either can show the parafoveal change first. Autofluorescence and multifocal electroretinography are adjuncts, not the core.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Autofluorescence**	It shows the parafoveal ring beautifully once damage is established.	It is an adjunct, and it changes late compared with OCT and fields.
B, C	**Colour plates, Amsler**	Both are cheap, quick, and test central function of a sort.	Neither is sensitive to parafoveal loss, which spares the very centre early.

3 · E Parafoveal thinning, fovea spared

Toxicity starts two to six degrees from fixation, thinning the outer nuclear layer and ellipsoid zone in a ring while the centre stays normal. The fundus still looks unremarkable at that stage.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Subfoveal ellipsoid loss**	The bull's eye name suggests the centre is the target of the damage.	The centre is spared until late. Central loss points to another disease.
C	**Drusenoid elevation**	It sits parafoveally, which matches where toxicity is looked for.	Drusenoid change is age-related deposition, not drug-related thinning.

4 · C Report this week for withdrawal

Structure and function agree, which is definite toxicity, and damage continues after the drug stops. Withdrawal is the prescriber's decision and it is needed within days, not months.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Patient stops it today**	It is fast and it puts the patient in control of her own risk.	Abrupt withdrawal without the prescriber risks a lupus flare.
D	**Repeat in six months**	Confirming a screening finding before acting is usually good practice.	Two agreeing tests already confirm it, and six months adds permanent loss.

5 · D Baseline on file, results to the prescriber

Screening detects change, which needs a documented starting point, and only the prescriber can act on the result. No baseline exists here, so establishing one is part of this visit.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Kept, released on request**	Records are confidential, so waiting to be asked sounds correct.	Shared care obliges active communication to the prescriber, not storage.
B	**Note of patient risk**	It documents that risk was discussed, which feels protective.	It transfers a clinical duty to the patient. The duty is to inform the prescriber.

IF-THEN RULES FROM THIS CASE

If the daily dose exceeds 5 mg/kg of actual weight, report it whatever the retina looks like. If structure and function agree on parafoveal loss, that is toxicity and the drug must stop.

Case S1-17 · why

1 · B Primary open angle glaucoma, worse OS

Open angles, pressures in the mid twenties, inferior rim thinning and a matching superior nasal defect on repeatable fields complete the diagnosis. Gonioscopy excludes the secondary causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Ocular hypertension**	Acuity is normal and she has no symptoms, so nothing seems lost yet.	A repeatable nasal step with matching rim thinning is established damage.
D	**Normal tension glaucoma**	Asymmetric discs with a family history is the usual normal tension story.	Pressures of 26 and 28 on normal corneas are not normal tension.

2 · E Prostaglandin analogue at night

A pulse of 54 with asthma rules out a beta-blocker whatever the drop technique. A prostaglandin gives the largest single-agent reduction on once nightly dosing, which also protects adherence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Timolol with occlusion**	Punctal occlusion is taught precisely to limit systemic absorption.	It reduces but does not remove absorption. Asthma and bradycardia are absolute.
A	**Brinzolamide**	It avoids the respiratory and cardiac risk entirely, which looks safest.	It lowers pressure less and dosing three times daily costs adherence.

3 · A, B, F Repeat fields, gonioscopy, corneal thickness

Treatment is measured against a baseline, so the field must be repeatable, the angle mechanism must be known, and pressure must be interpreted against corneal thickness.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Imaging over gonioscopy**	Angle imaging is objective, repeatable and comfortable for the patient.	It cannot grade pigment, synechiae or recession, and it does not replace the lens.
D, E	**Cell count, autofluorescence**	Both are quantitative tests that fill out a thorough looking record.	Neither informs glaucoma diagnosis, target setting or progression.

4 · E Add a second agent, review at six weeks

A 25% reduction from 26 and 28 sets targets near 19 and 21, and the left eye with the field defect has not reached its. Adding a second agent is the next step, checked at six weeks.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Continue, review later**	The pressure has fallen and a single drop is easier to sustain.	The worse eye misses its target, and six months lets damage accumulate.
C	**Repeat fields first**	Progression is judged on fields, so more data before acting seems careful.	Fields cannot change at twelve weeks. Pressure is the actionable number.

5 · E Rim thinning matched by a field defect

Ocular hypertension is raised pressure without damage. Structure and function agreeing in the same hemifield, the inferior rim and the superior field, is the damage that names the disease.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Cup-disc 0.8**	A large cup is the image everyone associates with the diagnosis.	Large discs carry large cups. Ratio alone cannot separate the two.
C	**Family history**	It raises risk severalfold and rightly changes how closely she is watched.	Risk factors decide surveillance, not diagnosis.

> **IF-THEN RULES FROM THIS CASE**
> If the pulse is under 60 or the chest is asthmatic, no beta-blocker at any concentration. If structural loss and field loss agree in the same hemifield, treat it as glaucoma.

Case S1-18 · why

1 · C Acute primary angle closure

A hyperopic eye with a very shallow chamber, a pressure of 54, corneal oedema and a fixed mid-dilated pupil is acute closure. The quiet chamber and clear lens exclude the secondary causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Acute anterior uveitis**	Pain, redness and blurred vision are the uveitis triad, and pressure can rise.	No cells and no keratic precipitates, and uveitis does not shallow the chamber.
B	**Plateau iris syndrome**	The cold remedy gives a dilating trigger, and plateau eyes close on dilation.	Plateau eyes have a chamber of normal central depth and a flat iris plane.

2 · E Medical pressure reduction, then reassess

At 54 the iris sphincter is ischaemic and pilocarpine works poorly, and laser is unreliable through a hazy cornea. Lowering pressure medically clears the cornea and makes iridotomy possible.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Intensive pilocarpine**	Constricting the pupil is the mechanism that should reopen the angle.	A sphincter ischaemic at 54 will not respond, and repeated doses add anterior rotation.
A	**Immediate iridotomy**	Iridotomy is the definitive treatment, so doing it now sounds decisive.	The oedematous cornea will not transmit the beam reliably or safely.

3 · A, B Iridotomy to both eyes

Iridotomy removes the pressure gradient that bows the iris forward. The fellow eye is narrow with the same anatomy and roughly half of such eyes close within five years, so it is treated too.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Long-term prostaglandin**	Drops lower pressure and feel like the standard glaucoma answer.	Drops do not address the mechanism, and the angle can still close on them.
D	**Laser trabeculoplasty**	It is laser to the angle, which sounds like the right target.	It needs a visible open angle to treat, and it does not relieve pupil block.

4 · B Topical agent with angle monitoring

A patent iridotomy has relieved the block, so the residual pressure comes from synechial and trabecular damage. That is treated medically first, with gonioscopy to track any further closure.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Second iridotomy**	If pressure remains high the first opening may look inadequate.	A patent iridotomy needs no second site. The problem is now the angle itself.
A	**Observe three months**	The crisis has passed and 24 is far from the pressure she presented with.	Synechiae progress silently, and this eye has already lost angle.

5 · E Convex iris, shallow central chamber

Pupil block bows the whole iris forward and shallows the chamber centrally as well as peripherally. Plateau eyes keep a normal central depth with a flat iris and a forward ciliary body.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Fixed mid-dilated pupil**	It is the signature sign of the attack and it is genuinely present.	It reflects sphincter ischaemia at high pressure, whatever closed the angle.
D	**Cold remedy trigger**	A dilating drug precipitates both mechanisms, so it feels diagnostic.	Plateau eyes close on dilation too. The trigger does not name the mechanism.

IF-THEN RULES FROM THIS CASE

If the cornea is oedematous at a very high pressure, lower the pressure medically before attempting laser. If one eye has closed, the fellow narrow eye gets a prophylactic iridotomy.

Case S1-19 · why

1 · D Neovascular glaucoma

Ischaemic retina four months ago, new vessels at the pupil margin and fibrovascular tissue bridging the angle with broad synechiae is neovascular glaucoma at the synechial stage.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Acute angle closure**	Ache, haloes, corneal oedema and a pressure of 44 read like an acute attack.	The central chamber is deep, and the closure is by membrane over three weeks.
B	**Ghost cell glaucoma**	There has been intraocular blood, and degenerate cells do block outflow.	Ghost cells need vitreous haemorrhage and give no iris or angle vessels.

2 · A, B Angle membrane with broad synechiae

Appositional closure is reversible on indentation and leaves the angle anatomy intact. Tissue bridging the angle with permanent adhesions is mechanical closure that no drop can reopen.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Vessels at pupil margin**	Rubeosis is the hallmark of the disease, so it feels like the answer.	It proves neovascularisation, not that the angle is already sealed.
D, E	**Oedema, three weeks of ache**	Both show how high and how sustained the pressure has been.	Neither examines the angle, and both occur with appositional closure.

3 · D Anti-VEGF and drops, laser to follow

Anti-VEGF regresses the angle and iris vessels within days while drops carry the pressure, and panretinal laser then removes the ischaemic drive permanently. The order matters, not the choice.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Panretinal laser alone**	It treats the underlying ischaemia, which is the root of the problem.	It takes weeks to work and the cornea is too hazy to laser adequately now.
A	**Trabeculectomy**	Pressure is 44 with a sealed angle, which sounds like a drainage problem.	Filtration fails in an actively neovascular eye. Regress the vessels first.

4 · D Cyclodiode laser for pressure and pain

With minimal vision, a painful eye and a neovascular angle, reducing aqueous production is the lowest risk route to comfort. Filtering procedures scar closed and carry a real risk of loss.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Drainage device**	Tubes are the recognised choice in neovascular glaucoma with vision.	Reserved for eyes with useful vision. This eye needs comfort, not a tube.
B	**Enucleation**	It is definitive for a blind painful eye and ends the problem.	Pain has not yet been treated. Enucleation is a last resort, not a next step.

5 · E VEGF from the ischaemic retina

Hypoxic retina secretes vascular endothelial growth factor, which diffuses forward into the aqueous and drives new vessels on the iris and in the angle. That pathway is why anti-VEGF works within days.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Uveal prostaglandin**	The eye is red and aching, so an inflammatory mediator fits the picture.	Prostaglandins cause inflammation and lower pressure. They grow no vessels.
D	**Ferritin from old blood**	There was retinal haemorrhage, so iron products are genuinely present.	Iron toxicity is a late siderotic process and grows no fibrovascular tissue.

> **IF-THEN RULES FROM THIS CASE**
> If an ischaemic eye develops a high pressure with iris vessels, treat the ischaemia and the pressure together. If the eye is painful with no useful vision, cyclodestruction before filtration.

Case S1-20 · why

1 · C Demyelinating retrobulbar optic neuritis

A young woman with subacute unilateral loss, pain on movement, dyschromatopsia, an RAPD and a normal disc has retrobulbar inflammation. The earlier leg symptom points to demyelination.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Anterior ischaemic**	It is the commonest optic neuropathy and shares the RAPD and colour loss.	It needs a swollen disc and it does not hurt on eye movement.
E	**Apical compression**	Compression also spares the disc early and causes progressive loss.	Compression progresses over months, without pain on movement.

2 · A Pain on movement with a normal disc

An RAPD, reduced acuity and dyschromatopsia occur in both, so neither separates them. Pain on movement with a flat disc in a woman of 29 is specific to inflammation of the nerve behind the globe.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Afferent pupil defect**	It is the sign that proves the nerve is at fault, so it feels decisive.	Any unilateral optic neuropathy gives it. It localises without discriminating.
D	**Ishihara 4 of 14**	Colour loss is taught as the marker of optic neuritis.	Ischaemic neuropathy desaturates colour in proportion to acuity too.

3 · D Contrast MRI, then neurology

Imaging counts the lesions that set the risk of multiple sclerosis and excludes compression, and that result drives disease modifying treatment. Steroids do not change the final acuity.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Steroids first**	Treating the inflammation immediately feels like the active choice.	Steroids only shorten the attack, and imaging must come before immunotherapy.
A	**Oral prednisolone**	It avoids admission and appears to give the same anti-inflammatory effect.	Standard oral dosing alone was associated with more recurrent attacks.

4 · C, D Recovery expected, steroids only speed it

The natural history is good, with most eyes back near normal acuity within three months, and steroid benefit is confined to the speed of that recovery. Residual colour and contrast loss is common.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Colour fully recovers**	Acuity recovers so well that colour is assumed to follow it.	Colour and contrast deficits persist in many eyes with 20/20 acuity.
B	**Poor acuity, poor outcome**	Presenting acuity predicts outcome in most other optic neuropathies.	Even eyes at counting fingers usually recover substantially here.

5 · E Traction on the inflamed nerve sheath

The superior and medial rectus origins share the dural sheath of the nerve at the apex, so every movement pulls on an inflamed, swollen sheath. This is why the pain is worse on adduction and elevation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Ciliary artery ischaemia**	It is the mechanism of the disease this case must be separated from.	Ischaemic neuropathy is painless, and the disc here is flat and normal.
C	**Raised orbital pressure**	Orbital congestion genuinely causes ache worsened by movement.	There is no proptosis, no chemosis and no restriction of movement.

> **IF-THEN RULES FROM THIS CASE**
>
> If a young adult loses vision in one eye with pain on movement and a normal disc, image the brain and orbits before reaching for steroids. If the disc is swollen instead, rethink the diagnosis.

Case S1-21 · why

1 · B Arteritic AION from giant cell arteritis

Chalky pallid oedema, vision at counting fingers and an RAPD point to arteritic infarction rather than a crowded-disc event. Jaw ache, scalp tenderness, ESR 78 and CRP 42 name the vasculitis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Non-arteritic AION**	Commoner, and it also gives sudden painless loss with disc oedema.	Oedema here is chalky and pallid, with systemic symptoms and an ESR of 78.
C	**Retinal artery occlusion**	Sudden profound loss in an elderly arteritic patient fits an arterial event.	No retinal opacification and no cherry-red spot; the lesion is a pallid swollen disc.

2 · C, D, E Jaw claudication, pallid oedema, raised markers

Both entities give sudden painless loss with an RAPD and disc oedema. Jaw claudication, chalky pallor of the swelling and raised ESR and CRP are the arteritic markers.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	**Small cup in fellow eye**	A crowded disc is a real risk marker, so it reads as supporting evidence.	A small cup argues for the non-arteritic form, not against it.
A, B	**RAPD, painless onset**	Both are present and both are abnormal, so they read as diagnostic.	Both occur in either form, so neither discriminates.

3 · B Steroid now, biopsy within two weeks

Sight in the fellow eye is what is at risk and it can go within days. Steroid starts on clinical suspicion with an ESR of 78, and biopsy stays positive for one to two weeks after treatment begins.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Wait for biopsy**	Confirming before committing an elderly patient to steroid feels careful.	The fellow eye can infarct while waiting, and biopsy yield survives early steroid.
D	**Prednisolone 20 mg**	It treats, and a lower dose limits harm in a 74-year-old.	Established visual loss needs high dose; 20 mg does not protect the other eye.

4 · B Short posterior ciliary artery occlusion

The laminar and prelaminar nerve is fed by the short posterior ciliary arteries. Granulomatous inflammation of these medium-sized vessels infarcts that watershed and gives pallid swelling.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Central retinal vein**	Vein occlusion also swells the disc and drops vision suddenly.	No dilated tortuous veins and no scattered retinal haemorrhages.
A	**Carotid stenosis**	Ocular ischaemic syndrome is real and she has vascular risk.	It gives retinal signs and gradual change, not chalky disc pallor.

5 · D Fellow eye symptoms with rising CRP

New obscurations in the uninvolved eye with a rising CRP mean active vasculitis and a second eye at risk. Steroid hyperglycaemia is managed alongside, not by cutting the dose.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Glucose 9.2**	A real steroid effect that needs treatment and looks like a reason to taper.	It calls for glycaemic control, not less steroid while sight is at risk.
B	**ESR 40, jaw ache**	A falling ESR reads as control, so the residual ache seems minor.	Symptoms hold the taper where it is, but no visual threat means no escalation.

IF-THEN RULES FROM THIS CASE

If disc oedema is pallid, or comes with jaw claudication or scalp tenderness, start steroid before the biopsy. If the fellow eye reports obscurations, escalate the same day.

Case S1-22 · why

1 · C Papilloedema, raised intracranial pressure

Positional headache, pulsatile tinnitus, obscurations and bilateral oedema with no venous pulsation are pressure signs. Preserved acuity with a sixth nerve deficit fits pressure, not neuritis.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Bilateral optic neuritis**	A young woman with swollen discs and headache is the neuritis stereotype.	Acuity is 20/20 with no RAPD and no pain on eye movement.
A	**Malignant hypertension**	It swells both discs and is the emergency nobody wants to miss.	BP is 128/78, with no arteriolar narrowing and no exudates.

2 · B, C, D MRI with venography, LP after imaging, stop the drug

Pressure has to be measured and secondary causes excluded before it is called idiopathic. Venous sinus thrombosis is the dangerous mimic, and tetracyclines are a recognised trigger.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Steroid taper**	Steroid does lower pressure briefly and the discs look inflamed.	It masks the picture, rebounds on withdrawal, and the weight gain worsens the disease.
A	**Prism for diplopia**	The diplopia is real and prism is the optometric reflex.	The sixth nerve deficit resolves with the pressure, so prism chases a moving target.

3 · A Sixth nerve stretch from raised pressure

A raised pressure gradient displaces the brainstem downward and tethers the sixth nerve where it crosses the petrous ridge. That makes it a false localising sign rather than a second disease.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Sixth nerve nucleus**	A nuclear lesion is the neat anatomical answer for an abduction deficit.	Nuclear lesions give a gaze palsy, and there is no gaze or facial sign here.
C	**Medial rectus restriction**	Thyroid myopathy limits abduction and is common in young women.	Nothing suggests restriction, and swollen discs with obscurations point intracranially.

4 · B Falling acuity with progressive field loss

Function decides. Loss of acuity and field over days is fulminant disease and needs shunting or optic nerve sheath fenestration, whereas headache and tinnitus alone do not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Headache at two weeks**	The symptom that brought her in has not improved, so treatment looks failed.	Headache lags and has other causes, so it is not the surgical trigger.
D	**Discs unchanged**	Persistent oedema on examination reads as uncontrolled pressure.	Oedema resolves slowly. Stable fields and acuity mean the nerve is coping.

5 · C Autofluorescent lumpy margin, vessels clear

Drusen elevate the disc without swelling the nerve fibre layer, so crossing vessels stay visible and the calcified bodies autofluoresce. Absent venous pulsation and a small cup occur in both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**No venous pulsation**	Its absence is taught as the sign of raised pressure.	It is absent in a fifth of normal eyes, so it cannot separate the two.
A	**Peripapillary bleed**	Haemorrhage looks like the active and dangerous option.	Drusen bleed too, from the same crowded disc margin.

IF-THEN RULES FROM THIS CASE

If both discs are swollen with normal acuity, measure the pressure and image the venous sinuses before calling it idiopathic. If acuity or field falls over days, refer for decompression.

Case S1-23 · why

1 · A Aneurysmal pupil-involving third nerve palsy

A complete external palsy with a 6 mm sluggish pupil and abrupt severe pain is compression until imaging proves otherwise. A posterior communicating artery aneurysm is what must be excluded today.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Microvascular palsy**	He is 58 and hypertensive, which is the vasculopathic profile exactly.	The pupil is 6 mm and sluggish, and the glucose is 6.1.
C	**Ocular myasthenia**	Ptosis with diplopia and no visual loss is the myasthenic pattern.	The pupil is involved and the deficit does not fatigue on upgaze.

2 · C Same-day CT angiography, neurosurgery

A pupil-involving palsy with pain is a warning leak until proven otherwise, and the rupture risk is measured in hours. CT angiography is fast, available and sensitive for aneurysms of this size.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**MRI within a week**	MRI is the better parenchymal study and it does get imaging done.	A week is longer than the interval in which these aneurysms rupture.
E	**Observe four weeks**	Microvascular palsies do recover in four to twelve weeks.	Observation belongs to pupil-sparing palsies, and only after imaging.

3 · A, B, F Pupil involvement, down and out, no fatigue

Myasthenia never touches the pupil and it fatigues, while myopathy spares the pupil and ignores nerve territory. A fixed down and out eye with a sluggish pupil is a nerve trunk lesion.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Ptosis 5 mm**	Ptosis is the headline sign and it is severe here.	Levator weakness occurs in nerve, junction and muscle disease alike.
E	**Pain behind the eye**	Pain suggests compression, so it feels like a localiser.	Pain tells you the cause is urgent, not where the lesion sits.

4 · C Superficial pupillomotor fibres

The parasympathetic fibres travel on the dorsomedial surface of the nerve, fed by pial vessels. Extrinsic compression reaches them first, while microvascular disease infarcts the core.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Vasa nervorum ischaemia**	It is the real mechanism of the vasculopathic palsy that spares the pupil.	Ischaemia hits the central fibres, which is why the pupil is spared.
D	**Ciliary ganglion inside**	The ganglion does sit close to the inferior division route.	It lies in the orbit behind the globe, not within the nerve trunk.

5 · D Risk review, prism or occlusion, six-week check

With imaging clear, a vasculopathic palsy is the working diagnosis and most recover by twelve weeks. Relief of diplopia is temporary while the angle changes, and the blood pressure is the treatable part.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Surgery at six weeks**	The deviation is disabling and surgery is definitive.	Wait at least six months for a stable measurement; most of this recovers.
B	**High-dose steroid**	Steroid is the reflex when a palsy fails to recover quickly.	No inflammatory markers, no orbital signs, and it worsens his risk profile.

IF-THEN RULES FROM THIS CASE

If ptosis and an ocular motor deficit come with a dilated sluggish pupil, image the vessels the same day. If the pupil is spared in a vasculopath, imaging still comes before observation.

Case S1-24 · why

1 · E Right Horner syndrome, preganglionic

Ptosis with a raised lower lid, a smaller pupil in the dark and dilation lag are sympathetic denervation. Neck pain, hoarseness and cough put the lesion on the preganglionic run through the chest.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Physiological anisocoria**	A fifth of people have it, and lid laxity at 52 is ordinary.	Physiological anisocoria is equal in light and dark and has no dilation lag.
C	**Partial third palsy**	Ptosis plus anisocoria is the third nerve pattern.	The wrong pupil is small here, and motility is full.

2 · A, B, F Dark anisocoria, dilation lag, raised lower lid

Sympathetic denervation weakens the dilator and Muller's muscle, so the deficit shows in the dark, dilation lags and the lower lid rises. Upper lid ptosis on its own has many causes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Upper lid 2 mm lower**	It is the sign that brought him in and it is on the affected side.	Aponeurotic and mechanical ptosis do the same thing without pupil signs.
D, E	**20/20, full motility**	Normal findings feel reassuring and therefore relevant.	They exclude other disease but they localise nothing.

3 · A Apraclonidine 0.5%, reversal confirms

Denervation supersensitivity of the dilator lets a weak alpha agonist reverse the anisocoria: the small pupil dilates and the ptosis lifts. Hydroxyamphetamine localises, but only after confirmation.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Hydroxyamphetamine**	It is the classic Horner drop and it does separate the levels.	It answers where, not whether, and it cannot be used on the same day.
B	**Pilocarpine 0.125%**	Dilute pilocarpine is a supersensitivity test, so the logic feels right.	That test is for a tonic pupil, where the sphincter is denervated.

4 · C Neck and chest apex imaging with carotid study

The sympathetic chain runs from hypothalamus to orbit, and a hoarse voice with neck pain in a smoker points at the apex and the carotid. Imaging has to cover the whole preganglionic route.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Brain MRI alone**	Central lesions do cause Horner syndrome and MRI is the neuro study.	The clues sit in the neck and chest, and a brain study misses both.
B	**No imaging**	Isolated Horner can be benign, and the signs are unchanged for weeks.	Hoarseness, neck pain and a smoking history make observation unsafe.

5 · C Abrupt onset with neck pain and tinnitus

Dissection presents with sudden neck or facial pain, Horner syndrome within hours and pulsatile tinnitus, and it can embolise within days. Hoarseness and weight loss point instead to the apex.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Weight loss, haemoptysis**	It is the most alarming option and it does signal serious disease.	It points at an apical tumour, which is urgent but is not a dissection.
A	**Longer dilation lag**	A bigger number looks like a worse lesion.	Lag grades the denervation, not the cause or the level.

IF-THEN RULES FROM THIS CASE

If anisocoria is greater in the dark and the pupil dilates late, call it sympathetic and confirm with apraclonidine. If the voice or the neck is involved, image the chest apex.

Case S1-25 · why

1 · E Ocular myasthenia gravis

Variable ptosis that worsens on sustained upgaze, a lid twitch on refixation, improvement after ice and a hypertropia that changes size are junctional fatigue. The pupils are always spared.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Aponeurotic ptosis**	Ptosis worse by evening is what patients with lid laxity describe.	Aponeurotic ptosis does not fatigue on upgaze or lift after an ice pack.
B	**External ophthalmoplegia**	It gives painless ptosis with motility loss and no pupil signs.	It is slowly progressive and symmetrical, and it does not vary hourly.

2 · A, E, F AChR antibodies, chest CT, nerve stimulation

Antibody testing confirms most cases, electrophysiology catches the seronegative ones, and a thymoma changes management in about one in ten. Orbital and vascular imaging answer a different question.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Orbital MRI**	Ptosis with a variable vertical deviation looks orbital.	Muscles and orbit are normal here; the fault is at the neuromuscular junction.
D	**CT angiography**	Ptosis with diplopia raises the aneurysm reflex.	The pupils are equal and brisk and the signs fatigue with effort.

3 · C Safety-netting for bulbar and respiratory signs

About fifteen percent of ocular cases generalise within two years, and a crisis is a respiratory emergency. She needs to know which symptoms mean the emergency department rather than the clinic.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Press-on prism now**	The diplopia is the complaint and prism is the optical fix.	A deviation that changes within the day cannot be prismed reliably.
B	**Start pyridostigmine**	It is the right drug class for the disease.	Prescribing sits with the neurologist, and it is not today's priority.

4 · C Postsynaptic receptor blockade

Receptor loss lowers the endplate potential, so repeated release fails to reach threshold as vesicle stores run down. Rest and cooling restore transmission, which is why ice lifts the lid.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Calcium channel antibody**	It is a real antibody-mediated junction disease with weakness.	That is Lambert-Eaton: strength improves with use and reflexes are lost.
E	**Mitochondrial deletion**	It is the mechanism of the disease this most resembles on inspection.	Mitochondrial ptosis is fixed and progressive, never fatiguable.

5 · D A fixed, poorly reactive pupil

The iris sphincter is not affected in myasthenia, so pupil involvement forces a different diagnosis. Orbicularis weakness and day-to-day variability are typical rather than exclusionary.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Orbicularis weakness**	Extra weakness sounds like a bigger and different disease.	Orbicularis involvement is common in ocular myasthenia and supports it.
C	**Normal reflexes**	Normal reflexes seem to rule out a neuromuscular disorder.	Reflexes are normal in myasthenia and lost in Lambert-Eaton.

> **IF-THEN RULES FROM THIS CASE**
> If ptosis varies within the hour, fatigues on upgaze or lifts after ice, test for myasthenia and image the chest. If the pupil is involved, it is not myasthenia.

Case S1-26 · why

1 · B Convergence insufficiency

An exophoria that appears only at near, a near point of 14 cm and positive vergence breaking at 12 are the findings that define it. Accommodation measures 11 D, which is normal at 19.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Accommodative insufficiency**	Blur at near after twenty minutes is the classic accommodative complaint.	Amplitude is 11 D, which is age-normal, and the exophoria is at near only.
D	**Basic exophoria**	There is an exophoria, and the near symptoms fit an exodeviation.	The distance cover test is orthophoric, so the deviation is not equal.

2 · D, E, F Remote near point, near exophoria, low vergence

Both conditions blur print, so the separation sits in the vergence data: a remote near point, an exodeviation confined to near, and a reduced positive fusional range. Amplitude is age-normal.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Amplitude 11 D**	It is the accommodative measurement, so it looks decisive here.	By Hofstetter the minimum at 19 is about 10 D, so 11 D is normal.
B, C	**Refraction, headache**	Both are in the record and both feel relevant to near work.	Neither separates the two, because both occur with either mechanism.

3 · A Office-based vergence therapy

Supervised in-office therapy with home reinforcement outperformed pencil push-ups and placebo in the CITT trial. Prism and plus adds treat the symptom without restoring fusional range.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Push-ups alone**	They are free, familiar and easy to instruct in a single visit.	In the trial push-ups alone performed no better than placebo therapy.
B	**Base-in prism**	It relieves symptoms immediately and is easy to dispense.	It leaves the vergence deficit untreated, so the demand returns.

4 · C Reassess, then prism if measures unchanged

Therapy is judged on the measures, not the calendar. If the near point and the fusional range have not moved, prism supports the deficit, whereas repeating a failed programme costs months.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Repeat unchanged**	Persistence is often the answer with therapy that works slowly.	Twelve weeks done well with no change in the measures is a failed course.
B	**Discharge**	The condition is benign and no pathology has been found.	The symptoms are disabling for a student and they persist untreated.

5 · A Written patient authorisation first

She is an adult, so the record is hers to release. Authorisation is written, specific to the recipient, and filed with the record; the request itself creates no permission.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**College request alone**	Educational requests feel administrative rather than clinical.	A third party gains no right of access by asking on its own letterhead.
D	**Parental authorisation**	Student status suggests a parent is still the responsible party.	At 19 she consents for herself, whoever pays the fees.

IF-THEN RULES FROM THIS CASE

If exophoria appears only at near with a remote near point, treat the vergence system before reaching for prism. If twelve weeks of therapy moves no measure, then add prism.

Case S1-27 · why

1 · C Refractive accommodative esotropia

Uncorrected hypermetropia of +5.00 D with 20 Δ at distance collapsing to 4 Δ esophoria under the full plus makes the deviation accommodative. The residual 8 Δ at near marks the excess.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Non-accommodative excess**	There is a genuine near excess of 8 Δ after correction.	That entity carries little hypermetropia and does not respond to plus.
E	**Infantile esotropia**	A large esotropia in a small child fits the label loosely.	Onset near three and a half years, and the response to plus, exclude it.

2 · A Full cycloplegic plus, constant wear

The deviation is driven by accommodative effort, so removing the effort with the full plus is the treatment. Six weeks lets alignment and acuity respond before anything is added.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Half the plus**	Undercorrecting a hypermetrope is a familiar habit from adult practice.	Any residual accommodative demand keeps driving the esotropia.
C	**Bifocal from the start**	There is a near excess, and a bifocal treats it directly.	Measure the residual near angle on full plus first, because it often settles.

3 · B, C, D High hypermetropia, plus response, near excess

The response to plus proves the mechanism, supported by significant hypermetropia and a deviation larger at near. Acuity and stereopsis measure the cost of the deviation, not its cause.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Acuity 20/30**	Reduced acuity in the deviating eye looks like the key abnormality.	It measures amblyopia risk, not what drives the deviation.
A	**Onset six months ago**	Age of onset is how esotropias are usually classified.	Onset narrows the list but proves nothing about accommodation.

4 · C Accommodative convergence through AC/A

Each dioptre of accommodation carries a fixed quantity of convergence. To clear +5.00 D of hypermetropia the child accommodates hard, and the linked convergence overwhelms fusional divergence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Low negative vergence**	Negative fusional vergence is what would hold an esodeviation latent.	That is the reserve that fails, not the drive that creates the deviation.
B	**Sensory deprivation**	There is 0.25 D of anisometropia and unequal acuity.	Deprivation esotropia needs poor vision in one eye, not 20/30.

5 · B +2.50 bifocal, segment splitting the pupil

A residual near esotropia on full distance correction is a high AC/A, treated by removing the near accommodative demand. The segment has to sit high enough to be used, at the pupil margin.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Extra 1.00 D of plus**	More plus relaxes more accommodation, so it looks like the same logic.	It blurs distance vision without touching the near excess.
A	**Surgery now**	Twelve prism dioptres at near is a real deviation.	Optical treatment is incomplete, and a bifocal usually aligns the near angle.

IF-THEN RULES FROM THIS CASE

If an esotropia shrinks under the full cycloplegic plus, accommodative drive is the disease. If the near angle persists on full plus, add a bifocal rather than more distance plus.

Case S1-28 · why

1 · E Traumatic right fourth nerve palsy

The three-step pattern is a right hypertropia worse in left gaze and on right tilt, with limited depression in adduction. Symptomatic extorsion and a 4 Δ fusion range date it to the injury.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Congenital fourth palsy**	Trauma often decompensates a long-standing palsy, and the pattern matches.	A congenital palsy carries a large vertical fusion range and no torsional symptoms.
B	**Skew deviation**	Head trauma with a vertical deviation does raise a central cause.	Skew gives intorsion of the higher eye and no dependence on head tilt.

2 · B, C, D Torsion, small fusion range, dated onset

The three-step results and the head tilt are shared by both forms, so they cannot separate them. Torsional symptoms, a small vertical fusion range and a dated onset belong to the acquired palsy.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A, E	**Three-step results**	They are how the palsy is identified, so they look like the answer.	Both forms give the same pattern, which names the muscle and nothing else.
F	**Habitual head tilt**	A tilt suggests a posture adopted over many years.	New palsies adopt a tilt within days, so a tilt dates nothing.

3 · D Temporary prism or occlusion, review at 3 months

Traumatic fourth nerve palsies improve for up to six months, so the measurement to operate on does not exist yet. Temporary relief keeps him functional while the angle changes.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Ground-in prism now**	It gives comfortable single vision and looks like definitive care.	The deviation is still changing, so the prescription will soon be wrong.
A	**Early surgery**	Three weeks of disabling diplopia feels like enough waiting.	Spontaneous recovery is common, so operate only on a stable measurement.

4 · D Intorters compete, superior rectus unopposed

Tilting to the right calls on the right intorters, superior oblique and superior rectus. With the oblique palsied the superior rectus elevates unopposed, so the hypertropia grows.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Extorters on right tilt**	Naming the other torsional pair sounds equally plausible.	Extorters act on tilt to the other side, which is why the deviation falls there.
A	**Vertical fusion range**	Fusion range is measured in this workup and it matters here.	Range is measured with prism, not by tilting the head.

5 · D Incyclotorsion of the higher eye

Skew is a central prenuclear imbalance and produces an ocular tilt reaction: the higher eye intorts and the head tilts towards the lower eye. A palsied superior oblique lets the eye extort.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Excyclotorsion**	Torsion is the discriminator, so any torsion looks like the answer.	Direction decides: extorsion of the higher eye is the fourth nerve palsy.
B	**Limited depression**	It is an underaction, and underactions occur in central disease too.	Depression failing in adduction is superior oblique weakness itself.

IF-THEN RULES FROM THIS CASE

If a vertical deviation follows the three-step pattern with symptomatic extorsion, it is an acquired fourth nerve palsy. If the torsion runs the other way, think skew and image.

Case S1-29 · why

1 · E 0.45 mm of axial elongation in a year

Progression is measured as axial growth, and 0.45 mm in a year at nine is roughly twice the expected rate. History and habits raise risk but they do not measure what this eye is doing.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Two myopic parents**	Parental myopia is the best-known risk factor for onset.	It predicts who becomes myopic, not how fast this eye is growing.
D	**Accommodative lag**	Lag theories of progression are still widely taught.	Lag has not predicted progression in trials, and 0.75 D is unremarkable.

2 · A Low-dose atropine 0.05% nightly

Only low-dose atropine here has slowed axial growth; orthokeratology and dual-focus soft lenses are the other proven options. Undercorrection accelerates progression and single vision is the comparator.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Undercorrection**	Less minus feels like less stimulus for the eye to grow.	Trials showed faster progression with undercorrection, not slower.
D	**Progressive addition lens**	Reducing the near demand is a plausible mechanism and is well marketed.	The effect measured in trials is small and does not change management.

3 · A Correct fully, start control, remeasure at six months

She is progressing fast enough to treat, so the visit ends with a chosen intervention, advice on outdoor time, and a measurement interval short enough to judge the effect.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Review in a year**	Annual review is the routine interval for a child in spectacles.	Another 0.45 mm of growth passes before anyone notices the failure.
C	**Atropine 1%**	The strongest concentration should give the largest effect.	It brings glare, near blur and marked rebound for little extra benefit.

4 · B Confirm compliance, then combine treatments

A quarter of a millimetre in six months is inadequate control, and the two commonest reasons are missed doses and too weak a stimulus. Combination therapy is the next step, not a stronger drop.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Atropine 1%**	Escalating the dose is the reflex when a drug underperforms.	Side effects rise steeply and rebound is worse, so combine instead.
E	**Remeasure in a year**	Trends need time, and a single measurement can mislead.	The trend is already unfavourable, and twelve months loses the window.

5 · D, E, F Disclose off-label use, consent, document alternatives

Off-label prescribing is lawful where it is clinically justified. The duty is disclosure of licensing status and alternatives, consent from the person with parental responsibility, and a record of both.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Notify the board**	Regulatory notification sounds like the safe and compliant step.	No such duty exists for prescribing within scope of practice.
C	**Child countersigns**	Involving the child respects her developing autonomy.	At nine she assents, while consent is the parent's to give and record.

IF-THEN RULES FROM THIS CASE

If axial length rises more than 0.20 mm in six months, treat rather than watch. If the drop is used off-label, record the licensing status, the alternatives and the consent.

Case S1-30 · why

1 · C 3.2 Δ net, base up before the right eye

Prism equals decentration in centimetres times lens power. Right 0.8 cm x 5.00 D gives 4.0 Δ base up, left 0.8 cm x 1.00 D gives 0.8 Δ base up, so the net imbalance is 3.2 Δ.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**4.0 Δ**	It is the prism induced by the right lens, correctly calculated.	The left lens induces 0.8 Δ the same way, and only the difference counts.
E	**4.8 Δ**	Adding the two prisms feels right when both eyes are affected.	Both bases are up, so the two prisms subtract rather than add.

2 · C Induced imbalance beats the fusion range

Vertical fusional amplitude is small, 2 Δ here, and the reading position generates 3.2 Δ. The deviation appears only in downgaze, which is why her distance vision is comfortable.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Decompensating phoria**	New spectacles often unmask a phoria, and the timing fits exactly.	The distance cover test is orthophoric and the symptom is position dependent.
E	**Superior oblique palsy**	Vertical diplopia worst on reading is the palsy's textbook symptom.	No torsion, no gaze or tilt pattern, and it began with the new lenses.

3 · A, E, F Near pair, slab-off, contact lenses

The imbalance is decentration multiplied by unequal power, so each fix removes one of the two: centre the lenses where she looks, cancel the difference, or move the correction to the cornea.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Higher fitting cross**	Adjusting the fitting height does sound like the lever to pull.	Raising the centres increases the drop in downgaze and the imbalance with it.
D	**Polycarbonate lenses**	Thinner lighter lenses do improve tolerance of an anisometropic Rx.	Material changes cosmetics and weight, never the induced prism.

4 · C Dedicated near pair, centres lowered

Centring each lens where she looks removes the decentration, and with it the induced prism, without touching her distance correction or her acuity in either eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Reduce the right power**	Equalising the powers does remove the imbalance arithmetically.	It leaves the right eye 1.00 D under-corrected and blurred at all distances.
D	**Base-down in the right**	Prescribing the opposing prism looks like a direct correction.	It corrects downgaze and breaks fusion at distance, where she is comfortable.

5 · D Calculate induced prism at the near position

With 4.00 D of anisometropia, any vertical drop below the optical centres creates imbalance, so the calculation belongs at the design stage, before the lenses are cut and glazed.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Vertex distance**	Vertex distance matters at these powers and is easy to check.	It changes effective power, not the prism induced by decentration.
A	**IPD to 1 mm**	Centration error is the problem, so a PD check feels right.	Horizontal centration is correct here, and the drop is vertical.

> **IF-THEN RULES FROM THIS CASE**
>
> If lens powers differ by more than 1.00 D, compute the induced prism at the reading position before dispensing. If it exceeds the vertical fusion range, centre a separate near pair.

Case S1-31 · why

1 · C Over-minus from a shorter dispensed vertex

A minus lens moved closer to the eye gains effective minus power, so -12.00 specified at 14 mm is too strong at 9 mm. Sliding the frame out restores the refracting vertex and the acuity.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Pantoscopic tilt**	Tilt does induce cylinder, and it grows fast with lens power.	Tilt-induced blur does not clear when the frame slides away from the eye.
B	**Aniseikonia**	Magnification effects are real at -12.00 and cause asthenopia.	Both eyes carry the same power, so the two retinal images match in size.

2 · D -11.25 DS at the 9 mm vertex

The lens sits 5 mm closer, so it needs less minus: F = -12.00 / (1 + 0.005 × 12.00) = -11.32 D, which is -11.25 to the nearest quarter dioptre.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**-13.00 DS**	It assumes a shorter vertex needs more minus, the commoner reflex.	Closer to the eye a minus lens gains effective power, so it needs less.
E	**-10.25 DS**	It is the right arithmetic for the corneal plane, a contact lens power.	The lens plane here is 9 mm, not 0 mm, so only 5 mm of the shift applies.

3 · D, E Clearer with the frame out, plus accommodative ache

Under-correction does not improve as the lens moves out, and it does not force accommodation. Both features here are the signature of too much minus in front of an eye working to clear it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
F	**14 mm refracting vertex**	It is where the error starts, so it feels like the key finding.	It gives the mechanism, not the direction the error runs in.
A	**Acuity 20/25**	Reduced acuity is the presenting complaint and is genuinely abnormal.	Over-minus and under-correction both blur distance by a similar amount.

4 · D Remake at -11.25 DS for the fitted vertex

The frame was measured correctly, the power was not. Respecifying the power for the plane it will be worn at keeps the short vertex, which a high myope wants for field and magnification.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Hold the frame at 14 mm**	It restores the conditions the refraction was measured under.	It gives away the field and the reduced minification a short vertex buys.
B	**Reading addition**	The ache is accommodative, and an add relieves accommodative strain.	It leaves the distance power wrong and asks a 24-year-old to stop accommodating.

5 · D -10.25 D at the corneal plane

At the cornea the whole 14 mm applies: F = -12.00 / (1 + 0.014 × 12.00) = -10.27 D. Order -10.25 and check the over-refraction on the eye.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**-12.00 D**	Copying the spectacle power is quick and works at low powers.	At -12.00 it over-minuses by about 1.75 D at the corneal plane.
B	**-11.25 D**	It is the right compensation, but for the spectacle plane at 9 mm.	A contact lens sits at the cornea, so the full 14 mm shift applies.

IF-THEN RULES FROM THIS CASE

If a high power is dispensed at a vertex other than the one it was refracted at, respecify the power for the plane it will be worn at. If the lens moves closer, minus needs less minus.

Case S1-32 · why

1 · E Sterile contact lens peripheral ulcer

A small peripheral infiltrate, clear cornea between it and the limbus, a quiet chamber, normal acuity and relief on lens removal is a hypersensitivity response, not an invading organism.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Bacterial keratitis**	Any infiltrate in a wearer who slept in lenses could be microbial.	Microbial ulcers are painful, central or paracentral, and the chamber reacts.
A	**Marginal keratitis**	The lesion sits peripherally, where staph hypersensitivity lands.	Lid margins are clean and there is no blepharitis to drive the reaction.

2 · D, E, F Clear limbal zone, small stain, quiet chamber

A sterile infiltrate is an immune response to lens-borne antigen: the epithelium stays largely intact, the stroma reacts locally, and the chamber stays quiet. Microbial invasion breaks all three.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Overnight onset**	Sleeping in lenses precedes both sterile and microbial disease.	Timing names the risk factor, not which of the two followed it.
C	**Acuity 20/20**	Preserved acuity feels reassuring and argues against infection.	A peripheral microbial ulcer can also spare the visual axis on day one.

3 · D Stop lenses, antibiotic cover, review at 24 hours

Removing the antigen source is the treatment. Cover is prophylactic while a small peripheral infiltrate is watched, and the 24-hour review is what separates a settling sterile lesion from an ulcer.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**Lubricants, one week**	Sterile infiltrates do settle on lens removal alone.	A week is too long to wait to find out the first diagnosis was wrong.
B	**Steroid and a refit**	Suppressing an immune infiltrate treats the mechanism directly.	Steroid without cover, and a lens back on a cornea not yet proven sterile.

4 · E Bigger, denser, defect wider than infiltrate, cells

Size, density, an epithelial break larger than the infiltrate beneath it, and any chamber reaction are the microbial markers. Progression on cover is the other indication to scrape.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**A second small infiltrate**	New lesions sound like spread, and satellites suggest fungus.	Multiple small peripheral infiltrates are the usual sterile pattern.
D	**Injection now diffuse**	The eye looks angrier, which is what the patient reports.	Injection tracks irritation. The infiltrate and the chamber have not moved.

5 · A Defer release, record that wear is unsafe

The duty to hand over a lens prescription runs from a completed fitting on a healthy eye. An active corneal infiltrate makes that lens the wrong device today, and the record has to say so.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Release with a caution**	It respects the release duty while passing on the clinical advice.	The supplier dispenses the power, not the caveat, and lenses arrive in days.
E	**Release as requested**	A verified fitting on file reads like a settled entitlement.	The entitlement is to a valid prescription, and validity turns on the eye today.

IF-THEN RULES FROM THIS CASE

If an infiltrate is small, peripheral, with a clear limbal zone and a quiet chamber, treat it as sterile and review at 24 hours. If it grows or the chamber reacts, culture it.

Case S1-33 · why

1 · C +6.25 D add, working at 16 cm

Kestenbaum: the starting add is the reciprocal of the acuity fraction, so 125/20 = 6.25 D. Its focal distance is 1/6.25 = 0.16 m, and the page has to be held there.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**+10.00 D at 10 cm**	It is the familiar answer, but for 20/200 rather than 20/125.	The reciprocal of 125/20 is 6.25, not 10.
A	**+4.00 D at 25 cm**	It matches the 25 cm she already uses and feels comfortable.	+4.00 magnifies too little to bring 1.6 M performance down to 1 M.

2 · D Scotoma to the right of fixation

Letter acuity measures one target at a time. A scotoma on the side the eye is moving into hides the next word, so saccades land in the blind area and the line is lost.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Acuity reserve**	Too small a reserve does slow reading, and hers is modest.	Reserve limits speed and endurance, not the ability to track a line.
A	**Poor illumination**	One ceiling light is genuinely inadequate for atrophy.	Light raises contrast and comfort, but it cannot fill a scotoma.

3 · A, B Task lighting at the page, eccentric viewing

Atrophy needs light on the page, and a scotoma over the incoming text needs a new preferred locus. Both act on what limits her, unlike a coating, a tint, or a lens held out of focus.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Yellow tint**	Tints are popular and can cut glare on a bright page.	They also cut light reaching a retina that needs more of it.
D	**3x hand magnifier**	3x is close to the magnification the arithmetic asks for.	Held at arm's length it is out of focus, and her tremor moves it.

4 · B Illuminated stand magnifier

A stand magnifier holds the lens at its focal distance for her, which a resting tremor cannot do with a hand-held lens, and its own light supplies what one ceiling fitting does not.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Hand magnifier**	It is portable, cheap, and the usual first device offered.	A resting tremor moves the lens off its focal distance continuously.
D	**+6.25 D at 25 cm**	The add is the one the arithmetic gives, and the habit is hers.	A +6.25 D lens focuses at 16 cm. At 25 cm the print is still blurred.

5 · A State the standard, document, name the duty to notify

The consultation is where this is settled: a plain statement that she is below the standard, the advice not to drive, her response, and who must notify. The record of that talk is the evidence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Notify without telling her**	Reporting duties exist and other road users are genuinely at risk.	Notification never precedes telling the patient, whoever has to do it.
C	**Daylight, familiar routes**	It sounds proportionate and preserves her independence.	An acuity below the standard is below it in daylight on a known road.

> **IF-THEN RULES FROM THIS CASE**
>
> If letter acuity is better than reading performance, look for the scotoma and the lighting before adding magnification. If acuity is below the driving standard, say so in the consultation.

Case S1-34 · why

1 · A Ethambutol toxic optic neuropathy

1500 mg for 60 kg is 25 mg/kg, and an eGFR of 42 slows clearance. Symmetric central depression, colour loss out of proportion to acuity, and no RAPD fit a dose-related toxic insult.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**B12 deficiency**	It produces the same painless, symmetric, central colour loss.	The dose here is 25 mg/kg with reduced clearance, which explains it.
B	**Optic neuritis**	Acquired colour loss with reduced acuity is the classic teaching set.	Neuritis is painful, usually unilateral, and gives an RAPD.

2 · A Red-green loss, from optic nerve disease

Köllner's rule pairs outer retinal disease with blue-yellow loss and optic nerve disease with red-green loss. Glaucoma and dominant optic atrophy are the standing exceptions.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
B	**Blue-yellow, outer retina**	It is the other half of the rule, stated correctly.	The maculae are normal and the field loss is central and symmetric.
E	**Red-green, gene linkage**	It reaches the right answer through a real piece of genetics.	Gene arrangement explains inherited defects, not acquired ones.

3 · E Stop ethambutol today, through the prescriber

Toxicity is dose and duration dependent, and recovery turns on how fast the drug goes. Stopping is the physician's decision, so the call happens today, with the dose per kg and the eGFR.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Reduce to 15 mg/kg**	Toxicity is dose related, so a lower dose looks like the fix.	No dose is safe once the nerve is affected, and three months is too long.
D	**Recheck in four weeks**	Confirming progression before acting feels careful.	Four more weeks of the drug is four more weeks of loss.

4 · D, E Write to the prescriber, and fix a measured baseline

The two jobs are to close the loop with the prescriber in writing and to fix a measurable baseline. Recovery is judged against recorded acuity, colour and fields, not against recall.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Hydroxocobalamin**	Vitamin repletion is standard in nutritional optic neuropathy.	This is a drug effect. There is no deficiency here to replace.
B	**Monthly ERG**	It sounds like objective monitoring of the visual pathway.	The ERG measures retina. The lesion here is in the optic nerve.

5 · D Unilateral RAPD with pain on eye movement

Toxic neuropathy is symmetric and painless, so an RAPD means the two nerves are unequally affected and pain points at inflammation. Pallor, dense colour loss and worse renal function all fit toxicity.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Temporal disc pallor**	A new abnormal sign after months reads like a different disease.	Pallor is the late appearance of the toxic insult already described.
E	**Bitemporal wedges**	Bitemporal loss usually means compression at the chiasm.	Ethambutol has a known chiasmal pattern with bitemporal defects.

> **IF-THEN RULES FROM THIS CASE**
>
> If colour and central field fall symmetrically with no RAPD, check every drug against weight and renal function. If a dose exceeds 15 mg/kg, the drug is the diagnosis until it is stopped.

Case S1-35 · why

1 · C Dilated examination with indentation today

Photopsia with new floaters is a symptomatic vitreous detachment, and axial myopia with recent surgery raises the odds of a tear. Only a dilated peripheral view with indentation settles it.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
D	**Within two weeks**	It books a proper examination, and most of these are uncomplicated.	A tear is treatable while the retina is attached, and that window is days.
E	**Emergency department**	It matches the urgency and gets her seen out of hours.	What she needs is dilated indentation, not an ultrasound scan.

2 · C 50%, one flagged call in two

Flagged calls are 27 true and 27 false, so 27/54 = 50%. Sensitivity 27/30 = 90% and specificity 143/170 = 84% describe the protocol; what a flag means also depends on prevalence.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
E	**90%**	It is the sensitivity, the figure the protocol is sold on.	Sensitivity is the hit rate among those with disease, not among those flagged.
B	**15%**	It is the prevalence, which sounds like the base rate answer.	Prevalence is the chance before the protocol ran, not after a flag.

3 · A, B, F Understanding, the named risk, return criteria

A refusal is informed when the patient has capacity, hears the specific consequence, and knows what would change the plan. A signature and a relative's view add nothing to any of the three.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Signed waiver**	A signature looks like the document that settles the question.	A waiver does not transfer the consequences of inadequate advice.
E	**Notes written next day**	The file is what the episode is judged on, so its timing feels central.	When the note was typed says nothing about what she was told.

4 · B Risks named, refusal, offer, return criteria

The entry has to show that the choice was informed and that the door stays open: what was disclosed, what she decided, the soonest slot offered, and the symptoms that mean she calls at once.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
C	**Patient declined**	It is true, brief, and what busy after-hours notes usually say.	It records the outcome without the disclosure that makes it defensible.
D	**Discharge letter**	Refusing urgent advice feels like a breakdown of the relationship.	Abandoning a patient mid-episode creates the liability it tries to avoid.

5 · E Same-day vitreoretinal referral

A field defect after days of photopsia means retina is off until proved otherwise, and macula-on repair is time critical. A superior nasal shadow places the break inferiorly and temporally.

OPT	DISTRACTOR	TEMPTING BECAUSE	RULED OUT BY
A	**Practice, next morning**	It gets her the dilated examination she should have had on day one.	The examination is no longer the question. A field defect needs a surgeon.
D	**Repeat the protocol**	The protocol is the practice's agreed triage tool for these calls.	Triage tools sort symptoms. A field defect has already answered it.

IF-THEN RULES FROM THIS CASE

If photopsia comes with new floaters, examine dilated with indentation within a day. If a refusal is reasoned, record the risk named, the offer made, and what should bring the patient back.

PART 10

Reference

The answer key, an eighteen-day schedule keyed to real case numbers, a cut-line recall deck, and the sources with their link status recorded.

ANSWER KEY

Letters were generated to a distribution, not chosen by an author. The audit that proves it is on page 4, printed from the same build.

EIGHTEEN DAYS

New cases on the left, recall on the right. If you are short of time, cut new cases and keep the recall column.

RECALL DECK

Cut on the dashed lines. Ten seconds per prompt, answered aloud. Misses go back into tomorrow's pile.

SOURCES

Every claim carries a state: verified, unconfirmed, or removed. Nothing in this manual is certified by its own author.

Answer key

Letters are generated to the distribution audited on page 4, not chosen by whoever wrote the item. Guessing one letter throughout scores close to a fifth of the paper, which is what it should score. Multiple letters in a cell mean an all-or-none item: every one, or no mark.

Session 1

CASE	1	2	3	4	5	6
S1-01	B	A, D, F	E	A	A	
S1-02	C	D, E, F	A	B	D	
S1-03	C	B	A	B	D, E	
S1-04	A	B, C, D	C	A	D	
S1-05	A	D	A, F	D	A	
S1-06	B	C	C	D, E, F	D	
S1-07	C	D	B	E	A, E	
S1-08	D	D	A, B	A	B	
S1-09	B	C, D, E	A	E	B	
S1-10	E	B	A, B, C	E	A	
S1-11	E	C	A, B, F	A	E	
S1-12	B	A, B, F	E	B	E	
S1-13	C	B	A, B	A	D	
S1-14	B	E	C, D, E	C	A	
S1-15	B	A, E	B	D	D	
S1-16	E	D, E	E	C	D	
S1-17	B	E	A, B, F	E	E	
S1-18	C	E	A, B	B	E	
S1-19	D	A, B	D	D	E	
S1-20	C	A	D	C, D	E	
S1-21	B	C, D, E	B	B	D	
S1-22	C	B, C, D	A	B	C	
S1-23	A	C	A, B, F	C	D	
S1-24	E	A, B, F	A	C	C	
S1-25	E	A, E, F	C	C	D	
S1-26	B	D, E, F	A	C	A	
S1-27	C	A	B, C, D	C	B	
S1-28	E	B, C, D	D	D	D	
S1-29	E	A	A	B	D, E, F	
S1-30	C	C	A, E, F	C	D	
S1-31	C	D	D, E	D	D	
S1-32	E	D, E, F	D	E	A	
S1-33	C	D	A, B	B	A	
S1-34	A	A	E	D, E	D	
S1-35	C	C	A, B, F	B	E	

Eleven days

One session, eleven days. Cases are answered cold, once. Everything after that is recall, not rereading. The right-hand column is the part people skip and the part that moves the score.

DAY	NEW	WORK	RECALL
Day 1	S1-01 to S1-05	Cases cold, keys the same evening	-
Day 2	S1-06 to S1-10	Cases cold, keys the same evening	S1-01 to S1-05, recall cards
Day 3	S1-11 to S1-15	Cases cold, keys the same evening	S1-01 to S1-10, recall cards
Day 4	S1-16 to S1-20	Cases cold, keys the same evening	S1-06 to S1-15, recall cards
Day 5	S1-21 to S1-25	Cases cold, keys the same evening	S1-11 to S1-20, recall cards
Day 6	S1-26 to S1-30	Cases cold, keys the same evening	S1-16 to S1-25, recall cards
Day 7	S1-31 to S1-35	Cases cold, keys the same evening	S1-21 to S1-30, recall cards
Day 8	Repair	The two weakest item-type columns	Condition cards and pharmacology
Day 9	Repair	Every item missed with high confidence	Rewrite each as one if-then rule
Day 10	Timed run	All 175 items in one 3.5 hour block	Score by item type, compare to the first pass
Day 11	Consolidate	Emergency gate and competing pairs, aloud	Formulas drilled, then Volume 2

BOTH VOLUMES, THREE WEEKS
Run this volume, then Volume 2 on the same eleven-day shape. Compare the two score sheets by item-type column, not by total. A total that rose while the treatment column fell is a warning, not progress.

IF YOU HAVE FEWER THAN ELEVEN DAYS
Keep the recall column and cut new cases, not the reverse. Ten cases reviewed to the point of recall beat thirty-five read once. With four days: the emergency gate, the competing pairs, and one timed half session.

Cut-line recall deck

Cut on the dashed lines. Ten seconds per prompt, answered aloud. Misses go back into tomorrow's pile.

Painful red eye in a contact lens wearer

Microbial keratitis until the slit lamp says otherwise. Measure it, culture if central, treat intensively, review inside 24 hours.

Never patch.

Pain out of proportion, water exposure, poor antibiotic response

Acanthamoeba. Culture, PCR, or confocal, repeated if suspicion persists. Ring infiltrate is late, not required.

Steroid makes it worse.

Lid swelling plus painful restricted motility

Orbital cellulitis. Emergency department, contrast CT of orbits and sinuses, intravenous antibiotics, ENT.

Colour and motility are the gate.

Over 50, profound loss, chalky pallid disc oedema

Arteritic AION. ESR, CRP, platelets, and immediate high-dose systemic steroid.

A normal ESR does not clear it.

Sudden painless profound monocular loss

Retinal artery occlusion, an acute stroke syndrome. Stroke pathway now, embolic source workup, GCA screen by age.

No office massage.

Flashes, shower of floaters, pigment in the vitreous

Retinal break until the periphery has been examined with indentation.

A normal central OCT proves nothing.

Deck, continued

New distortion with subretinal fluid in an older patient

Neovascular AMD until proven otherwise. Urgent retina referral for anti-VEGF.

Good acuity is a reason to treat quickly, not to wait.

Bilateral myopic shift with uniformly shallow chambers on a new drug

Ciliochoroidal effusion, commonly topiramate. Stop the drug with the prescriber, suppress aqueous, cycloplege.

Not pupillary block: no iridotomy, no miotics.

Acute ptosis and miosis after neck pain

Painful Horner: carotid dissection until imaged. Emergency head and neck vascular imaging.

Do not wait to finish localisation drops.

Recurrent chalazion at the same site with lash loss

Sebaceous carcinoma. Full-thickness biopsy with conjunctival mapping, examine nodes.

Not a third curettage.

Multiple-response item

Count the requested number, judge each option alone, choose patient-specific over generally true.

Partial sets score zero.

High-confidence miss

Top of the error log. Name the false rule, write the replacement as one if-then sentence, retest tomorrow from a changed stem.

> **FINAL 48 HOURS**
> Red flags, drug contraindications, formulas, image algorithms, and logistics only. No new content. Between the two sessions, do not autopsy Session 1: eat, hydrate, reset, and start the second session on pacing.

Sources and status

Official NBEO documents control scope and weighting. Everything below was checked on 15 August 2026, and link status is recorded so a reader can tell what was actually opened.

Official

STATUS	DOCUMENT
READ	NBEO. Part II PAM/TMOD exam information. optometry.org/exams/part-ii-pam-tmod. Source for item count, case size, session structure, timing, TMOD breakout, and delivery.
RESOLVES	NBEO. Part II Content Matrix, PDF dated May 2026. Link returns 200; content not machine-read for this edition, so domain ranges remain unconfirmed.
RESOLVES	NBEO. Part II PAM Content Outline and TMOD Content Outline, PDFs dated May 2026.
RESOLVES	NBEO. PAM Sample Cases and Test Day Instructions, PDFs dated May 2026.
DEAD	Pearson VUE tutorial deep link. Did not resolve; NBEO states the tutorial is being updated. Reach it from the exam page.

Clinical anchors

Mac Grory B, et al. Management of central retinal artery occlusion. *Stroke* 2021;52:e282-e294. · Marmor MF, et al. Recommendations on screening for chloroquine and hydroxychloroquine retinopathy, 2016 revision. *Ophthalmology* 2016;123:1386-1394. · Herpetic Eye Disease Study Group. *N Engl J Med* 1998;339:300-306. · Beck RW, et al. Corticosteroids in acute optic neuritis. *N Engl J Med* 1992;326:581-588. · AREDS2 Research Group. *JAMA* 2013;309:2005-2015. · CDC. Sexually transmitted infections treatment guidelines, 2021. · American Diabetes Association. Standards of care in diabetes.

Learning method

Dunlosky J, et al. Improving students' learning with effective learning techniques. *Psychol Sci Public Interest* 2013;14:4-58. · Roediger HL, Karpicke JD. Test-enhanced learning. *Psychol Sci* 2006;17:249-255.

KNOWN LIMITS OF THIS EDITION

Domain ranges and item-type bands are transcribed, not confirmed against the current matrix. Both sessions are complete: 70 cases and 350 items, every one with a generated answer letter and a distractor table. Clinical anchors are exam-oriented simplifications: verify labelling, dose, interactions, and local law before treating a patient.

ORIGINALITY AND INDEPENDENCE

Every case, option, rationale, table, and diagram here is original. Nothing reproduces NBEO secure items or copyrighted case images. This is an independent resource, not endorsed by NBEO, and no resource can guarantee a passing score.

How this was made

Recorded because the previous edition's defects were invisible to its reader, and because a claim you cannot check is worth less than a method you can.

Build

Content	Cases are structured data. Each item carries its options with the correct one flagged in place, an explanation, and a distractor table naming what makes each wrong option tempting and the finding that rules it out.
Answer letters	Generated after authoring, to an even distribution with no run longer than three and every letter live in both sessions. Options are rotated rather than shuffled, so ordered options stay ordered.
Layout	Every page is a fixed letter-sized box. A cost model estimates the height of each block and packs content into sheets, and a headless browser then measures the real render and fails the build if any page is clipped.
Numbering	Page numbers, the table of contents, both case indexes and the printed answer key are all generated after layout, so they cannot drift away from the content.
Print	Printed through the browser's own engine at letter size with the document owning its page geometry. The exported file is then checked for page count, page box and embedded fonts.

Typography

Display	Instrument Serif, for the page titles only.
Text	Spectral at 10.5 point, down to 8 point in the distractor tables and nowhere smaller. Chosen because it has static weights: the print engine silently substitutes a fallback for variable-axis fonts, which loses the body face in the exported file without any warning on screen.
Labels and numbers	IBM Plex Mono, for anything that has to be scanned rather than read.
One deliberate exception	The Greek delta in prism notation is set in a fallback face, in both weights, because Spectral has no Greek coverage. That was established by exporting the file with the character removed and comparing the embedded fonts. The alternative was to write prism notation incorrectly.
Colour	Petrol keys structure and navigation, red keys danger and prohibition, ochre keys a caution or a rule to memorise. Red is never used decoratively, so a red strip on a page always means the same thing.

Known limits of this edition

Unconfirmed figures	Per-domain item ranges and item-type percentage bands are transcribed from the prior edition. The official Content Matrix resolves but was not machine-read for this build. Both are marked unconfirmed wherever they appear.
Clinical simplification	Drug figures are exam anchors. Verify labelling, dose, interactions and local law before treating anybody.
Independence	Every case, option, rationale, table and diagram is original. Nothing reproduces secure NBEO items or copyrighted case images. This is an independent resource, not endorsed by NBEO, and no resource can guarantee a passing score.

Built 15 August 2026. 105 pages, 70 cases, 350 items.